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Life Cycle Assessment of an SOFC/GT Process

Pernilla Olausson

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Life Cycle Assessment of an SOFC/GT Process

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For the last few years much effort has been put into the research on different kinds of fuel cells, since these are considered to be both an efficient and environment friendly way to convert energy. The fuel cell studied here is the solid oxide fuel cell (SOFC) that works at a high temperature (800-1000 °C) and today achieves a stand-alone electric efficiency of approximately 50% (LHV). When integrating the SOFC in a gas turbine process (SOFC/GT process) an efficiency of 70-75% can be reached. The SOFC and the SOFC/GT process are considered to be environment friendly regarding the discharges during operation. Especially formation of nitrogen oxides (NO_x) is low since the SOFC temperatures are low compared to NO_x formation temperatures.

To study the whole environmental impact of the SOFC/GT process a life cycle assessment (LCA) is carried out to find the "hot spots" in the process' life cycle. Since the SOFC/GT process is under development today the collected data are mainly from literature and articles based on laboratory results. When performing the LCA only the SOFC-module and the gas turbine are included. A collection of data of all processes included, extraction of minerals, processing of raw material, production of the components, operation of the SOFC/GT process and transports between all these processes. These data are then added up and weighted in impact categories to evaluate the total environmental impact of the SOFC/GT process. All these steps are performed according to the ISO 14 040-series.

The stand-alone most contributing phase during the life cycle of the SOFC/GT process was found to be the production of the SOFC. All processes during the production of the SOFC are carried out under laboratory circumstances, which require more energy and materials than if the processes were commercialised and optimised. For the SOFC/GT process to be competitive with other energy converting processes regarding the discharges of emissions to the air, the use of energy and materials must be lowered and a lifetime of at least 40 000 hours must be achieved. Also a sophisticated waste management system ought to be developed and effort put into recycling of the SOFC, which is not the case today.

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Preface

When I started to work on this thesis I knew very little about solid oxide fuel cells and how to perform a life cycle assessment (LCA). Today after many late nights in front of the computer and a lot of e-mails I have learned what a big task it is to perform an LCA and my knowledge of fuel cells has increased. To sum up this period it has been very interesting and instructive.

First I would like to thank Maria Sunér and Eva-Marie Rönnquist at Sycon Energy Consulting for their help and support during my work with the LCA. I also would like to thank Jens Pålsson and Azra Selimovic, at the Department of Heat and Power Engineering at Lund Institute of Technology for their help with the SOFC/GT process.

I thank my examiner Professor Tord Torisson, at the Department of Heat and Power Engineering at Lund Institute of Technology and Lars Sjunnesson at Sydkraft for giving me the opportunity to perform this thesis.

I also would like to thank Carsten Bagger at Risoe National Laboratory in Denmark and all people at Sycon, who have helped me during the work.

Finally I would like to thank my mother and all my friends for their support during the rough times.

Abstract

For the last few years much effort has been put into the research on different kinds of fuel cells, since these are considered to be both an efficient and environment friendly way to convert energy. The fuel cell studied here is the solid oxide fuel cell (SOFC) that works at a high temperature (800-1000 °C) and today achieves a stand-alone electric efficiency of approximately 50% (LHV). Because of the high temperature after the energy conversion in the fuel cell, the excess heat can be used for electricity generation by afterburning and expansion of the exhaust gases leaving the SOFC in a gas turbine. When integrating the SOFC in a gas turbine process (SOFC/GT process) an efficiency of 70-75% can be reached. Many developers claim that the SOFC and the SOFC/GT process are environment friendly regarding the discharges. Especially formation of nitrogen oxides (NO_x) is low since the SOFC temperatures are low compared to NO_x formation temperatures.

To study the whole environmental impact of the SOFC/GT process a life cycle assessment (LCA) is carried out to find the "hot spots" in the process' life cycle. Since the SOFC/GT process is under development today the collected data are mainly from literature and articles based on laboratory results. When performing the LCA only the SOFC-module and the gas turbine are included. Short descriptions of the components; pre-reformer and desulphurization system are also included to get a comprehensive view of the process.

The collected data of all processes included, extraction of minerals, processing of raw material, production of the components, operation of the SOFC/GT process and transports between all these processes, are added up and weighted in impact categories to evaluate the total environmental impact of the SOFC/GT process. All these steps are performed according to the ISO 14 040-series.

The stand-alone most contributing phase during the life cycle of the SOFC/GT process was found to be the production of the SOFC. The pyrolysis process of the required raw materials and the sintering of the sub components of the SOFC require an extended amount of energy. Of course both these processes are carried out under laboratory circumstances, but even when halving the use of energy this phase is much more dominant than for other processes assessed at Sydkraft. Much effort must be put into this phase, which Risoe does, for the SOFC/GT process to be competitive with other energy converting processes regarding the discharges of emissions to the air.

When the SOFC/GT process is considered as waste the turbine and the fuel process components (pre-reformer and desulphurization system) can be recycled but the SOFC is hard to split up in its sub components again. Even the waste obtained during the production of each sub component can not be recycled today and are just placed at landfills. A sophisticated waste management system ought to be developed and effort put into recycling of the SOFC.

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1 Introduction

In the world today, the need for energy is increasing, mostly in the developing countries, but also in the industrial countries. During the last years much effort has been put into the research on different kinds of fuel cells, since these are considered to be both an efficient and environmentally friendly way to convert energy. One of these fuel cells is the solid oxide fuel cell (SOFC) that works at a high temperature (800-1000 °C) and today achieves a stand-alone electric efficiency of approximately 50% (LHV) [1]. Because of the high temperature after the energy conversion in the fuel cell, the excess heat can be used for steam generation of water, heat exchange or as the case is here, for electricity generation by after burning and expansion of SOFC exhaust gases in a gas turbine. When integrating the SOFC in a gas turbine process (SOFC/GT process) an efficiency of 70-75% can be reached if the SOFC is pressurised [1]. The other advantage for this system is the low emission level, especially the NO_x-level, which is hardly noticeable.

1.1 Background

At the Department of Heat and Power Engineering at Lund Institute of Technology, (LTH), in Lund, Sweden research on integration of a solid oxide fuel cell (SOFC) into a gas turbine (GT) process is carried out. This research is done together with the power utility of Sydkraft AB in Sweden. Since this kind of power plant can be of interest for Sydkraft in the future, and since the total environmental impact of such an integrated process is rather unknown, a life cycle assessment (LCA) of the SOFC/GT process is to be done in this thesis.

1.2 Objective

The purpose of this thesis, is to look at the total environmental impact that the SOFC/GT process is causing during its whole life time. A tool that can be used to do this, is the life cycle assessment, which is an ISO 14 040 procedure. When doing an LCA, the material flow of the product or service is investigated, characterised and classified, which leads to interpretation of the environmental impact and finding environmentally dangerous "hot spots" for the total system. The LCA will have a similar disposition as other life cycle assessments performed at Sydkraft to follow Sydkraft's "general detail report". This makes it easier for Sydkraft to use this report since there is a continuity.

1.3 Method

Since the solid oxide fuel cell and the SOFC/GT-process are yet not commercialised, most of the data used here are from literature studies and laboratory results from developers. The collected data are then processed in the life cycle assessment program, EcoLab, to estimate the total environmental impact that the SOFC/GT process has.

1.4 Assumptions

The assumptions made of the collected data in the LCA are described in detail in chapter 4.6.2. The most significant assumption made here is the material composition of the turbine and the total weight of the gas turbine. Also almost all data have been found in literature where all tests of the components have been executed under laboratory circumstances. Maybe these results have lead to an increase in the use of material and energy during the production of the components. But since the production of especially the SOFC is not yet commercialised these results have been assumed to be close enough for this first LCA of the SOFC/GT process.

1.5 Short description of the program EcoLab

EcoLab is an LCA program that is manufactured by the Swedish company Nordic Port. In this program databases are built up through input of new substances and activities into the present database. An activity is for example production of a component that requires a certain material input, energy input and waste output. The input of this data is collected during the inventory analysis and is then fed into the program. After the activities are built up they are connected to each other in a flow chart scenario (see figure 1-1) and then added up into inventory results. The results are then assessed by entering the classification indexes. These indexes of emissions are weighting parameters that have been estimated for each emission, depending on their impact media (see chapter 6.2). When the creation of activities and the entering of indexes are finished the inventory data results and the impact assessment results are presented in tables and the assessment is done. The assumptions of the data in the LCA is further described in chapter 4.6.2.

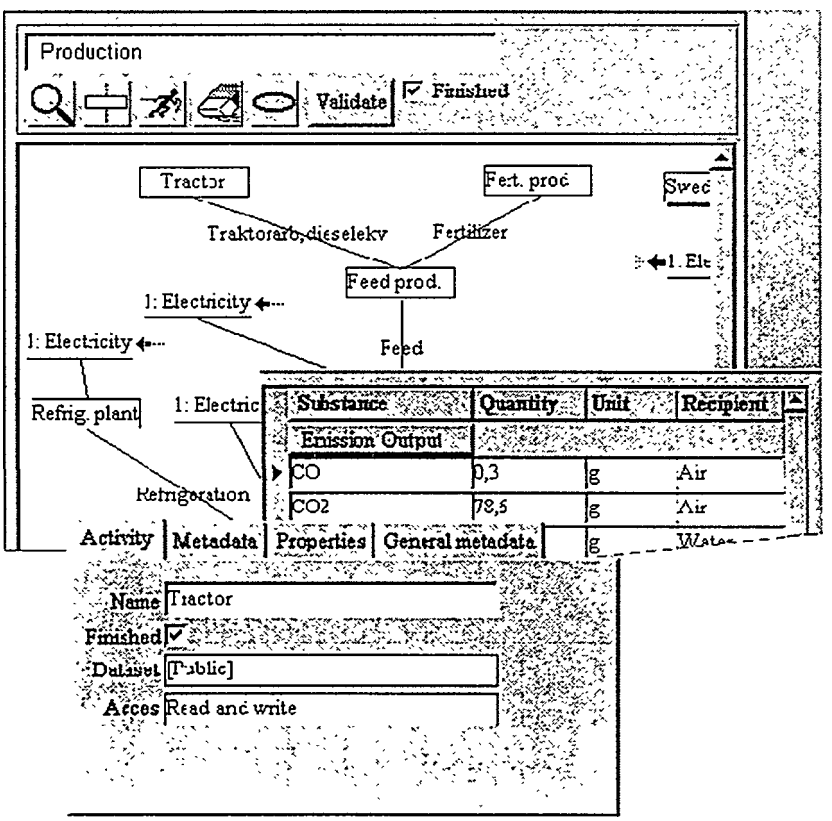


Figure 1-1. Presentation of a flow chart scenario in EcoLab for created activities [1]

2 Technical description of the SOFC/GT process and its components

The SOFC/GT process that is investigated here, is a combination of a SOFC-module with a gas turbine, where the process diagram is presented in figure 2-1.

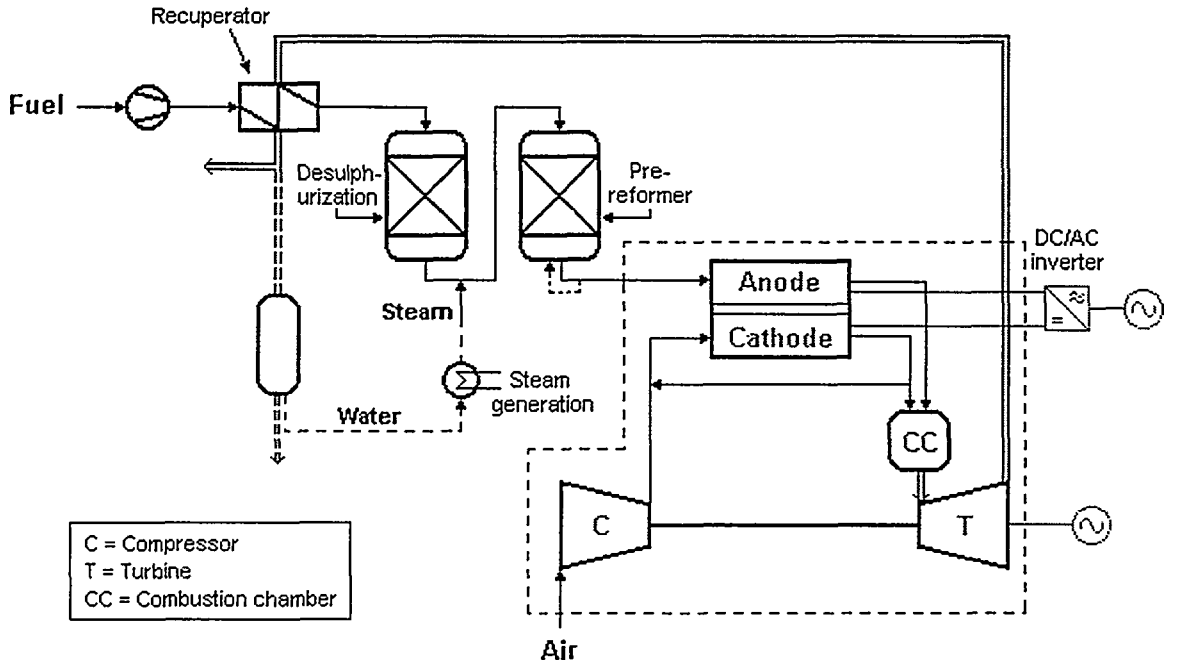


Figure 2-1. Process diagram of SOFC/GT process.

The dotted line around the SOFC-model (presented as anode and cathode in figure 2-1) and the turbine is the system boundary for the process that will be studied in the life cycle assessment. Main components of the process, shown in figure 2-1 and further described in chapter 2.2 - 2.5, are:

- SOFC-module, represented as anode and cathode in figure 2-1
- Gas turbine
- Pre-reformer - external steam-reformer
- Desulphurization system

Other components that are present in the process are the recuperator, fuel compressor, DC/AC inverter and other electronic equipment.

The SOFC/GT process is one of the processes within the fuel cell utilisation that today is considered very promising, especially in the production of electricity. A future scenario for the SOFC/GT process, when compered with other stationary power units, is represented in figure 2-2.

In the diagram, the curve that represent SOFC/GT processes, probably covers a too large power output area. Today most developers talk about the SOFC power ranges of 0,1-1 MW

and the development of SOFC/GT processes goes up to 5 MW [3]. The demonstration SOFC processes that have been evaluated the last years, only have the power range of a couple of 100 kW. The area in which the SOFC/GT process in the future will be most competitive, probably is in the local distribution of electricity. To compete with the utility size combined cycles will be very hard, because of the cost and the low SOFC life time.

In the middle of 1999, Siemens Westinghouse will be testing the worlds first SOFC/GT process with a 200 kW SOFC integrated with a 50 kW GT (two-shafted) [3].

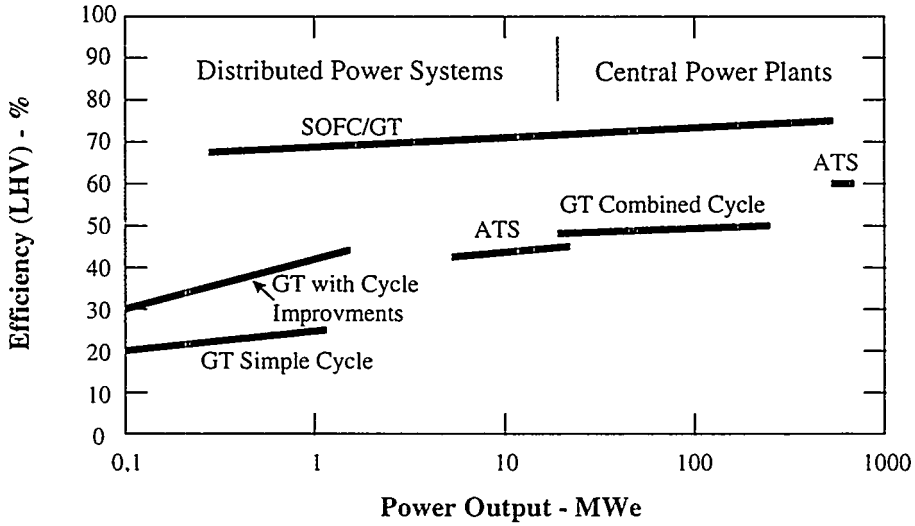


Figure 2-2. A comparison between different stationary power units [2].(ATS = Advanced Turbine Systems)

2.1 Process data

The SOFC/GT process in this study has a SOFC-module consisting of 20 000 unit cells, produced at Risoe National Laboratory in Denmark and a gas turbine assumed to be produced by Volvo Aero Turbines in Sweden. Process data for the SOFC/GT process studied here are:

The fuel cell's total power output:	226,1 kW (net)
The power output from the turbine including the compressor work:	202,2 kW (net)

The fuel's low heating value (LHV) [3]:	14,02 MJ/kg
The processes total efficiency [3]:	73,16 %

The total power output (net) of the process are 428,4 kW

The mass flows into the process are; 0,5071 (17,509 mole/s) of air and 0,04185 kg/s (2,808 mole/s) of reformed fuel into the anode [3]. The reformed fuel, which has been processed in the pre-reformer, is assumed to have a steam to carbon-ratio of 2,5, i.e. 2,5 mole steam is mixed with 1 mole of methane before entering the pre-reformer [3]. Since the Danish natural gas contains 90.4 mole % of methane [19] the total requirement of natural gas into SOFC/GT process is 0.858 mole/s (=0.0156 kg/s). The reformed fuel has the following composition entering the anode in mole % [3]:

CH_4	CO	CO_2	H_2	H_2O
17,10	2,94	4,36	26,26	49,34

The total mass flow of the exhaust fumes leaving the turbine outlet after the total combustion is 0,549 kg/s. The amount of reformed gas entering the anode are 2,808 mole/s and 0,685152 mole of these is carbon. The amount (mole) of carbon dioxide leaving the process is the same as the amount of carbon entering, i.e. 0,685152 mole/s. The air entering the combustion chamber contains of 0,3 mole % carbon dioxide that also contributes to the exhaust gas. Since the total mass flow of the exhaust gas is 0,549 kg/s the mass flow of carbon dioxide must be 0,03247 kg/s = 116,9 kg/h.

2.2 SOFC

All fuel cells are converting chemical energy directly to electricity and heat, without the middle stage of combustion and mechanical work that for instance is the case in an ordinary gas turbine process. The electricity that is produced with the SOFC is direct current (DC), which must be inverted to alternating current (AC) before it is let onto the general electricity grid.

Figure 2-3 shows how the fuel cell works under load. The fuel cell contain an anode (fuel electrode), a cathode (air electrode) and an electrolyte. At the cathode, oxygen is reduced to oxygen ions by picking up electrons from the external circuit. The oxygen ions are transported through the electrolyte to the anode oxidise the fuel and emitting electrons to the external circuit (the load). The result of this procedure is that a direct current is established from the anode to the cathode through the external load.

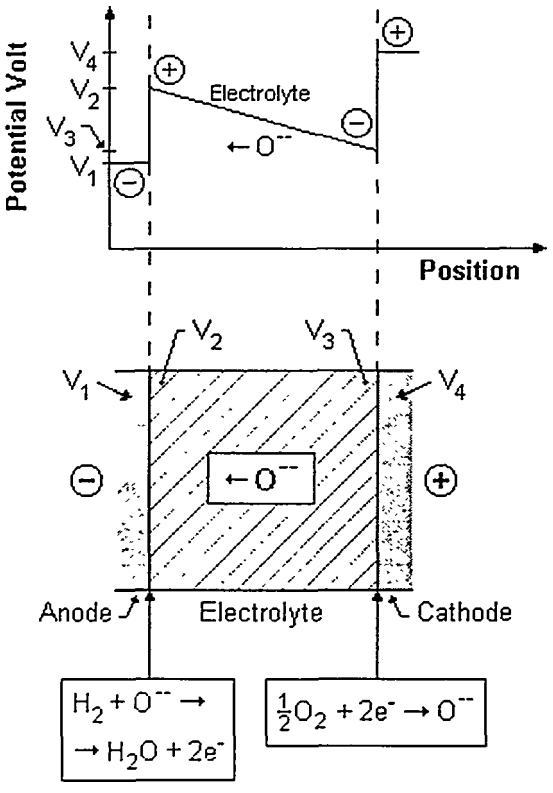


Figure 2-3. SOFC under load [4]

As shown in figure 2-3, the principal course of the potential through the fuel cell when it is under load, i.e. current is drawn from the cell, is increasing from the anode to the cathode. The potential that is measured is the one between V_4 and V_1 and it is also these two values that are being used when calculating the power of the fuel cell. When the cells are formed into a stack, the potential is being built up from fuel cell "one" to the "last" fuel cell in the stack by a series connection. The interconnect is used as an electric conductor between the anode and the cathode of the two after each other following cells.

The fuel used in most types of fuel cells, as well as in the SOFC, is the primary feedstock hydrogen. Since pure hydrogen gas is expensive, the fuel to be used in commercialised solid oxide fuel cells will be natural gas, which contains methane, that through cracking in a steam reformer forms hydrogen. As oxygen carrier air is usually used.

2.2.1 SOFC materials

The SOFC is a construction of different layers of materials in a solid state. Since the SOFC-components are in contact with each other, almost on an atomic level, there are great requirements on the material constants, especially the thermal expansion coefficient (TEC), to be as similar as possible. If the TEC varies too much between the material layers, this could lead to crack formation in the ceramic materials when fluctuating in temperature, i.e. thermal cycling. Another requirement of the materials in the SOFC is that their crystal structures have to be stable through the whole thermal cycle. Nor are the components that are in direct contact with each other not to react chemically, which also is a requirement during the operation time. To achieve these requirements above, the materials chosen in the SOFC, are not the most common materials, which will be described for each component below.

The SOFC, state-of-the-art, operates at a temperature of 1000 °C to generate power, despite the intrinsic high resistivity. Today a thin electrolyte film for the SOFC is fabricated in laboratories around the world. This type of cell has thin membranes and is used at operating temperatures around 600-800 °C. The lower temperature leads to increased flexibility in materials, longer cell life and a reduced cost, because metallic materials may be used for interconnecting and for heat exchangers.

2.2.1.1 Electrolyte

The material used in the electrolyte today is yttria stabilised zirconia (YSZ), which is a composition of 8-9 mole % of Y2O3 in ZrO2. This composition gives the electrolyte the required properties, which make it a good ion conductor and a good electronic insulator. If fabricated right, the electrolyte is also gas tight, which leads to an avoidance of spontaneous combustion within the cell [4]. A new material composition under development for the new thin-film electrolytes is Gd-doped ceria, $(Ce_{0.8}Gd_{0.2})O_{1.9}$ [5].

The thickness of the electrolyte today is 50-200 µm depending on the design. If lowering the operating temperature, which is desirable, the ionic conductivity in the YSZ decreases. To compensate this decrease, the thickness of the electrolyte can be reduced to approximately 20 µm when sophisticated fabrication methods are used [5]. This thinner geometry may also leads to a decrease of the intrinsic resistivity.

2.2.1.2 Cathode

The cathode has a porous (20-40 %) [4] design to facilitate the oxygen transportation. Because of the oxygen atmosphere the cathode material must be stable in this environment. The material must also have a good electrocatalytic activity as well as a good electronic conductivity. To achieve these requirements a ceramic material, strontium doped lanthanum-manganate, $La_{1-x}Sr_xMnO_3$ (x=0,1-0,2), has proven to be suitable [4]. This is today the most popular composition used by different developers.

In Risoe, the cathode layer is fabricated in two steps. The layer nearest to the electrolyte is the electrochemically active one and the second layer in contact with the interconnect is a current collecting layer. A total thickness of 60-80 μm is reached when both layers are completed [6]. A further description of the fabrication of the planar SOFC at Risoe can be found in chapter 5.2.3.5, "Production of a unit cell, including transports".

2.2.1.3 Anode

Also the anode is made porous (20-40 %) [4] to facilitate the transportation of fuel gas and product gases. The material used, must of course be stable in a reducing environment as well as being a good electrocatalyst. An anode made of 100 % nickel would be an outstanding electrocatalyst of hydrogen, but the TEC of nickel is too high to match with the other layers in the cell. The porous Ni-structure also would collapse fast due to creep. Hence porosity would disappear and the function would cease. In order to lower the TEC and stabilize the reduced structure of Ni, nickel has been mixed inhomogeneously with YSZ to nickel-zirconia cermet. The proportion of nickel is often set to approximately 40 vol% of the total solids [4].

In Risoe also the anode layer is fabricated in two steps with the electrochemically active layer nearest to the electrolyte and the a current collective layer nearest to the interconnect. A total thickness of 60 μm is reached, when both layers are completed [6].

2.2.1.4 Interconnect [4].

It is required of the interconnect that it should be a good electronic conductor, a good ionic insulator, gas tight and stable at both oxidising and reducing atmospheres at approximately 1000 °C. Materials that can fulfil these requirements today are based on $LaCrO_3$. The interconnect material composition, which is often used is $La_{1-x}M_xCrO_3$ (M = Sr or Ca, x = 0,1-0,3). An alternative, when operating at a lower temperature of 800-850 °C, is the exchange of lanthanum-chromite to stainless steel. In Risoe the thickness of the interconnect for 1000 ° C cells is 3 mm including gas channels, i.e. the most material demanding component.

2.2.2 Summary of the SOFC components' materials

A summary of the materials and their thermal expansion coefficient of the SOFC components is displayed in table 2-1.

Table 2-1. SOFC components' materials.

Component	Material	TEC*E6, 1/K [4]
Electrolyte	YSZ	10,5
Cathode	La(1-x)Sr(x)MnO(3)	12
Anode	NiO-YSZ	11,5
Interconnect	La(1-x)M(x)CrO(3)	10

2.2.3 SOFC-module

The SOFC-module is a bundle of unit cells formed to the required power size (kW). There are two kinds of SOFCs that will be described here, tubular and planar SOFC, figure 2-4 and 2-5. The tubular fuel cell is developed by Siemens Westinghouse and the planar fuel cell with cross flow is developed by the Risoe National Laboratory in Denmark. The reason that these two types are selected, is because Siemens Westinghouse's tubular SOFC is the fuel cell that is approaching commercialisation and the planar fuel cell from Risoe is the one investigated at LTH. The planar SOFC is also the most investigated type in Europe. Though they have a different design, they work in similar ways. The most likely lifetime for the SOFC today is approximately 10 000 hours, which will be the figure used here. The goal however of the lifetime of the SOFC is at least 40 000 operating hours at full load, which is required for commercialisation.



Figure 2-4. Bundle of SOFCs from Siemens Westinghouse [7].

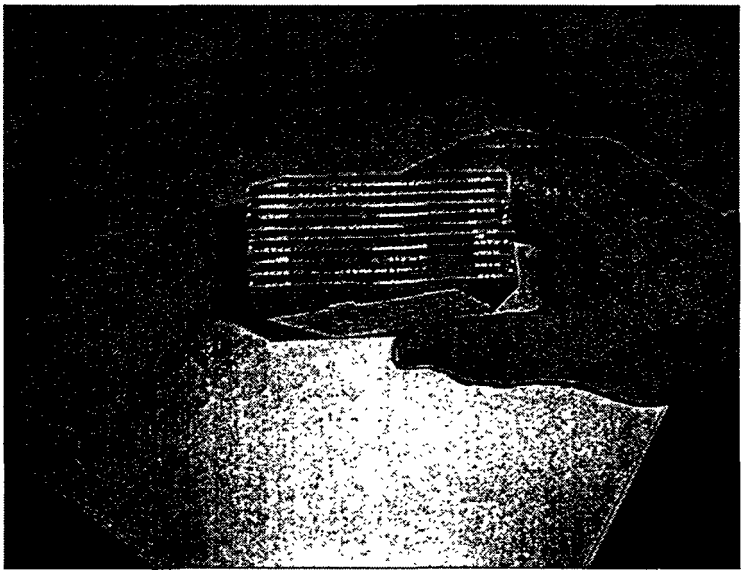


Figure 2-5. SOFC-module of 100 W from Risoe National Laboratory in Denmark.

2.2.4 Tubular SOFC

Siemens Westinghouse is one of the companies today, that puts most effort into the research on the tubular SOFC. The geometry of this tubular SOFC is shown in figure 2-6 a) and b).

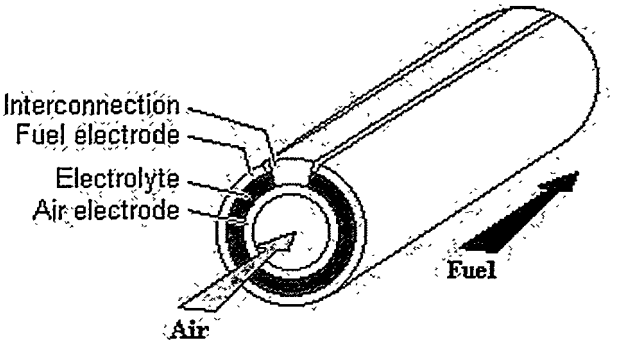


Figure 2-6 a). Siemens Westinghouse tubular SOFC [7].

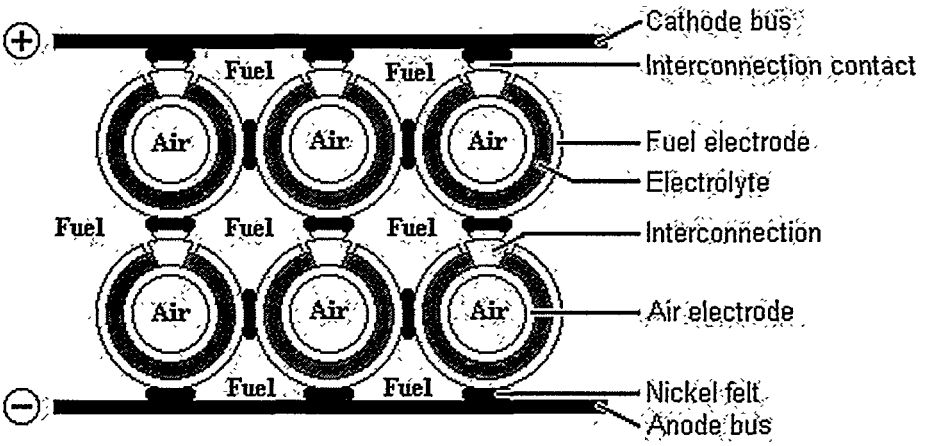


Figure 2-6 b). Bundle of Siemens Westinghouse SOFC [8].

Developers of SOFC have their own speciality in material composition, fabrication methods etc, which often is confidential. In the Siemens Westinghouse tubular design the air flows through an inner tube, while the fuel flows outside the whole cell. Every fuel cell is 0,5 m long and has a power of 25 W and the fuel cells are connected to each other by the interconnect. As shown in figure 2-7, the tubular SOFC from Siemens Westinghouse has a combustion chamber directly after the compartment, where chemical reactions take place.

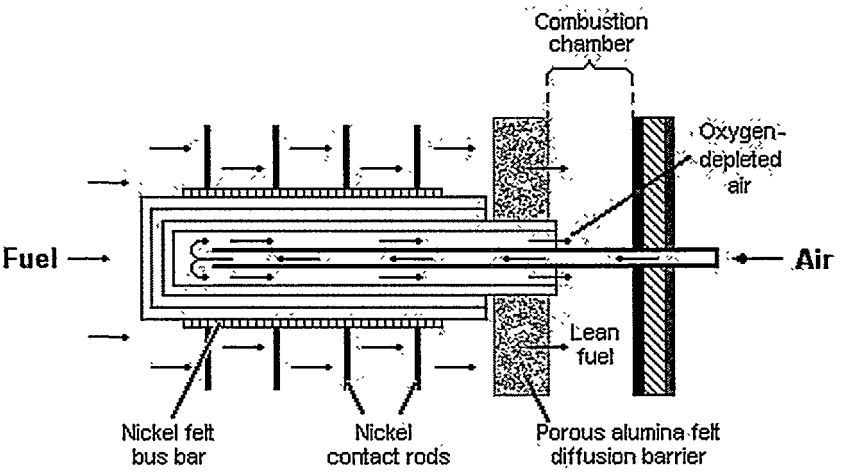


Figure 2-7. Tubular SOFC from Siemens Westinghouse [4]

Since the tubular fuel cell from Siemens Westinghouse already has a combustion chamber, the combustion chamber in picture 2-1 is not necessary when using this type of SOFC.

2.2.5 Planar SOFC with cross flow

The planar SOFC is developed in Europe, USA, Australia and Japan and is constructed as shown in figure 2-8. The planar model is cheaper to manufacture due to simpler manufacturing methods.

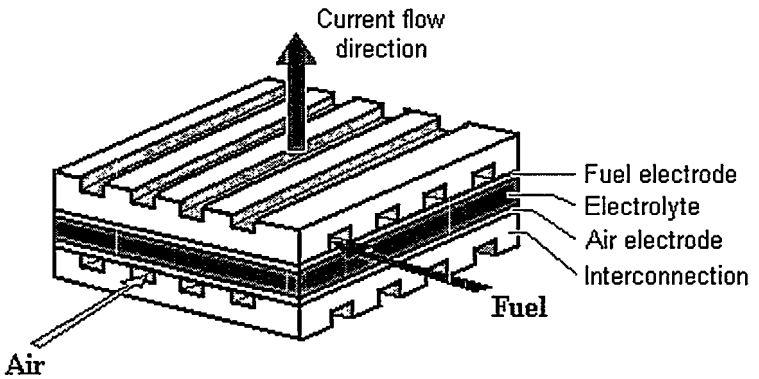


Figure 2-8. Planar SOFC with cross flow [9].

The SOFC-module, which is to be investigated in the life cycle assessment (LCA), is the module tested in Risoe during the second half of 1995. This 1st generation laboratory fuel cell had a cost of 21 000 DKK/kW when installed. Today a metallic interconnect is used, which reduces the cost of both production and materials. Today Risoe produces laboratory planar SOFCs when installed have a cost of 3 000 DKK/kW. The EU goal for the year of 2005 is set to 500 •/installed kW (≈ 3 700 DKK). With better fabrication methods of Risoe's new planar SOFC a cost of approximately 1 500 DKK/installed kW is estimated [6].

The materials of the Risoe 1st generation laboratory planar SOFC 1995 are [10]:

- Electrolyte: Yttria-stabilised zirconia, YSZ (8 mole % of Y_2O_3 in ZrO_2)
- Cathode: Doped lanthanum, $La_{0,85}Sr_{0,15}Mn_{1,1}O_3$ with YSZ, YSZ-LSM
- Anode: NiO-YSZ
- Interconnect: $La_{0,8}Sr_{0,2}Cr_{0,97}V_{0,03}O_3$ (called LSCV)

The geometry and the fabrication of this planar SOFC are further described in chapter 5, Inventory analysis.

2.3 Gas turbine

In a development project, initially involving ABB, Vattenfall and Volvo Aero Turbines (VAT) the High Speed Generation (HSG) technology was brought into daylight. This system contains a single shaft machine with the compressor, turbine and electrical generator mounted on the same shaft see figure 2-9 and 2-10.

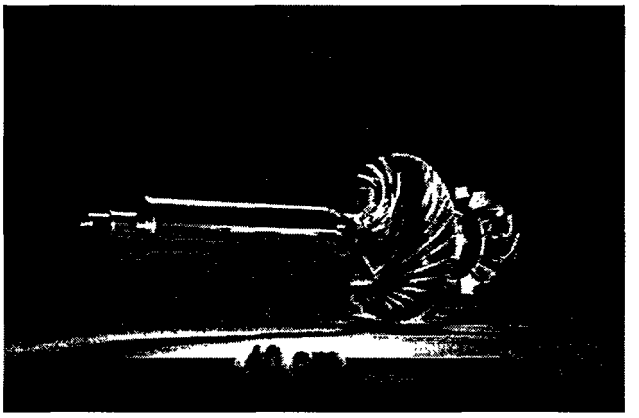


Figure 2-9. Single shaft constellation in microturbine [12].

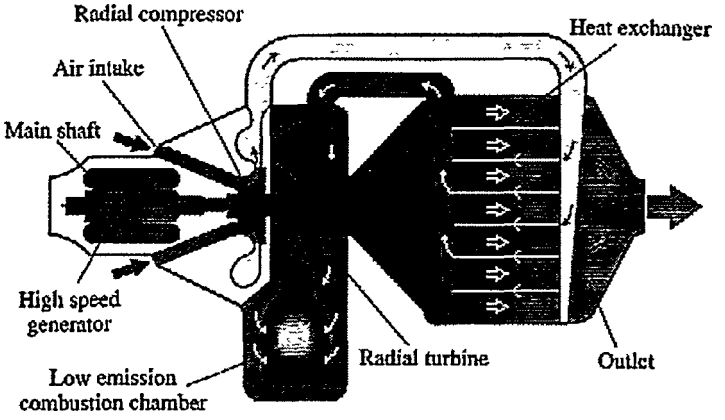


Figure 2-10. Working principles of HSG gas turbine from Volvo Aero Turbines [13]

The compressor is a single stage centrifugal compressor with a construction that avoids losses of the total pressure. The pressurised air then flows through the recuperator, which leads to a considerable raise of the temperature. The air then enters the low-emission combustion chamber, where the air is mixed with the fuel. The turbine used, is a radial type with uncooled

vanes and the gas that expands has a total temperature loss of approximately 300 °C [14]. The turbine drives the compressor and the high-speed generator at a rotation speed of 90 000 rpm (1500 Hz) at full load. The high-speed generator is a synchronous AC-machine, where the magnetic core in the generator is a strong 2-pole permanent magnet enclosed in a steel case [14].

In 1997, Vattenfall started a demonstration plant, with these HSG components, which has shown very low emissions, see table 2-2.

Table 2-2. Measured emissions at full load [14]

Substance	Amount	Unit
Carbon monoxide	8	mg/MJ fuel
Nitrogen oxides	14	mg NO2/MJ fuel
Total hydrocarbons	< 4,5	mg/MJ fuel

These low emission rates are due to the low turbine inlet temperature and the high air/fuel ratio during the combustion. Other advantages with this HSG-technology are the possibility of variation of fuels, low noise levels and high efficiency [14].

Here the SOFC/GT process has a turbine inlet temperature around 880 °C, pressure approximately 5 bar and the net power is set to 202,2 kWe. The HSG technique is assumed to be used in the future also in turbines around this size and the data collected for the 50 kWe turbine are enumerated for the 200 kW turbine. The combination of a micro turbine together with the SOFC seems to be one of the most environment friendly solutions regarding the emissions during the operation time. The micro turbine is perhaps today not totally applicable, but after some reconstruction of today's model it seems to be possible to use this type of gas turbine in the SOFC/GT process.

2.4 Pre-reformer

The reason for using a pre-reformer together with the fuel cell, is to convert the fuel into hydrogen that easily releases electrons when reacting with the oxygen in the fuel cell. A complete internal reforming in the fuel cell leads to a low temperature in the fuel inlet area of the fuel cell, when cracking the hydrocarbons. Another problem that can occur with internal reforming is the possible formation of carbon in the anode chamber when the hydrocarbons are cracked [15]. These two effects contribute to a faster degradation of the fuel cell than when a pre-reformer is used. The prominent disadvantage of the pre-reformer is the economical cost, due to an additional component.

2.4.1 Chemical reactions in the pre-reformer

The pre-reformer used here is a steam reformer, where methane and the higher hydrocarbons in the natural gas are mixed with the water steam before entering the reformer and then undergo the following reactions [17]:

$$CH_4 + H_2O \Leftrightarrow CO + 3H_2 \qquad (2\text{-}1)$$

$$C_nH_m + nH_2O \rightarrow nCO + (n+m/2)H_2 \qquad (2\text{-}2)$$

these reactions are accompanied by the following water-gas shift reaction [17]:

$$CO + H_2O \Leftrightarrow H_2 + CO_2 \quad (2\text{-}3)$$

One possibility to get the feedwater, which is pre-heated to steam if necessary, for the steam reformer is to separate the water in the exhaust gas, after the recuperator, as shown in figure 2-1. This will indeed decrease the input of water into the process.

2.4.2 Characteristics of the pre-reformer

The pre-reformer is often a nickel-catalyst, where the conversion-rate of methane depends on, among other things, the area of the catalyst. In the steam reformer the operating temperature is between 500-900°C and the pressure should be between 1-15 bar [16]. Here about 70 % of the methane is converted into hydrogen and carbon dioxide, but this conversion-rate is depending on the user's requirements and into what type of process the reformer is integrated in. As shown in figure 2-11, the conversion of methane depends on both the pressure and the steam to carbon (s/c) ratio.

The pre-reformer is designed as a heat exchanger, which uses firing of hydrogen or reduced fuel from the anode as a heat source. At the firing the hydrogen is usually drawn from the gas outlet of the reformer and in the case of using reduced fuel from the anode this "fuel gas" is drawn from the anode outlet. There are bottom-, side- and top-fired steam reformers, where the bottom-fired is mostly used when integrated into a fuel cell process. One example of a bottom-fired reformer, fabricated for integration with fuel cells, is the heat exchange reformer (HER) from Haldor-Topsoe in Denmark, see figure 2-12. This construction has both co- and countercurrent heat exchange, which results in a constant metal temperature of the reformer tube [17].

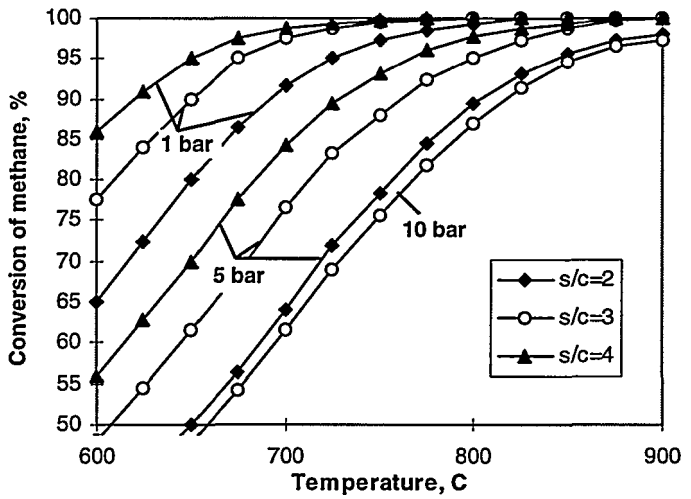


Figure 2-11. Conversion of methane at equilibrium for different pressures and steam to carbon ratios [17].

As shown in equation 2-2, the stoichiometric ratio of steam per carbon is 1.0. Tests have shown that this ratio is not the most effective one. At least a steam to carbon ratio of 2 is necessary when using methane as feedstock.

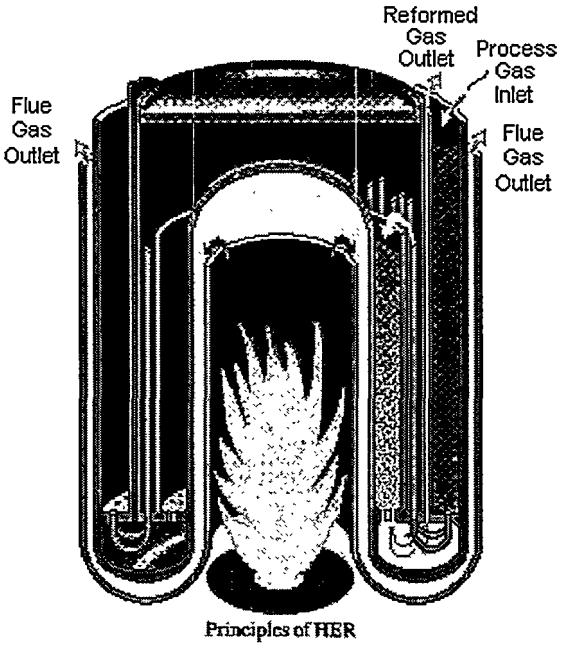


Figure 2-12. Heat exchange reformer, bottom fired from Haldor Topsoe [18].

2.4.3 Poisoning [16]

The most common problem in the steam reformer is sulphur poisoning. If the feedstock contains around 10 ppb a desulphurization system is not needed. If the sulphur contents are higher than 10 ppb the nickel catalyst will be poisoned, which leads to a loss in activity and higher temperatures at the walls of the catalyst. This rise in temperature can eventually lead to carbon formation on the catalyst surface, which leads to an earlier exchange of the catalyst than necessary.

2.5 Desulphurization system

The sulphur compounds in the natural gas can be H_2S, organic compounds of sulphur or both. Depending on which type of sulphur is present, different reduction methods are used, zinc oxide absorbent system for H_2S and hydrodesulphurization of the organic compounds.

The gas data specification for Danish natural gas is shown in table 2-3. These data are of interest because of the location of a possible SOFC/GT process, purchased by Sydkraft AB that would be stationed in Sweden. This means that the natural gas used for this process would be the natural gas distributed in Sweden. The tracers in table 2-3 is the maximum values according to the "2 Gasfamilie Gruppe H in Technische Regeln des DVGW für die Gas beschaffenheit". As can be seen in the table 2-3 the sulphur contents are both H_2S and organic, which require both the zinc absorbent system as well as the hydrodesulphurization (HDS) system.

Table 2-3. Gas data specification for Danish natural gas, 1998 [19].

Gas data:

Main components		Average value mole %
Methane	CH4	90,40
Ethane	C2H6	5,34
Propane	C3H8	2,03
I-Butane	C4H10	0,38
N-Butane	C4H10	0,53
I-Pentane	C5H12	0,15
N-Pentane +	C5H12	0,09
Hexane +	C6H14	0,08
Carbon dioxide	CO2	0,70
Nitrogen	N2	0,31

Tracers:

Oxygen	O2	<= 3,0 vol.%
Total sulphur	S2	<= 120 mg/m3, <= 150 temporary
Merceptan sulphur	CH3SH	<= 6 mg/m3
Hydrogen sulphur	H2S	<= 5 mg/m3

2.5.1 Zinc oxide absorbent system

The zinc absorbent system works as a sponge, where the bed of zinc oxide absorbs the H_2S. The reaction that takes place is:

$$ZnO + H_2S \rightarrow ZnS + H_2O$$

This reaction is very dependent on pore and lattice diffusion, since the formed water must diffuse away and the sulphide ion must diffuse into the lattice at the same time as the oxide ions diffuse to the surface [16]. Because of the bigger size of the sulphide ions and because the change in structure of the hexagonal zinc oxide to the cubic zinc sulphate, the bed ends up being less porous than before. These effects can contribute to that all zinc oxide does not react with the sulphide.

One zinc oxide absorbent system that can be used for the SOFC/GT process, is the HTZ-series from Haldor Topsoe A/S in Denmark. The HTZ-3 is a catalyst where the purity of ZnO is > 99,9%, has a recommended operation temperature of 300-400 °C and an operation pressure of 10-50 bar [20]. ZnO reacts above this temperature but at the temperature 700 °C the formation of zinc metal starts [16]. The pressure does not affect the reaction as much as the temperature does. The catalyst is a bed of cylindrical extrudates, which can absorb up to 507 kg S/m3 HTZ [20].

If the SOFC/GT process in this thesis has a total operation time of 10 000 h at maximum load, the total gas volume flowing through the system is approximately $6,7*10^5$ m^3 of natural gas. In every m^3 of natural gas there are approximately 5 mg of H2S as shown in table 2-3. A total

mass of 3,35 kg S is then to be absorbed in the zinc oxide bed, which requires a total volume of 6,6 litre zinc oxide.

2.5.2 Hydrodesulphurization system

When reducing organic sulphides from the natural gas, a catalyst made of highly porous alumina carrier, impregnated with nickel or cobalt molybdate is the most common. The reason why the zinc oxide bed does not work for organic sulphides, is because these molecules are still thermally stable at the operation temperature of the bed. By adding hydrogen to the catalyst the organic sulphides react and become hydrocarbons and sulphur hydrogen.

$$\text{Organic sulphur compound} + H_2 \rightarrow \text{Hydrocarbon} + H_2S$$

In the gas storage in Sweden the organic sulphur compound is mercaptan sulphur, CH_3SH, which leads to the following reaction [16]:

$$CH_3SH + H_2 \rightarrow CH_4 + H_2S$$

To achieve a good result in the catalyst, both residence time and temperature are very determining factors. Typical operating temperatures are 350-400 °C with a pressure of 25-35 bar. The lower molecule weight the organic sulphides have, the faster the desulphurization is. Some organic sulphides with low molecule weight are possible to crack in the zinc oxide bed, but this leads to carbon deposition [16].

The hydrodesulphurization system must of course be followed by the zinc oxide bed to eliminate the sulphur hydrogen. After these desulphurizing treatments the gas can be delivered to the pre-reformer or directly to the fuel cell if no external reformer is present.

2.5.3 Desulphurization system for the SOFC/GT process

If the SOFC/GT process is to be located in Sweden, one alternative of desulphurization system, such as the one in figure 2-13, is possible. The desulphurization process needed, includes both the hydrodesulphurization of hydrocarbon sulphides and the zinc oxide absorbent system, since the natural gas in Sweden contains both H_2S and CH_3SH. One alternative of this system, which Haldor Topsoe in Denmark fabricates, is the combination of the TK-251 (HDS) with the HTZ-3 (ZnO).

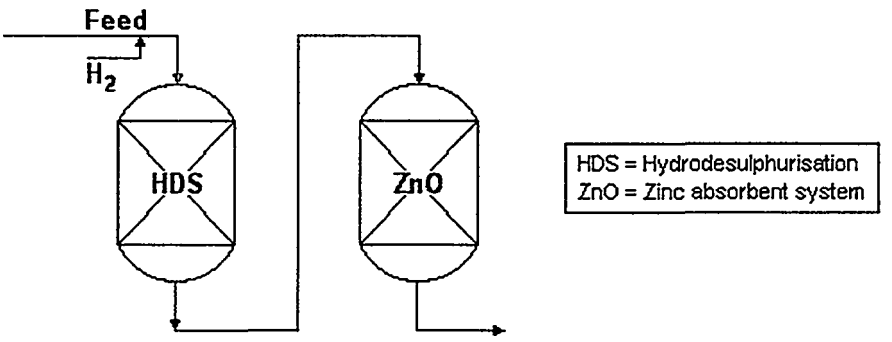


Figure 2-13. Desulphurization system for the SOFC/GT process [20].

3 Life cycle assessment (LCA)

Life cycle assessment (LCA) is a method that has become more and more common in companies, as an environmental evaluation method. LCA is method to follow a product's (or service's) complete environmental impact "from cradle-to-grave". Already in the late 1960s some energy flow assessments were done, but it was not until at the end of 1980, that the LCA model used today was formed. The LCA method has been discussed and there has been an uncertainty concerning the interpretation of the results. This uncertainty has lead to difficulties in the comparison between the products' environmental impact.

During the last years an ISO-standard for LCA has been developed and the first principles and framework of the ISO 14 040 series, which deal with LCA, were ready in the fall of 1997. In the ISO 14 040 series the following "chapters" are included:

- ***ISO 14 040:*** Environmental management - Life cycle assessment - Principles and framework
- ***ISO 14 041:*** Environmental management - Life cycle assessment - Goal and scope definition and inventory analysis
- ***ISO 14 042:*** Environmental management - Life cycle assessment - Life cycle impact assessment
- ***ISO 14 043:*** Environmental management - Life cycle assessment - Life cycle Interpretation

While performing an LCA, all the phases of the product's life cycle are studied, from the extraction of the raw material to the final disposal of the product. The product's or sub-product's production processes and the operation phase of the product are also included in the study. Between these steps, transportation of different kinds often occurs. The total environmental impact of these steps is then added up and evaluated in the LCA. When interpreting the results in the LCA, it is possible to find "hot spots" in the product life, which can be very useful when trying to diminish the environmental impact of the product.

3.1 Method

According to ISO 14 040, the LCA is divided into four main phases [21]:

1. Goal and scope definition (regulations according to ISO 14 041)
2. Inventory analysis (regulations according to ISO 14 041)
3. Impact assessment (regulations according to ISO 14 042)
4. Interpretation (regulations according to ISO 14 043)

Since an LCA is an iterative process, the goal and scope definition are updated during the whole assessment, fig 3.1. This means that the goals and scopes can be changed during the process, when limitations and assumptions are modified.

3.1.1 Goal and scope definition

In the first phase, goal and scope definition, the objective of the assessment is described as well as the limitations and the assumptions of the LCA. The functional unit that will be used in the study is defined. Depending on the system that is to be investigated a functional unit is chosen as a reference that relates the inputs and outputs of the system. Other important definitions are system boundaries and the required data quality.

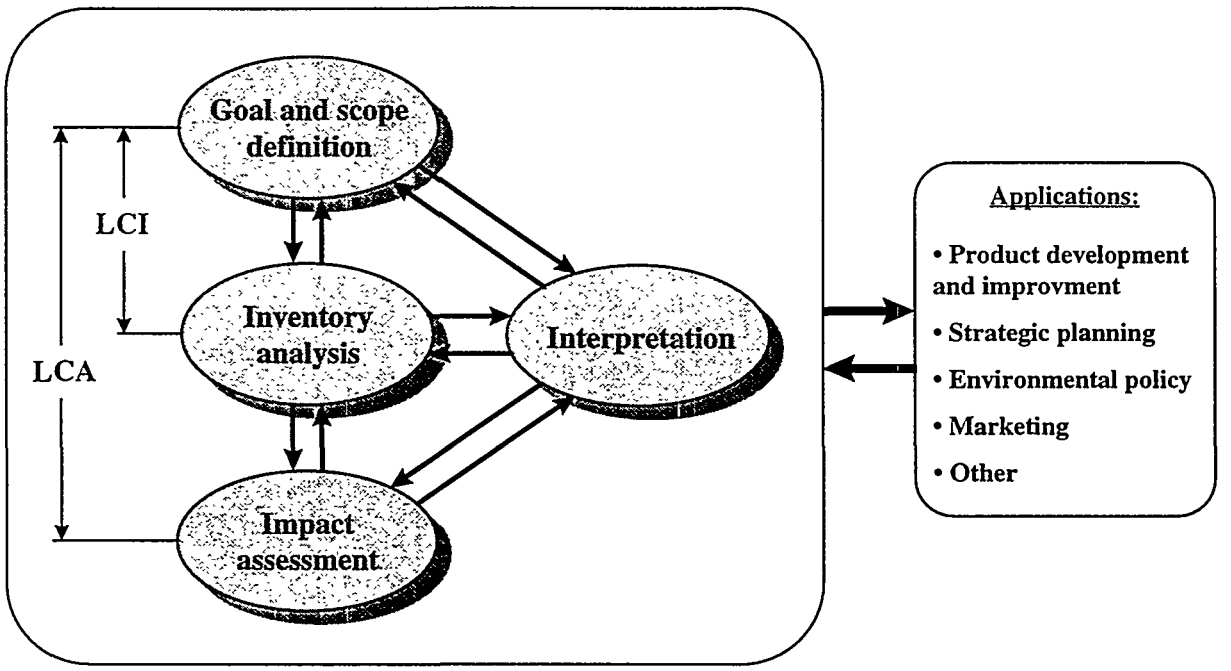


Figure 3-1. The four phases' iterative process.[remodelling of 21]

3.1.2 Inventory analysis

The second phase in the LCA is the inventory analysis. This is often the most time consuming phase in the study, since the collection of all the data takes time. The collection of data results in quantified material flows to and from the system boundaries. According to ISO 14 040, this second phase is divided into three steps [21]:

1. Description of flow charts according to the system boundaries that have been decided in phase 1.
2. Collection of data. In this step it is also decided how old the data are allowed to be, which technical level that is to be used (the best, average or the worst), etc.
3. Calculations of the collected data.

If the LCA would be stopped at this stage and no impact assessment is made it is not a life cycle assessment, it is a life cycle inventory (LCI). To accomplish an LCA the third step, impact assessment must be carried out.

3.1.3 Impact assessment

The last phase is the impact assessment. This phase is divided in three sub-phases; classification, characterization and valuation. Classification means that all the calculated data from the inventory analysis are sovred into environmental impact categories, such as global

warming, acidification etc. Examples of impact categories according to the Nordic Guidelines[1] [22] are presented in table 3-1. In the characterization, the inventory data are weighted for the emissions under each impact category and calculated. These weighting parameters are also called equivalents and they describe to what extent a certain emission contributes to the environmental impact in question. Finally in the valuation, the total environmental impact is assessed. Here a weighting between the impact categories is made to further aggregate the data.

Table 3-1. Examples of impact categories, which can be included in the classification according to the Nordic Guidelines [22].

Impact categories
1. Resources - Energy and material
2. Resources - Water
3. Resources - Ground
4. Effects on health - Toxic effects (effects in the work environment are excluded)
5. Effects on health - Non-toxic effects (effects in the work environment are excluded)
6. Effects on health in the work environment
7. Global warming
8. Decomposition of stratospheric ozone
9. Acidification
10.Creation of photochemical oxidants
11.Eutrophication (overfeeding)
12.Eco-toxic effects
13.Influence of the biological multiplicity
14.Inflows not traced back to the "cradle"
15.Outflows not followed to the "grave"

Number 14 and 15 in table 3-1 are not really effect-categories, but are included while they are used in the analyses later. They are also important in the interpretation.

3.1.4 Interpretation

The last step in an LCA is the interpretation of the results. This interpretation can be made out of the inventory results or out of the classification, characterization or valuation results. When interpreting the results it is important that all calculations and assumptions are clear and well defined to get the best and most accurate interpretation possible.

[1] Nordic Guidelines, a project initiated in the Nordic countries in 1991, with a final report in 1995 (Nordic Guidelines on Life-Cycle Assessment, Nord 1995:20).

4 Goal and scope definition

4.1 Goal of the study

The goal of this study is to perform an LCA of the two components; the SOFC-module and the gas turbine, in the process described in chapter 2. The LCA is done according to the ISO 14 040-series. This LCA will provide Sydkraft AB and LTH with an understanding of the total environmental impact from the SOFC/GT process.

4.2 Scope

The LCA includes the 226 kW SOFC-module, developed by Risoe National Laboratory in Denmark (Risoe), and the 202 kW gas turbine, assumably developed by Volvo Aero Turbines in Sweden. The operational data used are presented in chapter 2.1.

4.3 Functional unit

In this case the product is energy/power, so the functional unit chosen is 1 kWh, i.e. the environmental load is referred to 1 kWh net delivered by the SOFC/GT process to the Swedish electric grid.

4.4 System boundaries

The components or processes included in the LCA are defined by the system boundaries. These boundaries are set, both towards other systems as well as within the system itself. Ideally all processes are included when performing an LCA, but this is not always possible depending on how much of the required data that is available. The required system boundaries are set within time, geographical coverage and technosphere-biosphere. The general process tree for the life cycle of the SOFC/GT process is shown in figure 4-1.

4.4.1 Definition of the life cycle

The life cycle includes mining of raw materials used during the production of the SOFC-module and the gas turbine, construction work, operation, demolition and finally the waste treatment. Between these phases, transportation is needed, which also is included here. In the operation phase, the construction and demolition of fuel refining establishments is included to some extent, using the LCA of the natural gas fuel production 1996 for Sydkraft's power plant Heleneholmsverket.

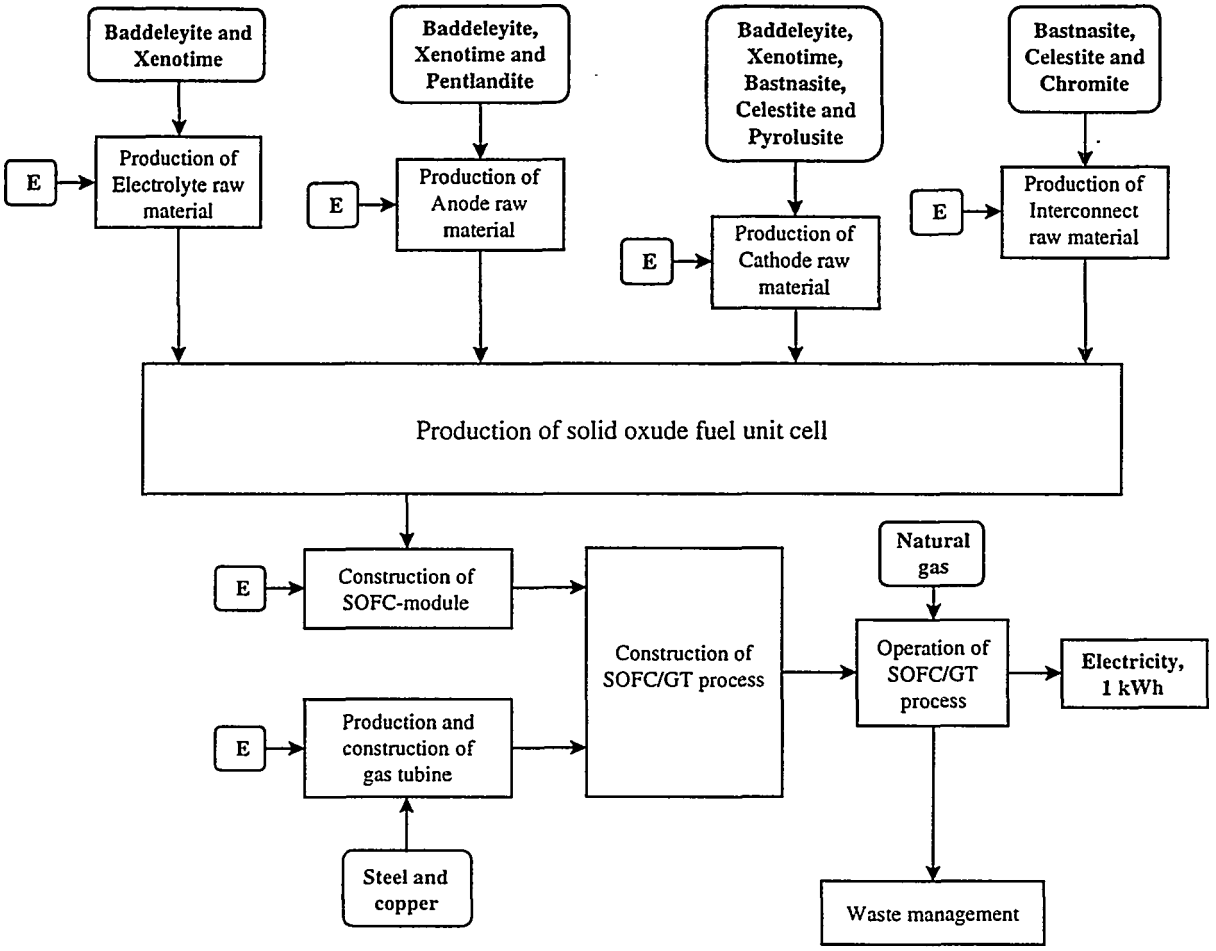


Figure 4-1. General process tree for the SOFC/GT process (E = energy).

4.4.2 Boundaries within the life cycle

In the inventory, the flows of materials and chemicals through all phases in the life cycle, are mainly from literature and from calculations based on literature. The judging of the environmental impacts of the process is perspicuous, i.e. in no way meant to cover everything. The issues of working environment and risks of catastrophes are not included. Also overhead, transport of labour and production of different chemicals used are not included.

4.4.2.1 Inflows not traced back to the cradle

Since the SOFC/GT process is under development all data required to do a complete LCA have not been found. All the raw materials used in the SOFC have been traced back to the cradle, but the chemicals used to produce these materials are not traced back with the exception of sulphuric acid, where data have been found. For the gas turbine, both steel and copper have been followed back to the cradle, which are considered to the only materials that the gas turbine consists of.

4.4.2.2 Outflows not followed to the grave

None of the waste products are followed to the grave during the production of the SOFC, since no information has been found about what really happens to the products or "waste-chemicals" after the production. For most of the material used in the SOFC, there are not any recycling methods today, so after the SOFC's lifetime deposition is assumed.

4.4.2.3 Electricity and fuel supply

When including the electricity in the life cycle the production of electricity is included (see figure 4-2). Electricity will therefore be presented as an internal parameter and only the raw material for the fuel production flows over the system boundary. The electricity used in the inventory is an "average" electricity mix in the country where the process/processes takes place. The composition of the different energy carriers used in the inventory and their emissions and waste products are further described in appendix C. Also the different fuel suppliers are included in appendix C.

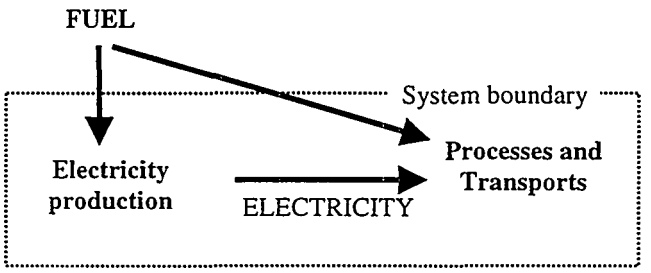


Figure 4-2. System boundary for the electricity production [23].

The energy used varies between the production phases of the SOFC components. For the mining, half of the energy is assumed to be diesel for the machine work and the other half is assumed to be electricity (average mix of electricity from the country where the process takes place) when processing the minerals into nitrates. This is an assumption that can be discussed, but since no distribution ratio has been found and in order to minimise the variables, this 50/50 %-ratio has been chosen.

4.4.3 Technosphere-biosphere boundaries

In this study the "cradle" is the mines, where the minerals are extracted. The "grave" is the landfill, where the waste products are deposited.

4.4.4 Geographical boundaries

For the SOFC several different suppliers and producers are used from several countries. The SOFC/GT process will however be used in the southern part of Sweden. The fuel there is natural gas from Denmark.

4.4.5 Boundaries in time

The SOFC/GT process will be tested in the summer of 1999. The data used in this study are from a laboratory test done in 1995 for the SOFC production at Risoe and from literature [9].

4.5 Data

4.5.1 Parameters

The parameters, accounted for in the results, are split up into categories. The categories chosen in this study are the following:

- Consumption of energy resources and internal use of energy
- Consumption of material resources
- Emissions and waste
  - to air
  - to water
  - solid waste
- Inflows not traced back to the "cradle"
- Outflows not followed to the "grave"

4.5.2 Units

The units used for emissions and raw materials are mg, g or kg. For energy, the units MJ, kWh or MWh are used.

4.5.3 Data quality

Since the SOFC/GT process is not yet at an operational stage, all parameters are collected from literature, articles, material from different companies and through interviews. Also most data collected are produced under laboratory circumstances, which would probably lead to an increase in both energy and material requirement, since smaller working machines during production are used in a laboratory. Also more wastage during these processes is probably excepted to produce very accurate prototypes. However, the data is considered sufficient enough for a first LCA of the SOFC/GT process.

The required natural resources , which are used to study the environmental impact is; baddeleyite, bastnasite, celestite, chromite, copper-ore, iron-ore, limestone, pentlandite, pyrolusite, rock and xenotime.

The energy resource is biomass, coal, crude oil, natural gas, natural uranium and water power. These energy resources are converted into electricity in some cases and in some cases they are refined into other fuels such as diesel. The energy resource during operational time is natural gas. Facts and figures for the energy resources and the energy carriers are found in appendix C. Concerning the amount of energy resources needed, a further description is to be found in appendix B.

The emission to air mainly comes from the use of energy, during the production of the components and during the operation of the process. The emissions that are presented in this study are: carbon dioxide (CO_2), methane (CH_4), nitrogen dioxide (N_2O), carbon fluorides (CF_4 + C_2F_6), aldehydes, benzene, carbon monoxide (CO), phenol (C_6H_6O), HC (hydrocarbons), dust, PAH (poly aromatic hydrocarbons), several metals and VOC (volatile organic compounds). Emissions during extraction of minerals and production of the SOFC are unknown. Only emissions due to energy use during these processes are estimated.

Emissions to water are also estimated from the energy need. All leakage and deposition of tailings during the mining-phase are unknown, which also is the case when producing nitrates. The emissions to water, according to energy need are; hydrochloric acid (HCl), ammonium hydrate (NH_3), nitrogen oxide (NO_x), sulphur oxide (SO_2 + SO_x), COD (chemical oxygen demand) and nitrogen compounds (N-tot).

Emission to the ground are mostly ashes and tailings from the extraction of minerals, but some radioactive waste occurs, due to the chosen energy carrier.

4.5.3.1 Data sources, construction work and demolition

Energy need for the construction work is not estimated. Only the need for concrete and reinforcing iron for the SOFC "box" has been approximated with help from Åke Carlsson and Per Carlsson at Sycon.

4.5.3.2 Data sources, operation

The data used during the operation phase are collected from the research project at LTH and are presented in chapter 2.1.

4.5.4 Data quality categories

The data that is collected during the way are divided into four categories:

Table 4-1. Description of data quality categories [23].

Categories	"Quality" and source
Measured	Measured data from the specificly studied process.
Estimated	Assumed or calculated data from the specificly studied process, alternatively valid for an equivalent process with the same product as the specified process.
Literature	Data collected from literature. Not valid for the specificly studied process, but could be an average of the branch or be valid for another process, which contributes with a similar product.
Standard	Standard data are used in cases when conveyances and distance of transport are unknown.

4.5.5 Data gaps

Since there is no possibility to measure any data during the operation and since some parameters during the production of the components are not available, these gaps can lead to uncertainties regarding the final results. This is known and will be taken into account, when interpreting the final results.

All data from the production phases of materials and chemicals are not included here, since this would lead to a to large task. The materials and chemicals used during the production phases of the components are not traced back to the "cradle" and not followed to the "grave".

The transports are also a bit of a data gap, regarding the SOFC. Where these transports take place during the production process and how many they are is not confirmed anywhere. Due to the many subcontractors it is also unknown how much of the partly processed material that is shipped during the transports and between which countries these transports take place.

For the gas turbine only the total mass of the drive line is know. Exactly what materials the gas turbine consists of is also unknown. The composition is chosen to steel and copper. Also energy and process materials/chemicals during the production of the gas turbine are unknown, only the processes for the production of the raw materials steel and copper are included in this study.

4.5.6 Data presentation

The collected data from literature and interviews are further calculated in appendix A and B to obtain the right quantities of energy and masses before using the program, EcoLab. The inventory data are then presented in tables for the two phases; construction and demolition phase and operation phase, see appendix D. During the construction and demolition phase there is a division between the sub components; total SOFC, electrolyte, anode, cathode, interconnect and gas turbine.

4.6 Method for inventory analysis

4.6.1 Allocation methods

No allocation methods are used here. There are not any known recycling methods for materials within the production of the SOFC-module. For the gas turbine there are recycling methods for both steel and copper but they have not been allocated. There is also only one product, electricity, produced by the SOFC/GT process and consequently the operational phase is not subject to any allocation.

4.6.2 Important assumptions in the inventory analysis

4.6.2.1 SOFC/GT process

The SOFC/GT process has an assumed life time of 10 000 operation hours at full load. This assumption is made after discussion with LTH. The goal for the system in the future is at least that it will have a lifetime of 40 000 hours, but today, after operating test processes, this is considered an optimistic length of lifetime for the SOFC. The gas turbine is assumed to have the same life time as the SOFC. The reformed natural gas that enters the SOFC is assumed to have been reformed with the s/c-ratio of 2,5.

4.6.2.2 SOFC-module

Calculations made for the quantity of masses are presented in appendix A; where the all the reactions are considered stoichiometrical. The energy calculations in appendix B include the assumption that half of the energy used during the production of raw materials for the SOFC have the energy carrier diesel, while the second half uses UCPTE* electricity. The extraction countries chosen here, are the ones that are given most prominence in the literature. Whether these countries really are the countries that supply the ores to ceramic producers is unknown. The production of the raw materials for the SOFC is assumed to take place in the USA. This assumption is made because there are production of the ceramics used for the SOFC in the USA. There are also other countries who produce the ceramics used, but to reduce the variables in the analysis this choice has been made.

No effort has yet been put into recycling of the materials used in the SOFC-module since recycled material today is more expensive than new materials. The waste product (SOFC) will be deposited on a landfill. Hopefully, recycling will be possible and cost effective in the future when the SOFC and SOFC/GT processes are commercialised.

4.6.2.3 Gas turbine

The gas turbine used in this study is probably the most uncertain component, as far as the data quality is concerned. The material composition is unknown, but since most gas turbines, including the generator, in some extent consist of steel and copper, the assumption is made that this turbine also contains these two materials. The assumption that only those two materials are present in the gas turbine and that the weights have been assumed to 140 kg steel and 20 kg copper can be discussed. The production of steel and copper is included, but no production data have been found from the production and assemblage of the gas turbine. Because of this lack of data this production is neglected, since it is assumed to be a minor contribution in the total production.

4.6.3 Recycling

In the production of the gas turbine, the assumption is made that some of the steel and the copper used is virgin and the remaining part is recycled. Analogously it is assumed that some of the materials will be recycled after the use of the gas turbine. The extent of recycling has been produced in a collaboration between Sydkraft and Elin Eriksson, CIT Ekologik [33].

Table 4-1. Extent of recycling for steel and copper at production and demolition. The estimations have been done together with Elin Eriksson, CIT Ekologik [33].

Material	**Extent of recycled material (%)**	
	Material input at production	Material output at demolition
Steel	25	70
Copper	50	80

* For explanation, see appendix C.

4.6.4 Reinvestments

Because of the assumed short lifetime of 10 000 hours, no reinvestments of the SOFC-module or the gas turbine are taken into consideration in this study. Nor is any use of materials during operation and maintenance, such as lubricating oil etc., included.

4.6.5 Transports

In this study the transports used, are boat (ocean going), boat (coast going), train (UCPTE), road (long distance > 100 km) and road (short distance < 100 km). The transports are divided into different groups because of their difference in emission factor, which is presented in appendix D. All conveyances are to be considered as fully loaded (100 %) when transporting the different materials to their next destination. When the conveyances are going back, 30 % of the "to"-distance is added to the total transport, but the load here is considered empty (0 %). The standard distances used for the transports within Sweden [24] are presented in table 4-2. The short distances presented in table 4-2 are used when transporting the waste to the landfill after the demolition.

Table 4-2. General standard data for the transport distances within Sweden [24]

Conveyance	*Ratio of load*	*Distance*
Road (truck), long distance	100 % load	300 km
Road (truck), long distance	0 % load	90 km
Road (truck), short distance	100 % load	30 km
Road (truck), short distance	0 % load	9 km

All transport distances covered outside the Swedish borders are measured air distances in a conventional atlas. The distances covered within land areas are assumed to be the distance from the border to the centre point in the country.

4.7 Reporting

The data collected in the inventory analysis are reported in tables for each sub component in chapter 5. The analysis of these data are reported in chapter 6 and the interpretation of the data is to be found in chapter 7.

4.8 Applicability of the study

The SOFC-module is from 1995 and research and since then much effort has been put into the development of this type of fuel. The gas turbine is a 1998 model and therefor more applicable today. In its entirety this LCA is considered to be the first attempt of an LCA of the SOFC (planar) and the SOFC/GT process and must be used accordingly.

4.9 Critical review

No critical review will be done here.

5 Inventory analysis

5.1 Introduction

In this chapter the collected data for the life cycle of the SOFC/GT process are presented. The goal is to find out more about the total environmental impact of the SOFC/GT process. This is done by collecting data for all production processes, operation and for the transports.

5.1.1 Process trees

The process trees are presented for each component in their sub chapter. The different phases for the components are represented by boxes which also is the case for transports. If the boxes have a full line the activity/product inside is traced back to the cradle or followed to the grave. If the line is dotted the activity or product is not traced back to the cradle or followed to the grave. The flow of materials between the boxes is presented as arrows.

5.1.2 Presentation of data

The collected data are presented in tables with the headlines product, raw material, used energy and emissions to air, water, ground and resources, see figure 5-1. All data for the production of the SOFC and its sub components are in relation to the production of 1 unit cell. For the production of the gas turbine, the construction and demolition and for the operation the data are presented as total mass flows and total energy used during the processes. For the energy requirements it is stated whether emissions at extraction (Ex) and final use (FU) or no emissions (none) are added to the life cycle. The emissions at extraction are from the phases; extraction, refining and distribution of the energy carrier to the process. Recourses are marked (r).

Process				
Production of SOFC				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
1 SOFC unit cell				
Raw materials				
YSZ	x g	Estimated	1,2,3	
.....				
Used Energy				
Diesel	y MJ	Literature	4,5	Ex/FU
Emissions to *air*				
CO2	z g	.....	.....	
Resources				
Xenotime	w g	Estimated	.....	

Figure 5-1. Example of table used to present the process data.

5.1.3 Units

The requirement of raw material is given in multiples of g (mg, kg, etc.). The energy requirement is given in multiples of J (MJ, etc.) or in kWh and the emissions to air, water and ground plus the amount of waste are given in multiples of g.

5.1.4 Energy carriers

The energy carriers differ between the processes depending on the production technique used and in which country the processes take place. The energy carriers used in the different processes are presented in appendix C.

5.1.5 Emission factors

The emission factors for the energy carriers are presented in appendix C.

5.1.6 Transport data

The transports used are road (long distance), road (short distance), train (with UCPTE electricity), boat (ocean going) and boat (coast going). The emission factors used in the database for the transports are presented in appendix D.

5.1.7 Inventory results

The inventory results are presented in tables and they are presented in relation to 1 kWh produced electricity by the SOFS/GT process. The inventory results are divided into two phases; the construction and demolition phase and the operational phase. The two phases are described below:

- *Construction and demolition*
  The construction phase includes the production of the components and preparation of the area where the SOFC/GT process is to be placed. All transports within this phase and the use of working machines are also included. The demolition of the process after the 10 000 hours includes the use of working machines and the transportation of waste material.

- *Operation*
  The operation phase includes the operation of the SOFC/GT process. Since the lifetime of the process is relatively short, an assumption has been made that no reinvestments are present.

5.2 Production of SOFC-module

The SOFC unit cells are produced at Risoe. The raw materials used to fabricate the sub components; electrolyte, anode, cathode and interconnect differ and the production of these raw materials can take place in most industrial countries. The USA has been chosen as production country for all raw materials. The most common countries where the mining of the

different minerals is performed are shown in table 5-1. These countries are chosen as suppliers to the USA.

Table 5-1, export countries of different minerals.

Mineral	Supply country	Product	Reference(s)
Baddeleyite	Australia	ZrO2	9, 27
Bastnasite	China	La2O3	9, 28
Celestite	China	SrCO3	9
Chromite	South Africa	Cr2O3	9, 28
Pentlandite	Canada	NiO	9, 27, 28
Pyrolusite	South Africa	MnO	9, 28
Xenotime	China	Y2O3	9, 28

When refining the minerals to raw materials, many different processes take place. These processes for each raw material are further defined in the process trees in chapter 5.2.1.

5.2.1 Process trees

The process trees for the production of raw materials and sub components are shown in figure 5-2 to 5-6. The process trees are based on Dr. Zapp's PhD thesis, "Ganzheitliche Material- und Energieflussanalyse von SOFC Hochtemperaturbrennstoffzellen" [9], but have been remodelled to suit this study. If more detailed process trees are of interest, this PhD thesis is recommended.

The outputs tailing and impurities, are names of different kinds of slag products, where tailing is coarse, mostly stone, material and impurities are more fine materials.

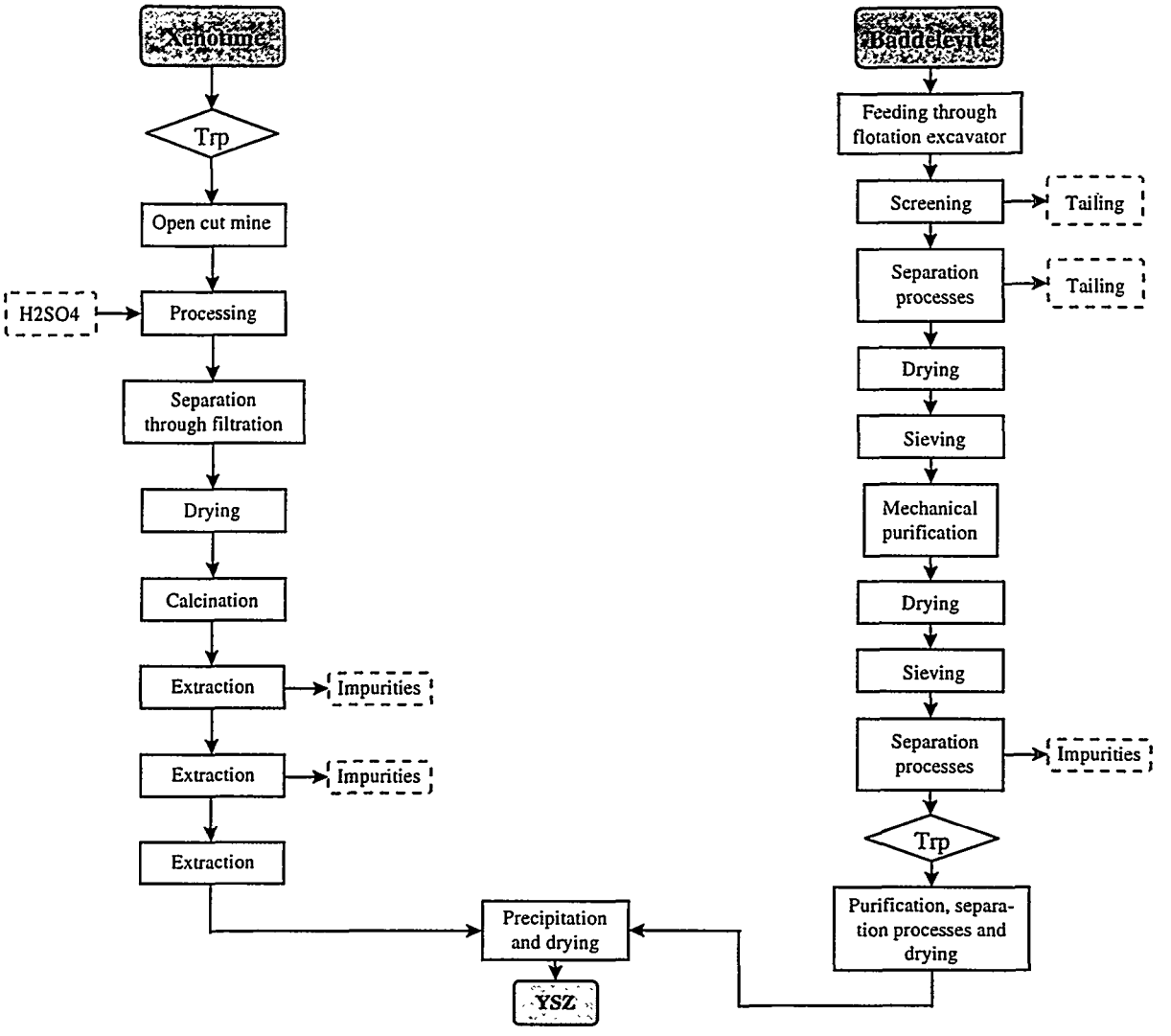


Figure 5-2. Process tree for production of raw material yttria-stabilised zirconia (YSZ) for the electrolyte [Remodelling of 9].

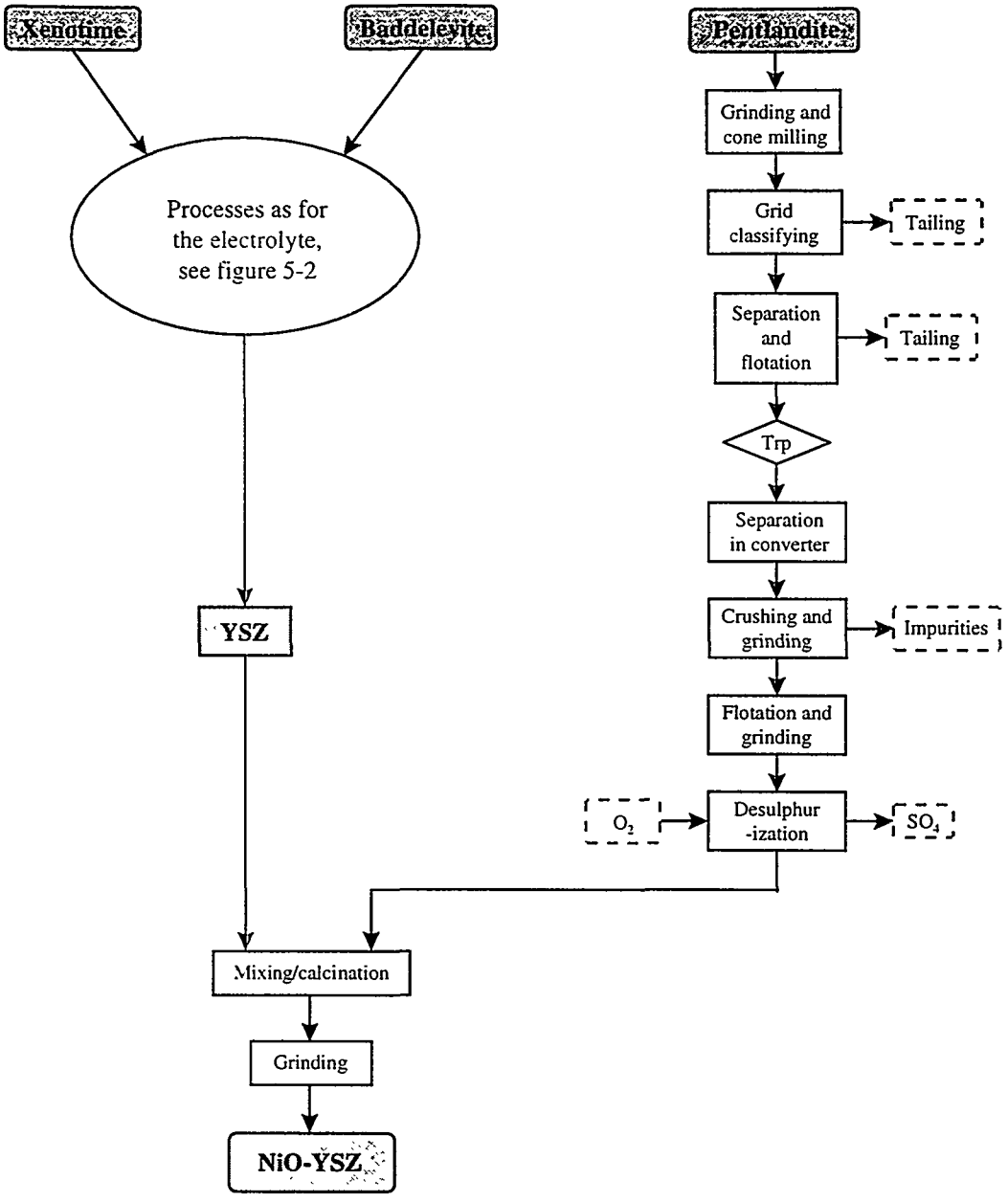


Figure 5-3. Process tree for production of raw material nickel oxide in yttria-stabilised zirconia (NiO-YSZ) for the anode [Remodelling of 9].

The process tree in figure 5-3 is not entirely applicable at Risoe. In Risoe the nickel oxide (NiO) and the YSZ are bought separately and mixed at Risoe because of the requirement of particle size and to be able to compose their own composition of NiO-YSZ.

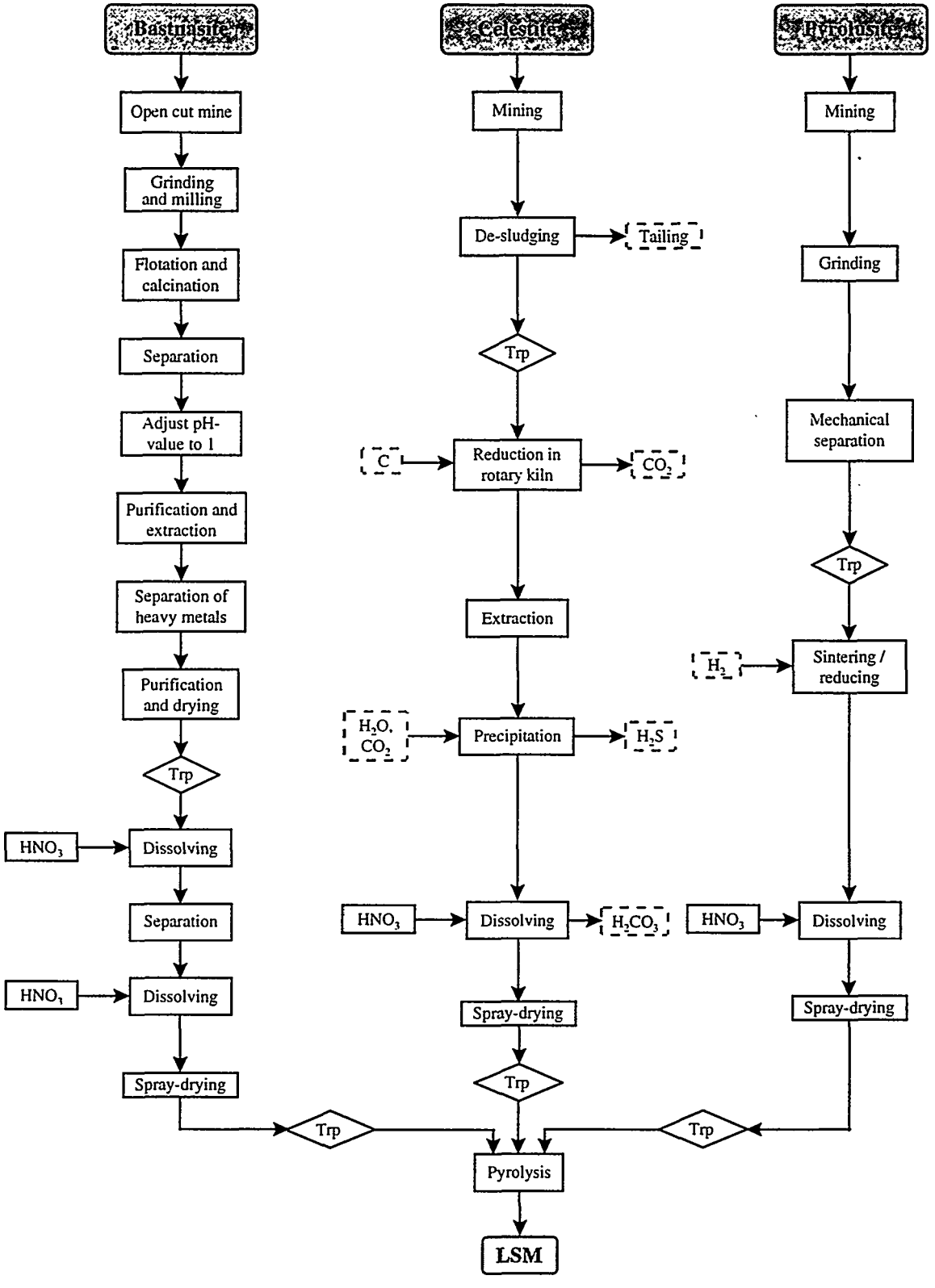


Figure 5-4. Process tree for production of raw material doped lanthanum (LSM) for the cathode [Remodelling of 9].

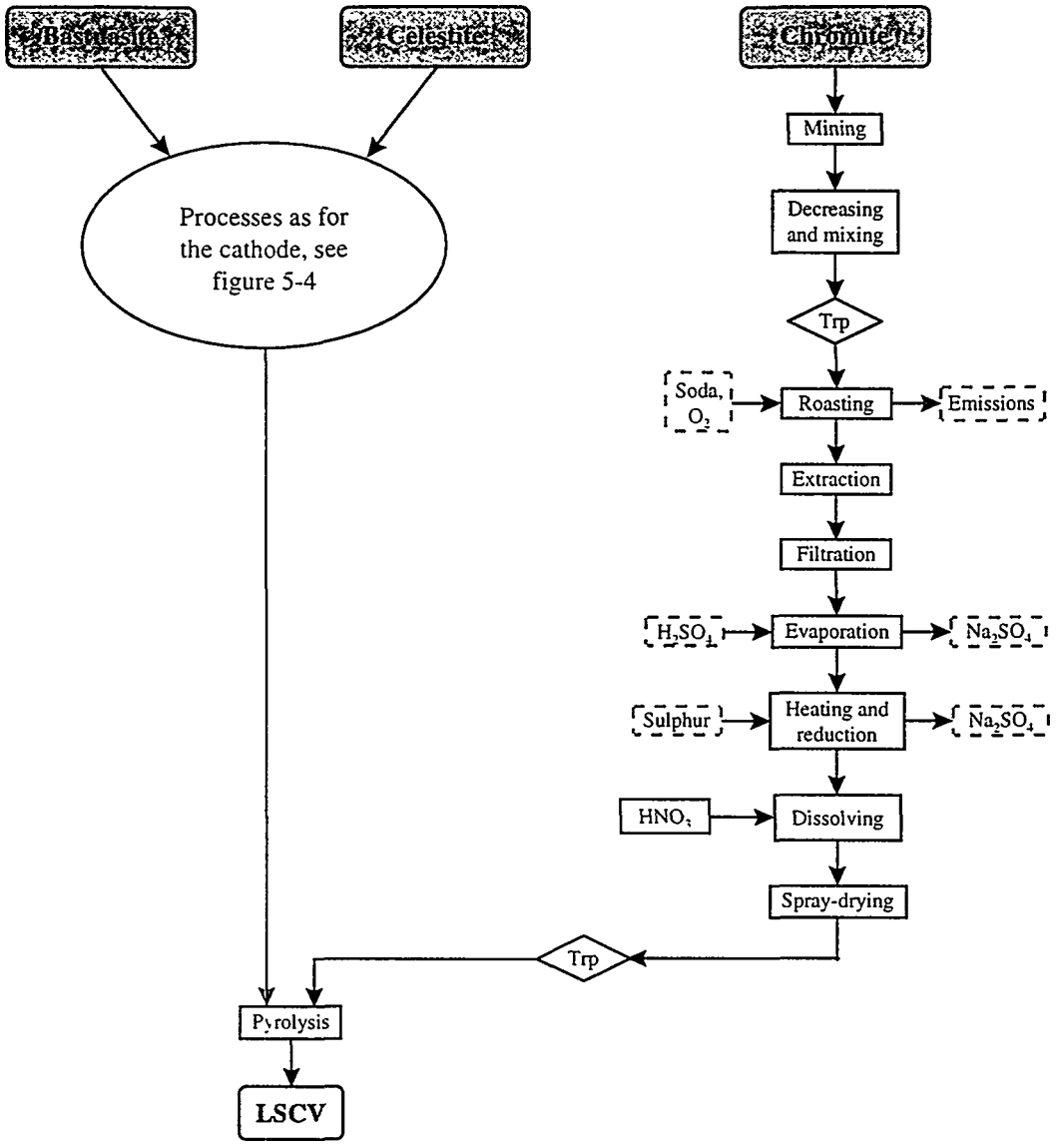


Figure 5-5. Process tree for production of raw material $La_{0.8}Sr_{0.2}Cr_{0.97}V_{0.03}O_3$ (LSCV) for the interconnect [Remodelling of 9].

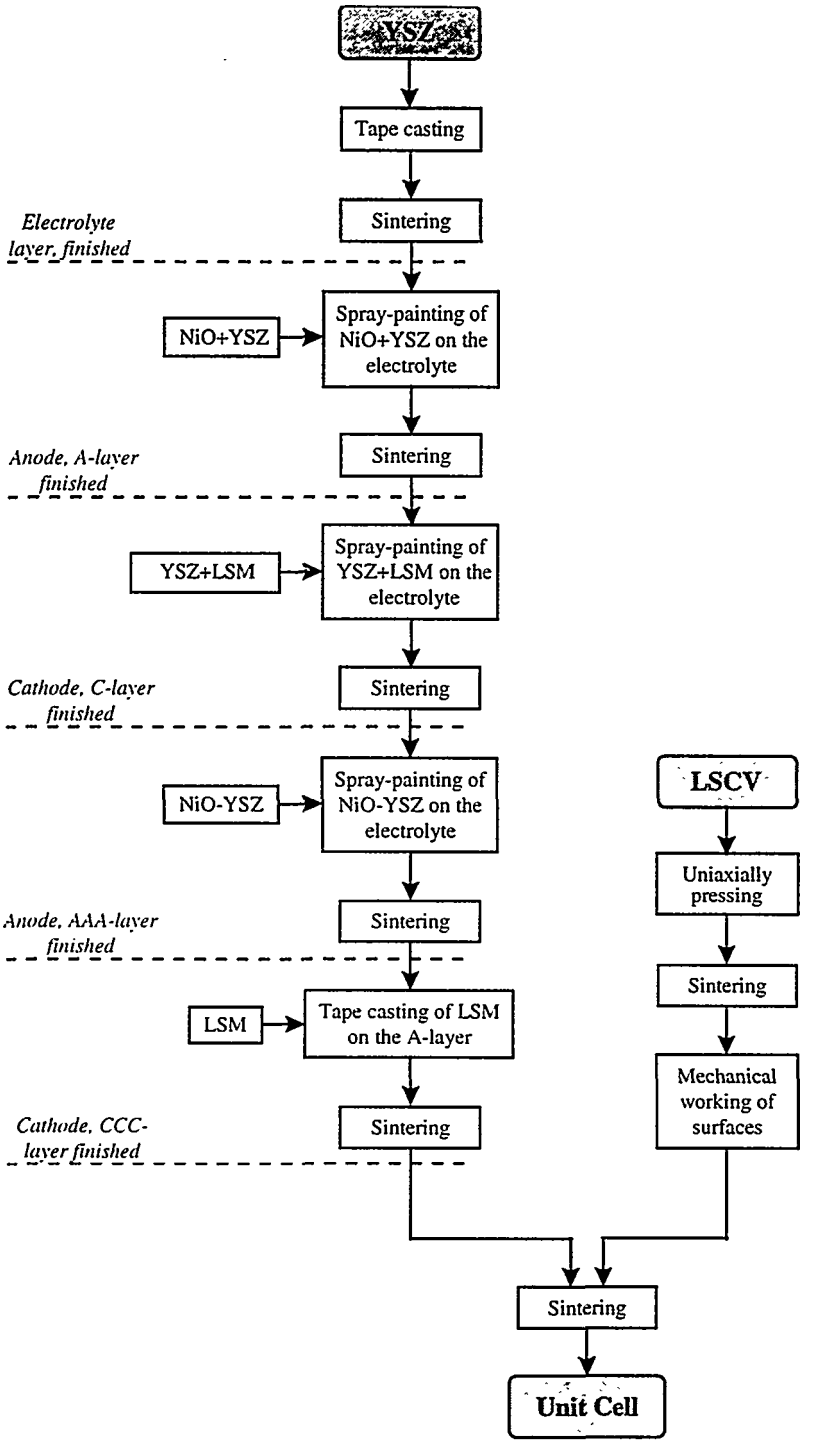


Figure 5-6. Process tree for production of a Risoe 1st generation laboratory planar SOFC unit cell for 1000 °C.

5.2.2 Mass flows

In the following tables the mass flows in and out of the processes are presented for each sub component. The transports are not included, since there is no mass change during transports. All mass flows are given for the production of one unit cell. The figures used in the mass flow tables are calculated in appendix A. These calculations are based on literature [9] except the output materials marked with a grey colour. The data are collected at Risoe [6].

In the tables 5-2 to 5-6 there are outputs named tailing and impurities. These are waste products during the processes gathered into one mass post because no data have been found regarding the distribution of the detailed waste products. In the literature [9] only the masses during chemical reactions within the processes are accounted for. To make the mass input and mass output of substances equivalent this post has been founded. Most of the contents in tailing and impurities are probably leftovers after processes such as flotation and separation, i.e. unwanted metals and rock within the mineral.

Table 5-2. Mass flows during the production of raw material (YSZ) for the electrolyte.

Output	**Amount [g]**
YSZ	16.0
Tailing and impurities	3437.62
Input	**Amount [g]**
Baddeleyite	3450
Xenotime	3.62

Table 5-3. Mass flows during the production of raw material (NiO-YSZ) for the anode.

Output	**Amount [g]**
NiO-YSZ	1.17
SO2	0.37
Tailing and impurities	204.17
Input	**Amount [g]**
Baddeleyite	112.13
Pentlandite	93.14
Xenotime	0.12
O2	0.32

The raw material used at Risoe in the cathode is LSM mixed with YSZ. This mix will from now on be presented as only LSM. The process trees used for the cathode is thereby both figure 5-2 and figure 5-4.

Table 5-4. Mass flows during the production of raw material (LSM) for the cathode.

Output	Amount [g]	Input	Amount [g]
LSM	1.30	Baddeleyite	28.03
YSZ	0.13	Bastnasite	2.28
CO2	0.11	Celestite	0.23
H2CO3	0.06	Pyrolusite	0.69
H2O	0.10	Xenotime	0.03
H2S	0.03	C	0.03
Nitrates*	2.29	CO2	0.04
Tailing and impurities	29.82	H2	0.01
		H2O	0.68
		HNO3	1.69

* Fumes that evaporates at the pyrolysis (unknown)

Table 5-5. Mass flows during the production of raw material (LSCV) for the interconnect.

Output	Amount [g]	Input	Amount [g]
LSCV	123.75	Bastnasite	212.45
CO2	36.52	Celestite	27.89
Fe2O3	21.61	Chromite	75.71
H2CO3	6.71	C	3.47
H2O	4.88	CO2	4.76
H2S	3.69	H2O	110.17
Na2SO4	76.87	H2SO4	26.54
Nitrates*	303.19	HNO3	197.79
Tailing and impurities	162.77	Na2CO3	57.36
		O2	15.15
		S	8.68

* Fumes that evaporates at the pyrolysis (unknown)

Table 5-6. Mass flows during the production of a Risoe SOFC unit cell.

Output	Amount [g]
Unit cell	83.00
YSZ	10.00
NiO-YSZ	0.27
LSM	0.33
LSCV	48.75
Input	**Amount [g]**
YSZ	16.00
NiO-YSZ	1.17
LSM	1.43
LSCV	123.75

5.2.3 Processes and transports for production of a SOFC unit cell

All transports have been estimated through measuring the way by air between the involved countries in a conventional atlas, since the ways of transports are unknown. Also where in the process-chain of the sub components the transports of material take place is unknown, wherefore the transported masses are assumed. When train is used it is assumed UCPTE electricity is used.

All energy used during the production of the SOFC and its sub components is calculated according to Dr. Zapp's PhD thesis [9] and the maximum value is used in this study. For the pyrolysis process used during the production of the cathode and the interconnect, the use of energy has been given by Risoe [6] for the production of 1 kg of raw material. All calculations are presented in appendix B.

5.2.3.1 Electrolyte, raw material production and transports

In the electrolyte, the two minerals used for the production of YSZ are baddeleyite and xenotime. The substance ZrO_2 is produced from the mineral baddeleyite, which is mostly to be found in the sand at beaches. The most common countries, where baddeleyite is extracted are Australia, India, South Africa and the USA [9]. In this study the extraction is assumed to take place in Australia, The production of the raw material ZrO_2 is assumed to take place in USA. The processes, which are used to produce ZrO_2 are roughly described in figure 5-2. No information about transports have been available and they are therefore assumed, see figure 5-2 and table 5-1.

The second substance that YSZ contains is Y_2O_3, which is produced from the mineral xenotime. Xenotime is open cut mined, mostly in Indonesia. Malaysia and China [9]. In this study it is assumed that the mineral is mined in China, as mentioned in table 5-1. The processes, which are used to produce Y_2O_3 are roughly described in figure 5-2. The transports here are also assumed, as for baddeleyite.

Table 5-7. Process data for the production of the electrolyte raw material, YSZ.

Process				
Production of raw material (YSZ) for the electrolyte				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
YSZ	16.0 g			
Raw materials				
See table 5-2				
Used energy				
Electricity, UCPTE*	1,685 MJ	Estimated	9	Ex
Diesel, working machines	1,685 MJ	Estimated	9	FU/Ex

The transported masses during the production of YSZ have been assumed. The mineral baddeleyite contains 10 % mineral concentrate, which has lead to the assumption that 10 % of

the required mass of baddeleyite is transported. The mineral xenotime contains 60,8 % of Y2O3, which is a relatively big part of the total weight. The assumption made in this study is that all of the xenotime is transported.

Table 5-8. Transport data within the production of the electrolyte raw material, YSZ.

Transport				
Transport of partly processed minerals from Australia to the USA				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
Baddeleyite	345 g	Boat, ocean going (100%)	21000	Estimated
		Boat, ocean going (0%)	6300	Estimated
		Train, UCPTE	7000	Estimated
Xenotime	3.62 g	Boat, ocean going (100%)	20000	Estimated
		Boat, ocean going (0%)	6000	Estimated
		Train, UCPTE	7000	Estimated

5.2.3.2 Anode, raw material production and transports

The raw material for the anode is nickel oxide, mixed with YSZ. The processes and countries where the extraction and the production of YSZ are performed, are the same as mentioned in 5.2.3.1. The nickel oxide is mainly processed through flotation of the mineral pentlandite. The processes used to produce NiO are shown in figure 5-3. Pentlandite is to a great extent found in Canada, but also Russia. Australia and South Africa have reserves of nickel ores [9]. In this study Canada is chosen as the country, where the pentlandite is extracted.

Table 5-9. Process data for the production of the anode raw material, NiO+YSZ.

Process				
Production of raw material (NiO-YSZ) for the anode				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
NiO+YSZ	1.17 g			
Raw materials				
See table 5-3				
Used energy				
Electricity, UCPTE	0.079 MJ	Estimated	9	Ex
Diesel	0.079 MJ	Estimated	9	FU/Ex
working machines				

The transported masses of baddeleyite and xenotime have been assumed in the same way as the masses for the electrolyte (see 5.2.3.1). After the first flotation after the mining of pentlandite the mass is reduced to 8 % and it is after this process that the transport is assumed to take place.

Table 5-10. Transport data within the production of the anode raw material, NiO-YSZ.

Transport				
Transport of partly processed minerals to the USA				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
Baddeleyite	11.21 g	Boat, ocean going (100%)	21000	Estimated
		Boat, ocean going (0%)	6300	Estimated
		Train, UCPTE	7000	Estimated
Xenotime	0.12 g	Boat, ocean going (100%)	20000	Estimated
		Boat, ocean going (0%)	6000	Estimated
		Train, UCPTE	7000	Estimated
Pentlandite	7.45 g	Train, UCPTE	4000	Estimated

5.2.3.3 Cathode, raw material production and transports

The processes used to produce the raw material, LSM, for the cathode are shown in figure 5-4. The minerals used are bastnasite, celestite and pyrolusite. Bastnasite is used to produce the nitrate $La(NO_3)_3*6H_2O$ and is to a great extent extracted in China. The production of the nitrate is assumed to take place in the USA as is also the production of all other minerals. There is no information available on transports.

The second substance needed to produce LSM, is the nitrate $Sr(NO_3)_2$, which is refined from the mineral celestite as shown in figure 5-4. The celestite is assumed to be extracted in China. The production of the nitrate is performed in the USA and the location of the transport is assumed.

Finally, pyrolusite is extracted to produce the nitrate $Mn(NO_3)_2*4H_2O$. The extraction of the mineral is assumed to take place in South Africa and the production of the nitrate is assumed to take place in the USA. The production line is shown in figure 5-4.

When the three nitrates have been produced they are transported to Denmark, where they are chemically converted to LSM. This is done in a pyrolysis process, where the three nitrates are dissolved with a complexing agent in water, dripped on a rotating hot surface, which makes the solvent (water) evaporate and the dissolved chemicals react and form LSM. LSM is then mixed with YSZ.

Table 5-11. Process data for the production of the cathode raw material, LSM.

Process				
Production of raw material LSM and YSZ for the cathode				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
LSM	1.3 g			
YSZ	0.13 g			
Raw materials				
See table 5-4				
Used energy				
Electricity, UCPTE	0.06 MJ	Estimated	9	Ex
Diesel, working machines	0.06 MJ	Estimated	9	FU/Ex
Danish average electricity	0.103 MJ	Estimated	6	Ex

For the minerals baddeleyite and xenotime the assumptions made in chapter 5.2.3.1 are used here as well. The mineral bastnasite contains 33,2 % La_2O_3 and it is this mass that is assumed to be transported. In celestite there is 95 % of $SrSO_4$, which is the start substance in the chemical reactions that follows in the production of the $Sr(NO_3)_2$. Therefore 95 % of the total mass of celestite is assumed to be transported. The last mineral, pyrolusite, contains 70 % MnO_2, and consequently 70 % of the total mass of pyrolusite is assumed to be transported.

Table 5-12. Transport data within the production of the cathode nitrates and YSZ.

Transport

Transport of partly processed minerals to the USA

Substance	Amount	Modes of conveyance	Distance [km]	Data quality/Ref.
Bastnasite	0.76 g	Boat, ocean going (100%)	20000	Estimated
		Boat, ocean going (0%)	6000	Estimated
		Train, UCPTE	7000	Estimated
Celestite	0.23 g	Boat, ocean going (100%)	20000	Estimated
		Boat, ocean going (0%)	6000	Estimated
		Train, UCPTE	7000	Estimated
Pyrolusite	0.48 g	Boat, ocean going (100%)	21000	Estimated
		Boat, ocean going (0%)	6300	Estimated
		Train, UCPTE	4200	Estimated
Baddeleyite	2.80 g	Boat, ocean going (100%)	21000	Estimated
		Boat, ocean going (0%)	6300	Estimated
		Train, UCPTE	7000	Estimated
Xenotime	0.03 g	Boat, ocean going (100%)	20000	Estimated
		Boat, ocean going (0%)	6000	Estimated
		Train, UCPTE	7000	Estimated

Table 5-13. Transport data within the production of the cathode raw material, LSM.

Transport

Transport of, nitrates and YSZ from the USA to Denmark.

Substance	Amount	Modes of conveyance	Distance [km]	Data quality/Ref.
Nitrates and	1.65 g	Boat, ocean going (100%)	11000	Estimated
YSZ		Boat, ocean going (0%)	3300	Estimated
		Train, UCPTE	3000	Estimated

5.2.3.4 Interconnect, raw material production and transports

In the production of the interconnect the three minerals bastnasite, celestite and chromite are required to obtain the nitrates wanted. The bastnasite and the celestite processes are the same as for the cathode. The chromite is needed to produce the nitrate $Cr(NO_3)_3*9H_2O$ and is assumed to be extracted in South Africa. The processes and assumed transports are shown in figure 5-5.

As for the cathode a pyrolysis process is carried out at Risoe in Denmark under laboratory circumstances.

Table 5-14. Process data for the production of the interconnect raw material, LSCV.

Process				
Production of raw material, LSCV				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
LSCV	123.75 g			
Raw materials				
See table 5-5				
Used energy				
Electricity, UCPTE	6.77 MJ	Estimated	9	Ex
Diesel, working machines	6.77 MJ	Estimated	9	FU/Ex
Danish average electricity	8.91	Estimated	6	Ex

The assumptions made for the transports of the minerals bastnasite and celestite in the cathode raw material process are the same assumptions made here for the interconnect. In the mineral chromite 80 % of Na2CrO4 can be obtained, which is the substance needed to produce $Cr(NO_3)_3$*$9H_2O$. The assumption is therefore that 80 % of the total mass of chromite is transported.

Table 5-15. Transport data for the production of the cathode nitrates.

Transport				
Transport of, to some extent processed minerals to the *USA*				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
Bastnasite	70.53 g	Boat, ocean going (100%)	20000	Estimated
		Boat, ocean going (0%)	6000	Estimated
		Train, UCPTE	7000	Estimated
Celestite	26.50 g	Boat, ocean going (100%)	20000	Estimated
		Boat, ocean going (0%)	6000	Estimated
		Train, UCPTE	7000	Estimated
Chromite	60.57 g	Boat, ocean going (100%)	21000	Estimated
		Boat, ocean going (0%)	6300	Estimated
		Train, UCPTE	4200	Estimated

Table 5-16. Transport data for the production of the cathode raw material, LSCV.

Transport				
Transport of nitrates from the USA to Denmark.				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
Nitrates	86.25 g	Boat, ocean going (100%)	11000	Estimated
		Boat, ocean going (0%)	3300	Estimated
		Train, UCPTE	3000	Estimated

5.2.3.5 Production of a unit cell, including transports

After the production of the electrolyte raw material, YSZ, and the anode raw material, NiO + YSZ, in the USA they are transported to Risoe for the production of unit cells.

Table 5-17. Transport data for deliverance of YSZ and NiO-YSZ..

Transport				
Transport of YSZ and NiO from the USA to Denmark.				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
YSZ	16 g	Boat, ocean going (100%)	11000	Estimated
		Boat, ocean going (0%)	3300	Estimated
		Train, UCPTE	3000	Estimated
NiO + YSZ (anode material)	1.17 g	Boat, ocean going (100%)	11000	Estimated
		Boat, ocean going (0%)	3300	Estimated
		Train, UCPTE	3000	Estimated

When all raw materials are transported to Risoe the production of unit cells begin. The first process is tape casting of the electrolyte, which is the bearing element in the fuel cell. After the tape casting of YSZ, a sintering process forms the electrolyte "film" at 1350 °C to a plate. The thickness of the electrolyte is 160 μm [6].

Both the cathode and the anode are made in two layers. The layer in contact with the electrolyte is the electrochemically active one (A on one side and C on one side) and the second layer, also in contact with the interconnect, is a current collecting layer (CCC and ACC), see figure 5-7. The second process step is the production of the cathode's and anode's electrochemically activ layers, which both are ceramic slurries that are spray painted on the electrolyte and sintered individually. After that the ACC layer is spray painted on to the A layer and then sintered (1300 °C). To put the CCC layer on the C layer, tape casting is used and then the layer is sintered at 1300 °C [6].

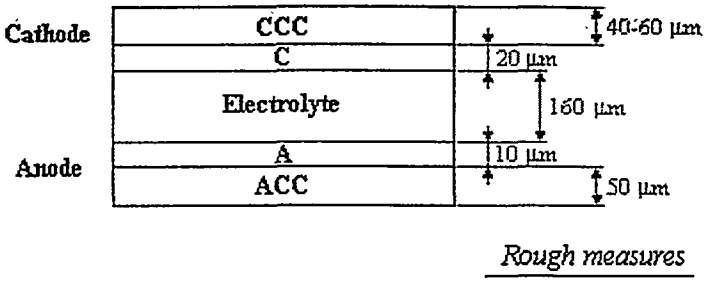


Figure 5-7. The layers in production of planar SOFC (Risoe) [6].

Finally the interconnects are produced. This is done by uniaxially pressing the powder to a plate and then sintering it. After this the interconnects are shaped by a cutter and other machines to give the fine surfaces, cross flow channels and the defined boarder lines that are needed. The process data of this process are presented below in table 5-18.

Table 5-18. Process data for the production of a unit cell.

Process				
Production of a unit cell				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
1 Unit cell	83.0 g			
Raw materials				
See table 5-6				
Used energy				
Danish average electricity	169 MJ	Literature	9	Ex

5.3 Production of the gas turbine

The gas turbine used in the SOFC/GT process is a micro turbine produced and developed by Volvo Aero Turbines. The mass flows and production methods for this single-shafted turbine are almost unknown. The data known are that the total weight of the drive line for a 40 kW turbine used in the Volvo ECC is 237 kg [21]. Only steel and copper are assumed to be input materials in the production process.

5.3.1 Process trees

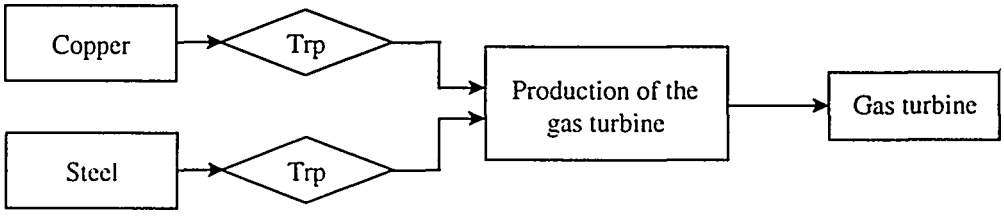


Figure 5-8. Process tree for the production of a gas turbine.

The processes used to produce steel and copper have been done before at Sydkraft and the results have been critically reviewed. The emission factors and the use of resources in these two processes are presented in appendix D and they are included in this study.

5.3.2 Mass flows

The assumption is made that the 237 kg drive line contains a gas turbine, a gearbox and axles between these two units. The second assumption done here is that the gas turbine (including the recuperator) stands for approximately 1/3 of the total weight, i.e. 80 kg. The materials used in the 202 kWe (net) gas turbine are set to twice this amount with the following composition, 140 kg steel and 20 kg copper. No waste material during the production of the gas turbine is included in this study.

5.3.3 Processes and transports for production of GT

Both iron-ore and copper-ore can be extracted and processed in Sweden and therefore the standard values used at Sydkraft, for the average transport distances in Sweden are used here [23]. The inventory results for the processing of steel and copper will be used here.

Table 5-19. Process data for the production of gas turbine

Process				
Production of gas turbine				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
Gas tubine	160 kg			
Raw materials				
Steel	140 kg	Estimated	13, 21	
Copper	20 kg	Estimated	13, 21	

Table 5-20. Transport data for the production of the gas turbine.

Transport				
Transport of steel and copper within the production of the GT				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
Steel	140 kg	Road, long distance (100 %)	300	Standard/23
		Road, long distance (0 %)	90	Standard/23
Copper	20 kg	Road, long distance (100 %)	300	Standard/23
		Road, long distance (0 %)	90	Standard/23

5.4 Construction and demolition of the SOFC/GT process

5.4.1 Preparation of the location for the SOFC/GT process

The two components, the SOFC-module and the gas turbine, are assumed to be fully assembled when they arrive at the location where they are to be stationed. Before they arrive, the ground must be flattened, the concrete foundations must be ready and this requires work by machines and transport of construction materials. The process is assumed to need a total area of 400 m^2. This whole area is estimated to be excavated, which leads to a total excavated material volume of 256 m^3 [28]. The total ground area that the SOFC unit requires is assumed to be 35 m^2 and since the turbine is rather small a total 25 m^2 are assumed to be enough for the turbine, recuperator, desulphurization system, etc. This leads to a total concrete foundation area of 60 m^2. The concrete foundations are estimated to be 0,2 m deep after a conversation with Åke Carlsson at Sycon. This leads to a total need of 12 m^3 of concrete. Other areas (340 m^2) within the location are assumed to be coated with asphalt. Within the concrete foundations a standard of 60-80 kg/m^3 concrete [28] of reinforcing iron is used. To prepare the location by excavating, coating with asphalt etc. a total need of 0,5 m^3 of diesel, used by working machines, is estimated [28]. Below in table 5-21 the total masses of required materials and energy used to prepare the location are presented.

Table 5-21. Material and energy use for preparing the location (estimated figures after literature [28]). Transports are included here.

Process				
Preparation of the SOFC/GT location				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
Prepared location	1			
Preparation				
Excavation	256 m3	Estimated	28	
Concrete	12 m3 = 27600 kg	Estimated	28	
Reinforcment iron	720 kg	Estimated	28	
Asphalt	6800 kg	Estimated	28	
Used energy				
Diesel, working machines	0.5 m3 = 17850 MJ*	Estimated	28	FU/Ex

* Calculations made with LHV = 42,5 GJ/ton, density = 840 kg/m3 [28]

Table 5-22. Transport data for the SOFC-module (from Denmark) and the gas turbine to the location in Sweden.

Transport				
Transport of SOFC-module and GT to the location				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
SOFC-module	1	Boat, coast going (100%)	20	Estimated
		Boat, coast going (0%)	6	Estimated
		Road, long distance (100%)	300	Standard/23
		Road, long distance (0%)	90	Standard/23
Gas turbine	1	Road, long distance (100%)	300	Standard/23
		Road, long distance (0%)	90	Standard/23

5.4.2 Demolition of the SOFC/GT process

The process data used here are assumed to be the same process data used at the preparation of the location, see table 5-21. However the transport data are changed. All materials of the SOFC/GT process are assumed to be transported to the local landfill within the community. All transports during the demolition phase are thereby done by road, short distance (see table 5-23) and the standard used at Sydkraft for short distance is used.

Table 5-23. Transport data for the demolition of the SOFC/GT process.

Transport				
Transport of SOFC/GT process during the demolition phase				
Substance	**Amount**	**Modes of conveyance**	**Distance [km]**	**Data quality/Ref.**
SOFC-module	1	Road, short distance (100 %)	30	Standard/23
		Road, short distance (0 %)	9	Standard/23
Gasturbine	1	Road, short distance (100 %)	30	Standard/23
		Road, short distance (0 %)	9	Standard/23

5.5 Operation phase of the SOFC/GT process

The fuel used for the SOFC/GT process is natural gas from Denmark, which is distributed in Sweden. The mass flow of reformed gas into the SOFC/GT process is 0.04185 kg/s, which requires 0.011 kg natural gas per second and the exhaust fumes out of the process is 0.549 kg/s. The assumed operation time is, as mentioned before, 10 000 hours, which leads to the process data described in table 5-24.

Table 5-24. Process data for the operation time, 10 000 h, of the SOFC/GT process.

Process				
Operation of the SOFC/GT process				
Product	**Amount**	**Data quality**	**Reference**	**Emission factor**
Electricity	4 293 561 kWh			
Used energy				
Natural gas	668175 m3	Estimated	3	FU/Ex
Emissions *to air after combustion*				
H2O	1 376 767 kg	Estimated	3	
CO2	1 168 744 kg	Estimated	3	

The total electricity production, 4 293 561 kWh is calculated with the efficiency 0,73 [3] for the 429 kW (net) SOFC/GT process with the operational time of 10 000 hours.

5.6 Inventory results

The inventory results are presented in its entirety in appendix D. The results are divided after components and their sub components.

- Production of the SOFC
- Production of the electrolyte
- Production of the anode
- Production of the cathode
- Production of the interconnect
- Production of the GT
- Operation phase of the SOFC/GT process
- Production of 1 m^3 natural gas
- Production of 1 kg sulphuric acid (HNO_3)
- Production of 1 kg copper (50/80)
- Recycling of 1 kg copper (50/80)
- Production of 1 kg steel (25/70)
- Recycling of 1 kg steel (25/70)

5.7 Discussion of the inventory results

When taking a closer look at the energy used during the production of the components in appendix D, the amount of energy used when producing the SOFC is considerably higher than for the gas turbine. The distribution of energy required to produce the components per produced kWh electricity is as shown in table 5-25.

Table 5-25. Energy used during the production of the components per kWh produced electricity.

Component	Used energy [MJ/kWh prod. el.]	Total energy requirement of the whole production [%]
Toatal SOFC-module	1,4467	99,990
Electrolyte (raw material)	*0,0451*	*3,117*
Anode (raw material)	*0,0013*	*0,090*
Cathode (raw material)	*0,0019*	*0,131*
Interconnect (raw material)	*0,1825*	*12,614*
Assemblage	*1,2159*	*84,038*
Gas turbine	0,00014	0,010

It is obvious that the interconnect is the stand-alone most energy demanding component to produce. This is not very surprising since this is the component in the SOFC that requires most raw material to produce. The need of raw material of the interconnect is approximately 87 % of the total weight of the SOFC unit cell. At Risoe today already a new type of interconnect is under development, a so called thin film interconnect. The use of energy will, because of this new type of interconnect, in all probability decrease a great deal, since the used energy during the production and the amount of used material are strongly connected.

Another interesting comparison that can be made in the inventory results are the emissions to air during the construction and demolition phase and the operation phase. The most distinguishing emission is carbon dioxide. In the total construction and demolition phase the carbon dioxide emission are 482 g/produced kWh electricity (99,8 % from the SOFC production). In the operation phase the contribution of carbon dioxide are 134,3 g/ produced kWh electricity (42 % is emission from the production of natural gas, see appendix D). In figure 5-9 the distribution of CO2 between the phases is presented. In figure 5-10 four other emissions to air for the phases are presented.

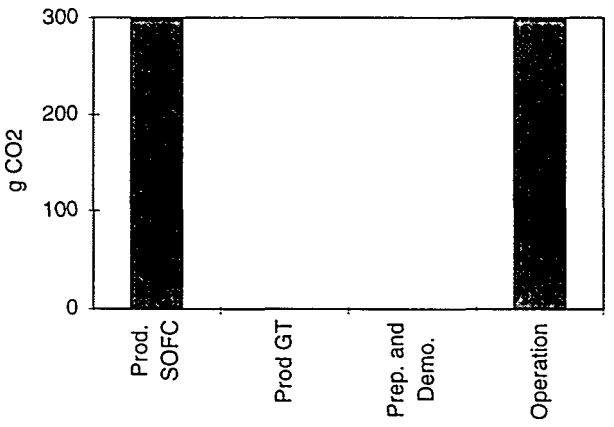


Figure 5-9. Distribution of carbon dioxide (CO2) for the phases during the production of 1 kWh electricity.

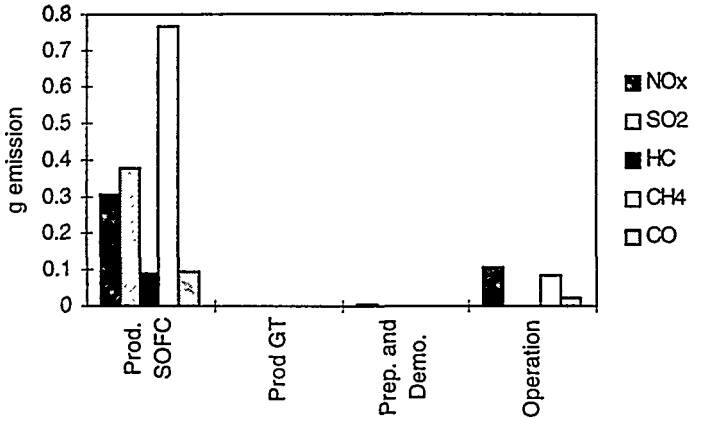


Figure 5-10. Distribution of emissions to air for the phases during the production of 1 kWh electricity.

As can be seen in the two figures above the production of the SOFC and the operation of the SOFC/GT process have an almost identical contribution of carbon dioxide to the air. For the other emissions to air, the production of the SOFC is the main contributor during the SOFC/GT process' life cycle.

6 Impact assessment

In this study the emissions are classified and characterised. The disposition of the impact assessment is as follows:

- Use of resources
  - *Non-renewable material resources*
  - *Renewable material resources*
  - *Recycled material resources*
  - *Non-renewable energy resources*
  - *Renewable energy resources*
  - *Recycled energy resources*

- Discharge of pollution
  - *contributing to the global warming*
  - *contributing to acidification*
  - *contributing to creation of photochemical ozone*
  - *contributing to eutrophication*
  - *discharge of toxic substances*

- Other information
  - *Solid waste*
  - *Recycling of material*
  - *Use of acreage*

The impact categories that have been chosen here are resources – energy and material, global warming, acidification, creation of photochemical oxidants and eutrophication. Also information on solid waste, recycling of material and use of acreage are chosen to be presented. These choices have been made, mainly because of the fact that the life cycle assessments performed at Sydkraft include these categories and this information, making these choices here as well creates a continuity among Sydkraft's reports.

6.1 Classification and characterization

The first step is the classification of the inventory data, i.e. the results from the inventory analysis. After this the data are characterised and the characterization is performed in three steps. The first step in the characterization is to decide which characterization factors to use and which characterization categories to assess. In the second step, a multiplication of the inventory data and the characterization factor is carried out, which leads to so called equivalent units. The final step is to add up the equivalent units to category indicators that leads to the potentially environmental impact for each environmental category. This last phase is presented in chapter 6.2. All impact categories that can be assessed according to the Nordic Guidelines are presented in table 3-1.

The use of resources, toxic substances, solid waste, recycling of materials and use of acreage, are not classified. They are just shown as inventory data.

6.1.1 Global warming potential (GWP)

The substances that have an impact on the global warming, have been chosen according to the Intergovernmental Panel of Climate Change (IPCC) [29]. The equivalent unit used is g CO_2 equivalents per g in a 100-year perspective, i.e. the total discharge of global warming gases is presented as the sum of GWP.

According to IPCC, the gases that contribute to the global warming are: carbon dioxide (CO_2), methane (CH_4), nitrogen dioxide (N_2O) and a number of gases belonging to the groups halogenated hydrocarbons (CFC, HCFC), bromined hydrocarbons, HFC etc. Since no halogenated hydrocarbons are present in the analysis of the SOFC/GT process, these have been left out in table 6-1. The characterization factor used for the GWP can be described as the heat absorption ability of the substance. The methane molecule then have an ability of heat absorption 21 times higher than the carbon dioxide molecule.

Table 6-1. Abstract from IPCC classification of the global warming, 100-year perspective, 1995 [29].

Classification	Characterisation factor
CF4	4500 g CO2 equivalents / g
CH4	21 g CO2 equivalents / g
CO2	1 g CO2 equivalents / g
N2O	310 g CO2 equivalents / g
CFC/HCFC	8500 g CO2 equivalents / g

6.1.2 Ozone Depletion Potential (ODP)

Ozone depletion gases are halogenated hydrocarbons and since non of these are present in this study, ozone depletion is calculated within this impact assessment.

6.1.3 Acidification potential (AP)

The gases that contribute to acidification are; sulphur dioxide (SO_2), nitrogen oxides (NO_x), ammonium hydrate (NH_3), hydrochloric acid (HCl) and other acids and sulphur compounds [31]. Here the classification and characterization factors are collected from Nordic Guidelines on Life-Cycle Assessment, technical report 10, table A17 [30]. In this table two columns are presented, minimal impact (min) and maximum impact (max.). In this study the "worst case" (max.) is used. The discharges are presented as the sum of the acidification potential, expressed as mole H^+/g substance. The weighting factor here is a measure of the assessed amount of produced mole of acid per g substance.

Table 6-2. Classification of acidification gases according to the Nordic Guidelines, technical report 10, table A17, max. [31].

Classification	Characterisation factor	
HCl	0.027	mole H+ / g substance
NH3	0.059	mole H+ / g substance
NOx	0.022	mole H+ / g substance
SO2	0.031	mole H+ / g substance

6.1.4 Photochemical Ozone Creation Potential (POCP)

Gases contributing to the creation of photochemical ozone are; NOx, volatile hydrocarbons and CO. Here the classification and characterization factors have been collected from CIT Ecologik's specification of indexes [31]. The discharges are presented as the sum of the photochemical ozone creation potential (POCP), expressed as g ethylene-equivalents/g substance. NO_x is not considered as a separate substance. It is considered as a background radiation and is deciding, which characterization factors that are chosen. Here a high background radiation is chosen, since this is suggested for European conditions [31].

Table 6-3. Classification of creation of photochemical ozone according to CIT's specification of indexes [31].

Classification	Characterisation factor
Aldehydes	0.443 g ethene-equivalents / g
Benzene	0.402 g ethene-equivalents / g
CH4	0.007 g ethene-equivalents / g
CO	0.032 g ethene-equivalents / g
Phenol	0.416 g ethene-equivalents / g
HC	0.416 g ethene-equivalents / g
HxCy (not CH4)	0.416 g ethene-equivalents / g
VOC	0.377 g ethene-equivalents / g

6.1.5 Aquatic Oxygen Depletion (AOD)

The contributing gases to aquatic oxygen depletion (AOD) are: NO_x, NH_3, volatile hydrocarbons, discharge of phosphor and nitrogen and discharging of oxygen consumption organic material (COD). This impact category is also called eutrophication. Nitrogen and phosphor are the two most important feeding-stuff in nature, but when there is a surplus of these the available oxygen becomes insufficient, which in the end impoverishes the growth of flora.

Here the Nordic Guidelines on Life-Cycle Assessment, technical report 10, table A18 is used for the classification and characterization factors. The column for N-limited in table A18 is used and the AOD potential is expressed as g O2-equivalents/g substance.

Table 6-4. Classification of substance, contributing to AOD, according to Nordic Guidelines, technical report 10, table A18 [30].

Classification	Characterisation factor
COD	1 g O2-equivalents / g
N	20 g O2-equivalents / g
NH4+	15 g O2-equivalents / g
NO3-	4.4 g O2-equivalents / g

6.2 Impact assessment results

All data used in this assessment are from chapter 5, where the inventory results in detail are presented in appendix E. The data for the energy carriers and transports are presented in appendix D. The data have been calculated with in the LCA-program EcoLab and are presented here after classification.

6.2.1 Use of resources

The resources used are presented in table 6-5 and 6-6 after quantity. Only material resource quantities bigger than 0,030 g (per produced kWh electricity) are accounted for. The resources are from the inventory data and the material resources are given in g and the energy resources in g or MJe (water power). The resources are calculated for 1 kWh produced electricity by the SOFC/GT process. The total results for both material and energy resources, during construction and demolition and operation, are described in appendix E. In the operation phase the fuel chain (production of natural gas) is included.

Table 6-5. Use of material resources per produced kWh of electricity, divided after construction and demolition, operation and the total usage.

Material resources	*Construction and demolition*	*Operation*	*Total*
Non-renewable material resources			
Baddeleyite	25,8470 g	0 g	25,8470 g
Bastnasite	1,5459 g	0 g	1,5459 g
Copper-ore (0,35 % Cu)	0,4509 g	0 g	0,4509 g
Limestone	0,0018 g	0,3098 g	0,3117 g
Pentlandite	0,6706 g	0 g	0,6706 g
Chromite	0,5451 g	0 g	0,5451 g
Celestite	0,2024 g	0 g	0,2024 g
Iron-ore	0,0258 g	0,0367 g	0,0625 g
Rock	0 g	0,0336 g	0,0336 g
Renewable material resources			
-			
Recycled material resources			
Copper	0,00233 g	0 g	0,00233 g
Steel	0,00815 g	0 g	0,00815 g

Table 6-6. Use of energy resources per produced kWh of electricity, divided after construction and demolition, operation and finally the total usage.

Energy resources	*Construction and demolition*	*Operation*	*Total*
Non-renewable energy resources			
Natural gas	3,1977 g	137,7113 g	140,9090 g
Coal	97,6921 g	0,0402 g	97,7323 g
Crude Oil	7,1988 g	0,7083 g	7,9071 g
Natural Uranium	0,000152 g	0,0000001 g	0,000152 g
Renewable energy resources			
Biomass	1,4995 g	0,00054 g	1,5001 g
Water power	0,01541 MJe	0,00002 MJe	0,01543 MJe
Recycled energy resources			
-			

6.2.1.1 Internal account of energy

The internal account of energy is not a part of the final results of the LCA, it is only used internally. This is because the used energy is based on the energy resources presented in table 6-6. The unit used in table 6-7, use of energy, is MJ and it is calculated per produced kWh electricity. All energy carriers are presented further in appendix D.

Table 6-7. Internal account of energy per produced kWh electricity

Energy carrier	*Construction and demolition*	*Operation*	*Total*	*Unit*
Electricity, Denmark av. (e)	1,2816	0,0004	1,2820	MJ
Natural gas (e)	0,0008	0,3112	0,3120	MJ
Diesel (e)	0,0701	0,0281	0,0983	MJ
Electricity, UCPTE (e)	0,07272	0,000002	0,0727	MJ
Diesel, boat (e)	0,02956	0,00002	0,0296	MJ
Oil EO1 (e)	0,00023	0,00001	0,00024	MJ
Electricity, Sweden av. (e)	0,00002	0,00004	0,00006	MJ
Oil EO5 (e)	1,1 E-06	1,3 E-06	2,4 E-06	MJ
LP-gas (propane) (e)	6,7 E-06	5,9 E-07	7,3 E-06	MJ
Electricity, Australia av. (e)	1,2 E-07	1,8 E-07	3,0 E-07	MJ
Gasoline (e)	3,0 E-08	4,0 E-08	7,0 E-08	MJ
Coal (e)	1,0 E-08	1,0 E-08	2,0 E-08	MJ

6.2.2 Discharge of pollution

The discharges of pollution are divided into global warming, acidification, creation of photochemical ozone, eutrophication and toxic waste. The contribution of the discharges are for 1 kWh produced electricity. Tables 6-8 to 6-11 are divided into the two phases construction and demolition and operation, together with the total results.

6.2.2.1 Global warming

The global warming potential (GWP 100), per produced kWh electricity, presented below in table 6-8, is given in g CO2-equivalents/g substance, in a 100 year perspective.

Table 6-8. The results for the global warming potential (GWP 100) per kWh produced electricity. The unit here is g CO_2-equivalents/g substance.

Substance	*Construction and demolition*	*Operation*	*Total*
CO_2	298,74445	297,30978	596,05423
CH_4	16,12286	0,00007	16,12293
N_2O	3,12918	0,18670	3,31588
$CF_4 + C_2F_6$	0,00005	0,00007	0,00012
Total	317,99654	297,49662	615,49316

6.2.2.2 Acidification

The acidification potential (AP) for the production of 1 kWh electricity, is given in mole*10^{-3} H^+/g substance.

Table 6-9. Acidification potential per kWh produced electricity. The unit here is mole10^{-3} H^+/g substance

Substance	*Construction and demolition*	*Operation*	*Total*
HCl	0,00001	0,00002	0,00003
NH_3	0,05444	0	0,05444
NO_x	6,93807	2,34830	9,28637
$SO_2 + SO_x$	11,67531	0,07983	11,75514
Total	18,66783	2,42815	21,09598

6.2.2.3 Creation of photochemical ozone

The Photochemical Ozone Creation Potential (POCP), for producing 1 kWh electricity, is given in mg ethylene-equivalents/g substance and is presented below in table 6-10. The results are calculated with the characterization factor of high background radiation of NOx.

Table 6-10. Creation of photochemical ozone per kWh produced electricity. The unit here is mg ethene-equivalents/g substance.

Substance	*Construction and demolition*	*Operation*	*Total*
Aldehydes	0,02316	0	0,02316
Benzene	0	0	0
CH_4	5,37429	0,57882	5,95311
CO	3,11766	0,77657	3,89423
Phenol	0,00058	0,00007	0,00065
HC	36.51514	0.53287	37.04801
VOC	0,79949	0,00005	0,79954
Total	45,83032	1,88838	47,7487

6.2.2.4 Eutrophication

The aquatic oxygen depletion potential, for the production of 1 kWh electricity, presented below in table 6-8, is given in mg O2-equivalents/g substance for the discharge of gases.

Table 6-11. Acidification potential per kWh produced electricity. The unit here is mg O_2-equivalents/g substance.

Substance	*Construction and demolition*	*Operation*	*Total*
COD	0,31831	0,03343	0,35174
N-tot	0,96967	0,11749	1,08716
Total	1,28798	0,15092	1,43890

6.2.2.5 Toxic substances

The toxic waste with the impact media air, is from the inventory data and has the unit g. It is divided into the categories dust, hydrocarbons, organic coal and metals. All emissions are further described in appendix E.

Table 6-12 Toxic waste per produced kWh electricity.

Toxic substance	*Construction and demolition*	*Operation*	*Total*
Dust			
Dust	0,1498 g	0,0001 g	0,1499 g
Hydrocarbons			
PAH	3,1 E-09 g	4,5 E-09 g	7,6 E-08 g
HC	0,0878 g	0,0013 g	0,0891 g
Organic coal			
VOC	0,0021 g	1,2 E-07 g	0,0021 g
Metals			
As	1,4 E-08 g	0,3 E-08 g	1,7 E-08 g
Cd	4,1 E-09 g	0,3 E-09 g	4,4 E-09 g
Cr	4,7 E-09 g	6,4 E-09 g	1,1 E-08 g
Cu	1,2 E-07 g	3,1 E-09 g	1,2 E-07 g
Fe	2,7 E-06 g	3,8 E-06 g	6,5 E-06 g
Hg	8,0 E-10 g	0 g	8,0 E-10 g
Mn	2,7 E-10 g	3,8E-10 g	6,5 E-10 g
Ni	2,9 E-09 g	4,1 E-09 g	7,0 E-09 g
Pb	2,1 E-07 g	2,0 E-08 g	2,3 E-07 g
V	6,5 E-09 g	9,3 E-09 g	1,6 E-08 g
Zn	2,2 E-07 g	1,5 E-07 g	3,7 E-07 g

6.2.3 Other information

In this chapter solid waste, recycling of materials and use of acreage are presented.

6.2.3.1 Solid waste

The solid waste is presented in detail in appendix E. All waste products named tailing and tailingxxxx in appendix E are presumed unknown waste products (see 5.2.2). In table 6-13 the waste products are divided into hazardous waste and other waste.

Table 6-13. Solid waste products per produced kWh electricity.

Toxic substance	*Construction and demolition*	*Operation*	*Total*
Hazardous waste	2,9 E-08 g	0,0507 g	0,0507 g
Radioactive waste	0,001534 ml	0,0000012 ml	0,001535 ml
Other waste	32,8406 g	0,0993 g	32,9399 g

6.2.3.2 Recycling of materials

All recycled materials are from the production of the gas turbine, which consists of copper and steel. The copper is recycled to 80 % and the steel is recycled to 70 % during the demolition. During the construction, the materials used have been recycled as follows: copper 50 % and steel 25 %. No reinvestments are made during the operation phase, so no material is recycled during the operation phase.

Table 6-14. Material to recycling (waste from the gas turbine process) per produced kWh electricity.

Recycled material	*Demolition*
Steel	0,003727 g
Copper	0,022825 g

6.2.3.3 Use of acreage

The direct use of acreage is the ground, where the SOFC/GT process is located. Here this acreage is 400 m^2 during the construction and demolition-phase. This leads to 0,00022 m^2 per produced kWh electricity.

7 Interpretation

In this chapter interpretations of the impact assessment results are made. The two analysis methods that are used are dominance analysis and sensitivity analysis. The first of these, the dominance analysis, is made to locate "hot spots" within the SOFC/GT process, i.e. find phases who contribute the most to the different impact categories. The sensitivity analysis is made to see the changes in the environmental impact that for instance a change of lifetime of the SOFC/GT process makes.

7.1 Dominance analysis of the production of the SOFC

In the dominance analysis of the production of the SOFC the four sub components electrolyte, anode, cathode and interconnect are investigated further. Also the assemblage of the four sub components to a unit cell is included in the analysis. The impact categories that are analysed are use of resources, global warming, acidification, creation of photochemical ozone and eutrophication.

7.1.1 Use of resources

The material resources used during the production of each sub component are shown below in table 7-1. The energy resources used during the production of the SOFC per kWh produced electricity are presented in table 7-2, where the sub components' energy use represents the energy used during the production of the raw materials. The final stage in the production, the production of solid sub components followed by the production of a unit cell (uc), is also presented in a column. The production of a unit cell will from now on be called assemblage.

Table 7-1. Material resources per kWh produced electricity used during the production of the sub components in the SOFC.

Material resource (mr)	Electrolyte use of mr [g]	Anode use of mr [g]	Cathode use of mr [g]	Interconnect use of mr [g]	Total use of mr [g]
Baddeleyite	25,6414	0,8073	0,2018	-	26,6505
Bastnasite	-	-	0,0164	1,5295	1,5459
Celestite	-	-	0,0017	0,2008	0,2025
Chromite	-	-	-	0,5451	0,5451
Pentlandite	-	0,6706	-	-	0,6706
Pyrolusite	-	-	0,0050	-	0,0050
Xenotime	0,0269	0,0008	0,0002	-	0,0279

Table 7-2. Energy resources per kWh produced electricity used during the production of the sub components in the SOFC. (Only Water power is in MJ, other energy resources are in g)

Energy resources (er)	Electrolyte use of er [g]	Anode use of er [g]	Cathode use of er [g]	Interconnect use of er [g]	Production of a SOFC uc, use of er [g]
Biomass	-	-	0,0009	0,0751	1,4239
Coal	0,5188	0,0233	0,0703	6,3302	90,7495
Crude oil	0,8115	0,0321	0,0232	2,2066	3,9167
Natural gas	0,1212	0,0053	0,0072	0,7507	2,3080
Natural uranium	0,000038	0,000002	0,000001	0,000112	0
Water power	0,0037 MJ	0,0002 MJ	0,0001 MJ	0,0109 MJ	0,0005 MJ

As presented in table 7-1 the material resources during the production of the SOFC are dominated by the mineral baddeleyite, which is required to produce the electrolyte. It answers for approximately 86 % of the total material resources. The baddeleyite is dominant since it only contains approximately 4 % of ZrO_2, which is the required substance. The energy resources needed to produce the electrolyte raw material are however not in majority in table 7-2. The interconnect is the post who among the sub components obviously is contributing most to the use of energy resources. This is because of its final weight (123 g) that answers for approximately 87 % of the total weight of SOFC. The minerals used to produce the interconnect also require a great amount of energy to be extracted and refined as shown in appendix B.

The post that contributes the most in table 7-2 is the production of a SOFC unit cell. The major energy resource is coal, which can be explained by the fact that most of the energy used during the production of the SOFC is Danish average electricity where coal accounts for 91 % of the energy resources.

7.1.2 Discharges of gases contributing to the global warming

The total contribution to the global warming during the production of the SOFC-module are 317 g CO_2-equivalents/g substance. The distribution of these between the production of raw materials of the sub components and the assemblage of the unit cell is as shown in figure 7-2.

The absolute major gas contributing to the global warming during the production of the SOFC is carbon dioxide (CO_2). Carbon dioxide answers for 94 % of the equivalents although its weighting index is only 1. The weighting index of methane (CH4) is 4500.

The interconnect is the sub component where the production of raw material contributes the most to the global warming. The pyrolysis process in the interconnect raw material production, carried out in Denmark, answers for approximately 50 % of the CO_2-equivalents/g contributed by the interconnect. The other 50 % mainly comes from transports, i.e. the fuel used during the transports, which also is the case during the production of the cathode raw material.

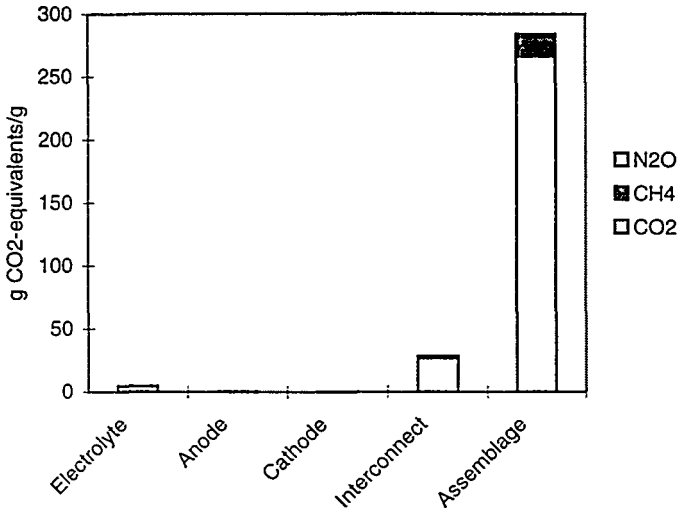


Figure 7-1. Distribution of CO_2-equivalents/g substance (per kWh produced electricity by the SOFC/GT process) between the production of raw materials to the sub components and the assemblage of the unit cell.

The assemblage of the unit cell is the major contributor to the global warming during the production of the SOFC. This is because of the requirement of energy. As shown in table 7-1, the production of a unit cell is the major post regarding the use of energy resources. When the energy resources, except renewable energy resources, are converted to electricity or heat, exhaust gases are produced and these contain gases contributing to the global warming.

7.1.3 Discharge of gases contributing to acidification

The total contribution to acidification during the production of the SOFC-module is 0,019 mole H^+/g substance. The distribution of these between the production of raw materials of the sub components and the assemblage of the unit cell is as shown in figure 7-2.

There are two major contributors to the acidification during the production of the SOFC: sulphur dioxide (SO_2) and nitrogen oxide (NO_x). The main contributor of sulphur dioxide is the Danish average electricity, which answers for approximately 70 %. Danish average electricity also answers for 54 % of the discharge of nitrogen oxide. The whole contribution of mole H^+/g during the assemblage of the SOFC comes from Danish average electricity.

The interconnect is once again the major contributor when comparing the production of the raw materials of the sub components. The major stand-alone contributing post during the production of raw material of the interconnect is the refining of the raw material in the USA (14 % of SO_2 and 12 % of NO_x). This is because of the great amount of raw material used in the interconnect (87 % of the total unit cell) that has to be processed. The transports contributes with approximately 7 % of SO_2 and 5 % of NO_x.

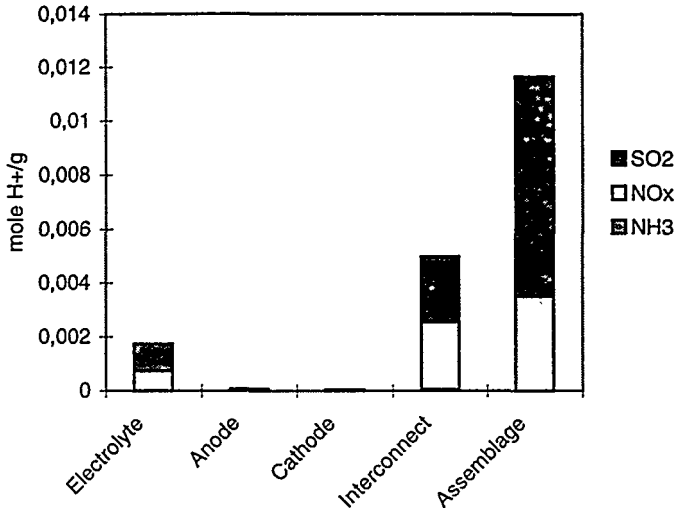


Figure 7-2. Distribution of mole H^+/g substance (per kWh produced electricity by the SOFC/GT process) between the production of raw materials to the sub components and the assemblage of the unit cell.

7.1.4 Discharge of gases contributing to creation of photochemical ozone

The total contribution to the creation of photochemical ozone during the production of the SOFC-module is 0,046 g ethylene-equivalents/g substance. The distribution of these between the production of raw materials of the sub components and the assemblage of the unit cell is as shown in figure 7-3.

As shown in figure 7-3 the gas contributing the most to the creation of photochemical ozone is the hydrocarbons (HC). In the assemblage of the SOFC unit cell methane also is a major contributor.

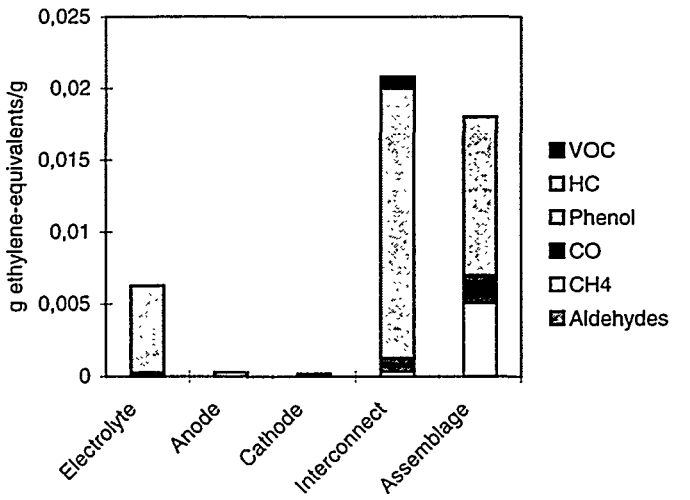


Figure 7-3. Distribution of g ethylene-equivalents/g substance (per kWh produced electricity by the SOFC/GT process) between the production of raw materials to the sub components and the assemblage of the unit cell.

The major contributor to the creation of photochemical ozone is the production of raw material for the interconnect, which answers for 46 %. During the production of raw material for the interconnect hydrocarbons are dominant. The major contributing phase of hydrocarbons is the refining of the raw material in the USA, which answers for 52 %. Once again the high quantity of mass of the raw material is the reason, which requires a large amount of electricity during the refining.

Also the electrolyte has a considerable pile in figure 7-3. The dominant contributor of hydrocarbons here is also the refining of the raw material in the USA (49 % of the total HC within the electrolyte).

7.1.5 Discharge of gases contributing to eutrophication

The total contribution to the eutrophication during the production of the SOFC-module is 0,0012 *g* O_2-equivalents/g substance. The distribution of these between the production of raw materials of the sub components and the assemblage of the unit cell is as shown in figure 7-4.

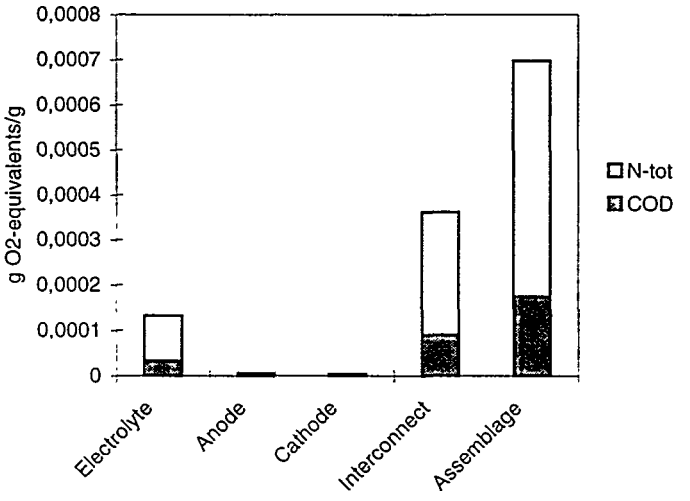


Figure 7-4. Distribution of g O_2-equivalents/g substance (per kWh produced electricity by the SOFC/GT process) between the production of raw materials to the sub components and the assemblage of the unit cell.

The major aquatic oxygen depleting substance during the production of the SOFC is N-tot (includes nitrogen, NH_4^+ and NO_3^-), which answers for 70 % of the g O_2-equivalents/g. Among the sub components the production of raw material for the interconnect again contributes most to the eutrophication. The stand-alone most contributing phase here is the extraction of the minerals and the refining done in the supply country. These processes answer for 70 % of the hydrocarbons and 63 % of the COD.

In the "assemblage-pile" the contribution of N-tot and COD all come from the Danish electricity. For the production of the raw material for the electrolyte the contributors of N-tot are the transports (54 %) and the Extraction and refining in the supply country (46 %). These two contributors are also dominant regarding the COD, 50 and 43 % respectively.

7.1.6 Toxic waste

During the production of the SOFC the contribution of toxic waste consists of dust, hydrocarbons (HC) and volatile organic compounds (VOC). The dust mainly comes from the assemblage of the SOFC. The main contributor of hydrocarbons is the production of the interconnect, which also is the case for the VOC.

7.1.7 Summary of production of the SOFC

The production of raw material for the interconnect is consequently the most contributing sub component within all impact categories. It is only the assemblage of the sub components that contributes more to the impact categories, but this is a part of the production of the interconnect. If the assemblage impacts are divided into four equal parts the following summarizing diagram (figure 7-5) of all discharges is received.

For the interconnect raw material production the following summarizing comments can be mentioned:

- The impact on global warming is mainly a result of the electricity used during the pyrolysis process in Denmark and the transports during the production.
- The two single major contributors to acidification are the refining of the raw material in the USA and the transports during the production of the raw material.
- In the creation of photochemical ozone the major contributor of hydrocarbons (HC) is once again the refining of the raw material in the USA.
- The main contributors to the impact on eutrophication are the extraction of minerals and refining of the raw material in the supply countries.

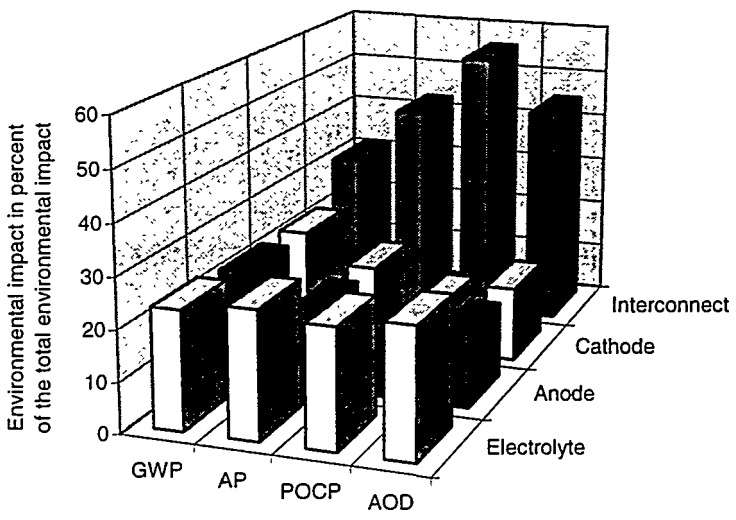


Figure 7-5. Summary of the environmental impact of the production of all sub components. The results are presented in percent of the total impact within the impact category.

7.2 Dominance analysis of the SOFC/GT process

When analysing the whole SOFC/GT process the results are divided into the following phases: preparation and demolition of the SOFC/GT process, production of the SOFC, production of the gas turbine and operation. The headlines used during the dominance analysis of the SOFC are also used here. The fuel chain is, i.e. production of natural gas, included in the operation phase.

7.2.1 Use of resources

As shown in table 6-5 the major material resources used are baddeleyite, bastnasite, copper-ore limestone and pentlandite. Baddeleyite, bastnasite and pentlandite are used during the production of the SOFC. The copper-ore and limestone are used during the production of the gas turbine. The amount of copper-ore is high because it only contains 35 % of copper. This can be compared with the smaller amount of iron-ore, which consists of 40-70 % iron.

Table 6-6 presents the used energy resources. The dominant energy resource is natural gas, which is quite obvious since it is used as feedstock for the SOFC/GT process. The immediate following resource in quantity is coal. Almost all coal used is in the production of the SOFC and it originally comes from the Danish average electricity used during the assemblage of the SOFC.

7.2.2 Discharges of gases contributing to the global warming

The total contribution to the global warming during the whole lifetime of SOFC/GT process is 617 g CO_2-equivalents/g substance. The distribution of the CO_2-equivalents/g between the phase preparation and demolition of the SOFC/GT process, production of the SOFC, production of the gas turbine (GT) and the operation are presented in figure 7-6.

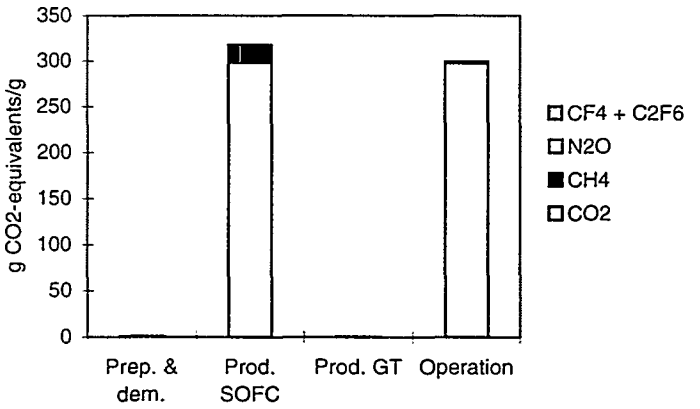


Figure 7-6. Distribution of CO_2-equivalents/g substance (per kWh produced electricity by the SOFC/GT process) between the preparation and demolition, production of the SOFC, production of the gas turbine (GT) and the operation.

Only very small amounts of methane (CH_4), nitrogen dioxide (N_2O) and CF_4+C_2F_6 are present. The main contributor to the global warming is carbon dioxide (CO_2). The production of the SOFC answers for 50 % of the total discharge of carbon dioxide. 89 % of this carbon

dioxide comes from the assemblage of the SOFC, which is a result of using Danish average electricity.

7.2.3 Discharge of gases contributing to acidification

The total contribution to acidification during the lifetime of the SOFC/GT process is 0,021 mole H^+/g substance. The distribution of the impact results of acidification is as presented in figure 7-7.

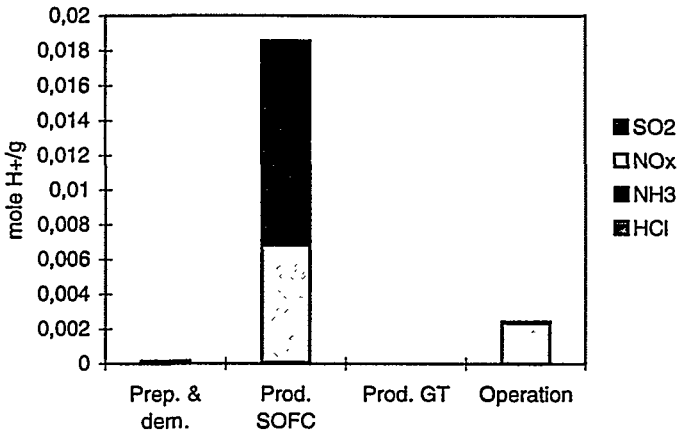


Figure 7-7. Distribution of mole H^+/g substance (per kWh produced electricity by the SOFC/GT process) between the preparation and demolition of the SOFC/GT process, production of the SOFC, production of the gas turbine (GT) and the operation.

The main contributor to acidification during the lifetime of the SOFC/GT process is the production of the SOFC where the discharge gas nitrogen oxide (NO_x) is in a majority. The assemblage of the SOFC answers for 50 % of the nitrogen oxide during the production of the SOFC, which originally comes from the Danish average electricity. Danish average electricity also is the main contributor to the discharges of sulphur dioxide (SO_2), which answers for approximately 74 %. In the operation phase most of the nitrogen oxide comes from the extraction of natural gas (= 90 %).

7.2.4 Discharge of gases contributing to creation of photochemical ozone

The total contribution to the creation of photochemical ozone during the lifetime of the SOFC/GT process is 0,048 g ethene-equivalents/g substance. The distribution of ethylene-equivalents between the phases preparation and demolition of the SOFC/GT, production of the SOFC, production of the gas turbine and the operation are as shown in figure 7-8.

Once again the production of the SOFC is the main contributor to the environmental impact category. The main contributors within the SOFC production to the creation of photochemical ozone are hydrocarbons (HC) and methane (CH_4). The use of Danish average electricity answers for 32 % of the hydrocarbons, which mainly is used for the assemblage of the SOFC. Another main stand-alone contributor is the electricity used at the refining of the raw materials in the USA, which answers for approximately 45 % of the hydrocarbons. The assemblage of the SOFC sub components to a unit cell is done with Danish average electricity and this process contributes to 94 % to the discharge of methane.

The extraction of natural gas answers for all hydrocarbons, carbon monoxide and methane that the operation phase contributes with to the creation of photochemical ozone.

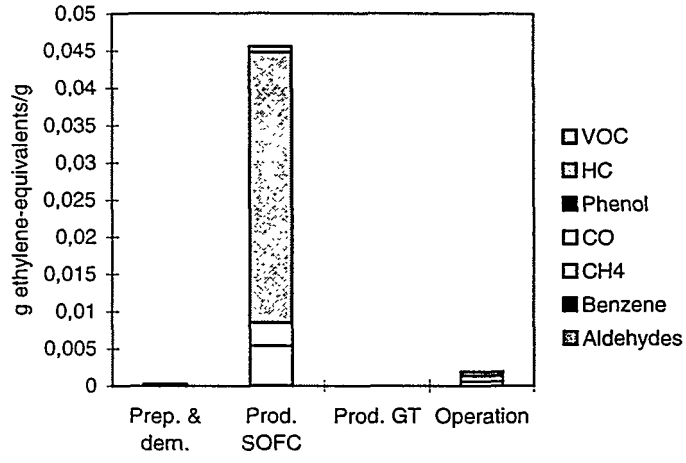


Figure 7-8. Distribution of g ethylene-equivalents/g substance (per kWh produced electricity by the SOFC/GT process) between the preparation and demolition of the SOFC/GT process, production of the SOFC, production of the gas turbine (GT) and the operation

7.2.5 Discharge of gases contributing to eutrophication

The total contribution to the eutrophication during the production of the SOFC-module is 0,0014 g O_2-equivalents/g substance. The distribution of O_2-equivalents between the phases preparation and demolition of the SOFC/GT, production of the SOFC, production of the gas turbine and the operation are as shown in figure 7-9.

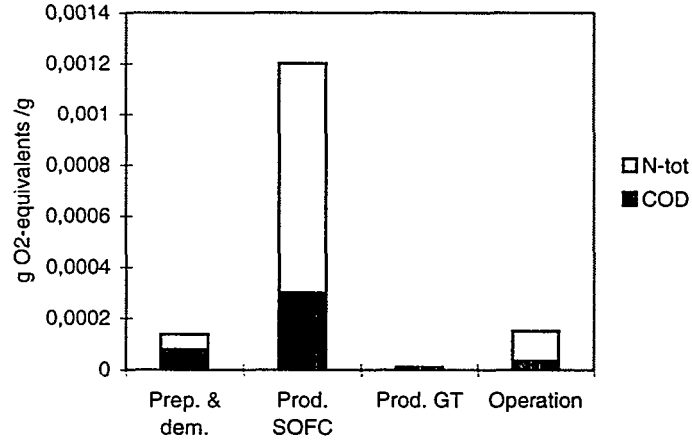


Figure 7-9. Distribution of g O_2-equivalents/g substance (per kWh produced electricity by the SOFC/GT process) between the preparation and demolition of the SOFC/GT process, production of the SOFC, production of the gas turbine (GT) and the operation.

The main substance that contributes to eutrophication is N-tot (includes nitrogen, NH_4^+ and NO_3^-), which answers for a total 76 %. During the production of the SOFC the assemblage of the sub components answers for 58 % of the N-tot. Also most of the chemical oxygen demand (COD) emissions comes from the assemblage, i.e. 58 %.

The extraction of natural gas answers for all N-tot and COD that the operation phase contributes with to the eutrophication.

7.2.6 Toxic waste

As presented in table 6-12 several toxic substances are discharged during the SOFC/GT lifetime. The main emissions are: dust, hydrocarbons and volatile organic compounds (VOC). All those toxic substances are mainly descended from the production of the SOFC.

7.2.7 Summary of all phases during the SOFC/GT process' life cycle

The production of the SOFC is the most contributing phase in most of the impact categories. Only in the impact category global warming the operation phase is dominant. In figure 7-10 the impact categories; global warming (GWP), acidification (AP), creation of photochemical ozone (POCP) and eutrofication (AOD) are summarised for the phases.

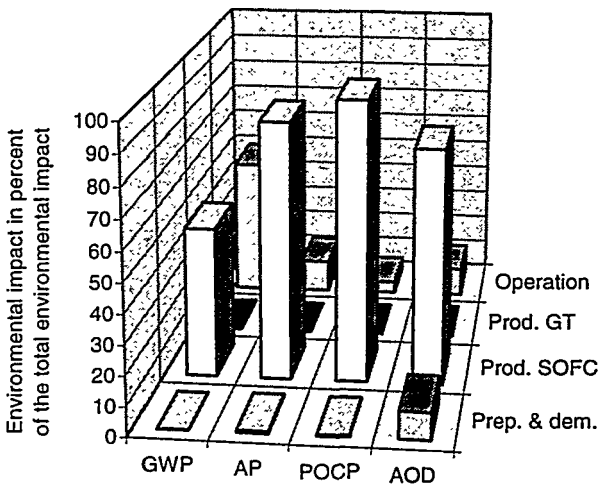


Figure 7-10. Summary of the environmental impact of all phases of the SOFC/GT process. The results are presented in percent of the total impact within the impact category.

Summary of figure 7-10:

- The production of the SOFC and the operation of the SOFC/GT process contribute with approximately 50 % each to the global warming. The carbon dioxide contributes most to the global warming with 97 %.
- The production of the SOFC is the dominant phase in all impact categories. The process, which is the main contributor in these categories is the assemblage of the SOFC in Denmark. When using the Danish average electricity, where the energy resource coal answers for 91 % of the mix, sulphur dioxide, different carbon molecules and nitrogen oxides are easily produced and dumped to the air through the exhaust gas.

7.3 Sensitivity analysis

One assumption that was made during the impact assessment was that the lifetime of the SOFC/GT process is 10 000 hours. If the goal is reached, i.e. 40 000 operation hours, all

impact assessment results during the construction and demolition phase are cut to a fourth. The impact that this has on the total process is shown in figure 7-11.

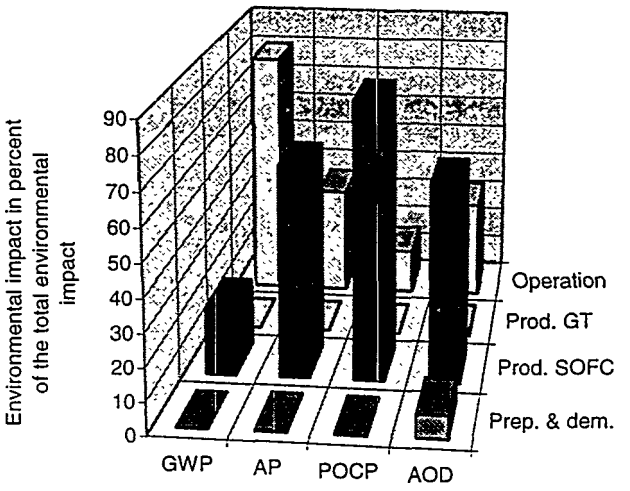


Figure 7-11.The environmental impact of all phases of the SOFC/GT process with the operating time 40 000 hours. The results are presented in percent of the total impact within the impact category

Although there has been a extensive decrease in all categories only the contribution to the global warming during the production of the SOFC falls below the results of the operation phase.

7.3.1 Exchange of electricity

In chapter 6 the pyrolysis processes and the final assemblage of the SOFC unit cell were carried out in Denmark, which implies use of Danish average electricity. This case will further be known as the reference case. The obtained changes that the cases below contribute to in the impact categories are shown in figure 7-12 to figure 7-15.

7.3.1.1 Case 1 – exchange of Danish average electricity to UCPTE electricity

The use of Danish average electricity is so dominant in all impact categories that an exchange of this to UCPTE electricity is made (case 1). In the UCPTE electricity coal only contributes with 0,256 MJ per MJ (Danish average electricity 0,91 MJ/MJ

7.3.1.2 Case 2 – halved use of UCPTE electricity

In case 2 the used energy is halved during the processes carried out in Denmark (assemblage and pyrolysis) but the electricity is still UCPTE.

7.3.1.3 Case 3 – minimum energy use during the production of SOFC raw materials

In case 3 the minimum energy is used during the production of the raw materials (see appendix B) otherwise the parameters are as in case 2.

7.3.1.4 Case 4 – exchange of UCPTE electricity to Swedish average electricity

The only change made in case 4 compared to case 3 is that the UCPTE electricity is changed to Swedish average electricity.

7.3.2 Results and interpretation

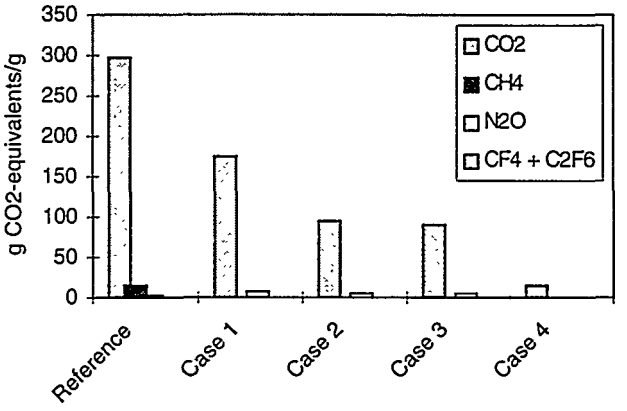


Figure 7-12. Comparison between the reference case and the contributions of the different cases to global warming in a 100-year perspective. The unit here is g CO2-equivalents/g substance.

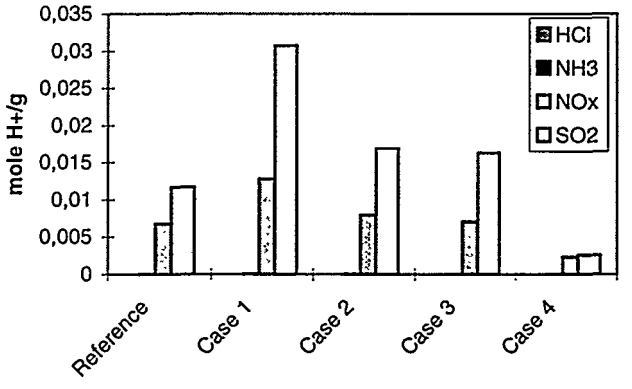


Figure 7-13. Comparison between the reference case and the contributions of the different cases to acidification. The unit here is mole H^+/g substance.

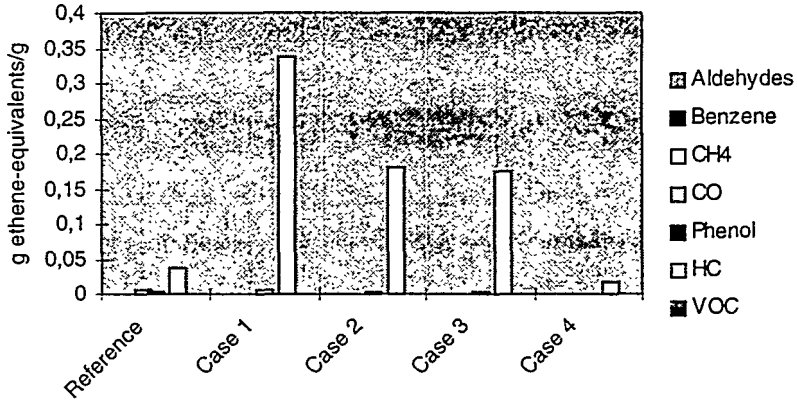


Figure 7-14. Comparison between the different cases contribution to the creation of photochemical ozone. The unit here is g ethene-equivalents/g substance.

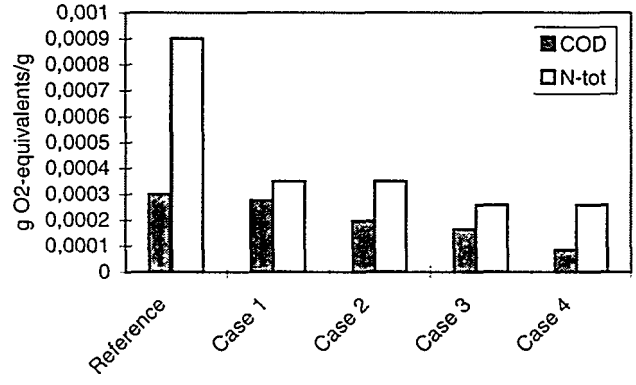


Figure 7-15. Comparison between the reference case and the different contributing to eutrophication.. The unit here is g O_2-equivalents/g substance.

When changing the Danish average electricity to UCPTE electricity (case 1) a significant change in the contribution of g CO_2-equivalents/g substance is obvious, see figure 7-12. In the reference case there was a contribution of 298 g CO_2 per produced kWh electricity. In the case 2 the amount of CO_2 has decreased to 175 g. A decrease of 40 % of the amount CO_2 is obtained. Also the contribution to eutrofication has decreased by a third. A disadvantage with the UCPTE electricity however is the production of hydrocarbons (HC), which contributes to the creation of photochemical ozone. The creation of photochemical ozone has increased almost 8 times in case 2 compared to the reference case. Also the contributions to acidification have increased.

When lowering the use of energy by half during the processes carried out in Denmark and still using UCPTE electricity (case 2), even lower contributions to the global warming are obtained. However, the contributions to the creation of photochemical ozone and acidification are still higher than in the reference case.

The estimated use of energy during the production of the raw materials, presented in appendix B, was in the study assumed to be the maximum value. When using the minimum value (case 3) only small decreases were obtained in all categories.

The last change from UCPTE electricity to Swedish average electricity (case 4) was made to see how a special constellation of energy carriers (approximately 50 % water power and 50 % nuclear power) could change the environmental impact. All categories were lowered significantly, but the big disadvantage when using the Swedish average electricity is the increase in use of natural uranium. The use of natural uranium in the reference case and when using UCPTE electricity are 0,00015 respectively 0,0014 per produced kWh electricity by the SOFC/GT process. When using Swedish average electricity the use of natural uranium is 0,0022.

8 Conclusions and discussion

The SOFC and the SOFC/GT process are known for their low emissions, especially the low NOx levels during operation. In this study the whole life cycle of the SOFC/GT process has been investigated to determine the total environmental impact of the process. Since the SOFC and the SOFC/GT process until today only have been tested under laboratory circumstances and sizes, only data available in literature have been used in this study. Several estimations and assumptions have been made in the assessment, so this study must be considered a first attempt to perform a life cycle assessment of the SOFC/GT process.

Since the first SOFC/GT process is to be tested by Siemens Westinghouse in the summer of 1999 no measured data has been available. The theoretical process studied here has an efficiency of 73 % and the reformed fuel has the composition described in chapter 2.1. The operation of this process has a complete combustion of all methane, carbon monoxide and hydrogen gas and when comparing the discharge of carbon dioxide with other processes at Sydkraft, the results must be considered promising. As shown in figure 8-1 it is very competitive with the utility size combined cycle regarding the discharge of carbon dioxide during the operation. The energy used during the construction phase in figure 8-1 is UCPTE electricity and half of the energy used in Denmark is assumed to be enough. The discharge of carbon dioxide during the construction phase however is very high compared to the other processes. In the processes "Gas (P+H)" and "Bio (P+H)" the amount of carbon dioxide are split between the power and the heat production, which leads to the low value. A further development within the production of the SOFC, which stands for 96 % of the amount of CO2, must be carried out to diminish the energy use.

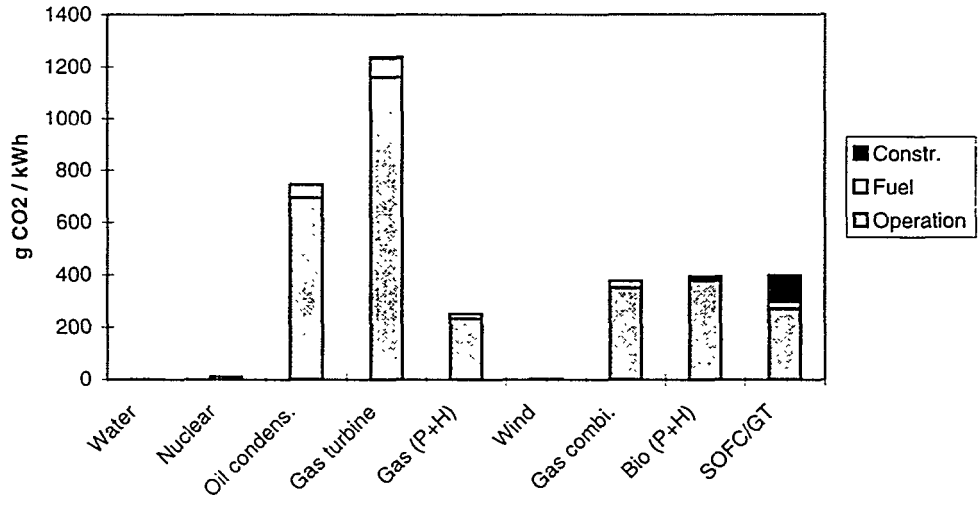


Figure 8-1. Comparison between processes regarding their discharge of carbon dioxide per kWh produced electricity [38]. (P+H = power and heat production)

The major assumptions that have been made during the LCA are the weight and composition of raw materials used during the production of the gas turbine. Nor has any use of energy during the production of the gas turbine been estimated. It is possible that these assumptions can have made the environmental impact of the gas turbine invisible, but according to the impact assessment it seems like the energy use during the production has to be very high to

reach any noticeable impact at all. The assumed lifetime of the SOFC/GT process, 10 000 hours at full load, has contributed with high emissions, especially discharges of carbon dioxide per kWh produced electricity. When investigating the possibility of a lifetime of 40 000 hours, which is the goal for many developers, the rate of emissions is lowered by a fourth during the construction and demolition phase. This is of course desirable in the future if the discharges during the production phase of the SOFC is not to exceed the contribution of emissions during the operation phase.

All data on mass flows and use of energy during the processes are collected from literature which is based on laboratory results. This has probably lead to a to high amount of both the requirement of material and use of energy. If the processes used are industrialised and made more efficient a halving of the energy and less wastage of materials seem possible. The development of thin film SOFC also points at the reduction of material use which consequently leads to less energy requirement during the production.

During the whole life cycle of the SOFC/GT process, excluding the operation phase, the production of the SOFC is the main contributor within all environmental impact categories. Especially the final processes in the production, the pyrolysis process and the sintering of the sub components, require much energy. Where the final assemblage of the sub components of the SOFC takes place is maybe the most determining factor when calculating which impact category that is effected. If the final assemblage is carried out in Denmark the discharge of carbon dioxide is for instance much higher than it is in Sweden. This because of the mix of energy carriers in the electricity. Depending on what these energy carriers are, different environmental impact categories are affected. A diminishing of all gases contributing to the different impact categories must be seen as the objective, but which environmental impact that is the worst and is to be considered the most hazardous to the environment can be discussed. All mixes of electricity have their contribution more or less to the impact categories.

When considering the recycling potential of the materials that the SOFC/GT process contains the components are split up in two groups. The conventional components, which are generally all but the SOFC, can be recycled fairly satisfactory. For instance the copper and steel used in the gas turbine are recycled and can be recycled again when the process is considered as waste. Also the catalysts used in the pre-reformer and the desulphurization systems are recycled. No ideas of what to do with the SOFC after its lifetime are known. Today the few fuel cells tested in laboratories are placed on landfills and since they contain vanadine, nickel, lanthanum etc. that are toxic to humans this could be a problem when the SOFC is further commercialised.

That the SOFC/GT process could work as a future power supplier is clear, especially as a local distributor. There are however problems that must be solved; the lifetime of the SOFC must be extended, some kind of recycling is necessary and the energy use during the production of the SOFC must be decreased. However, all development of the SOFC indicates a major decrease in the requirement of material which consequently leads to a decrease in energy use. A future LCA of this process should be performed on measured data and when more efficient production methods have been developed. This study must be considered as a first attempt of performing an LCA of the SOFC/GT process.

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Appendix A - Calculation of mass quantity, required during the production of the SOFC raw materials [9]

The calculations done in this appendix is following the calculations done in the report by Petra Zapp [9]. The calculated data is for one unit cell.

After speaking with Carsten Bagger [6] at the National Laboratory in Risoe the following quantity o masses were given:

Electrolyte:	m(tot) = 16 g Yttria-stabilized zirconia (YSZ)
Anode:	m(tot) = 1,17 g (0,65 g NiO + 0,52 g YSZ)
Cathode:	m(tot) = 1,43 g (1,3 g LSM* + 0,13 g YSZ)
Interconnect:	m(tot) = 123,75 g LSCV**

*) LSM = La(0,85)Sr(0,15)Mn(1,1)O3

**) LSCV = La(0,8)Sr(0,2)Cr(0,97)V(0,03)O3

All the mass quantities above are the total sum of the needed material during the production of the components, including the wastage.

Electrolyte

The raw material used in the electrolyte is yttria stabilized zirconia (YSZ). As given in the figures above the quantity needed is 16 g, for unit one cell. The distribution is 8 mole % Y2O3 in ZrO2.

Substance	Mole fraction	Mass fraction	Mass [g]	Nbr
Y2O3	0,08	0,1375	2,20	1
ZrO2	0,92	0,8625	13,80	2

Nbr 1: Zirconia

The substance ZrO2 is a 4 mass % part in treated "mineral concentrate". This leads to a total need of 345 g of mineral concentrate. Since the mineral baddeleyite, generally contains 10 mass % of treated mineral concentrate, the total mass quantity of ***baddeleyite is 3,45 kg.***

Nr 2: Yttrium

In the mineral Xenotime contains 60,8 mass% of Y2O3, so the total need of ***xenotime is 3,62 g***.

Summary of electrolyte

To produce the required material, which the anode contains, the following minerals and chemicals are to be used:

Substance output	Mass [g]
Y2O3	13,8
ZrO2	2,2
Tailing and slag	3437,62

Substance input	Mass [g]
Baddeleyite	3450
Xenotime	3,62

Anode

The distribution in mass quantity in the anode is the following:

Substance	Mass [g]
NiO	0,65
ZrO2	0,4485
Y2O3	0,0715

For the zirconia and the yttria, the same calculations done for the electrolyte is carried out here, which lead to the following quantities:
Baddeleyite: 112,13 g
Xenotime: 0,1176 g

Since the substance NiO doesn't exist naturally in any mineral a chemical reaction is needed:

$$2Ni3S2 + 7O2 \rightarrow 6NiO + 4SO2$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
Ni3S2	240,25	0,697	
O2	31,999	0,325	
NiO	74,709		0,65
SO2	64,059		0,372

Ni3S2 is extracted to 57,6 % from a converting mass, containing approximately 70 % of nickel. The mass quantity needed of this converting mass is 1,729 g

The converting mass is in an earlier stage enriched from 30 to 50% => m = 2,881 g

This nickel content in the flotation mass above (30%) has been improved from 11,6 % to 30 % => m = 7,451 g. This flotation mass is to be found to 8 % i the mineral pentlantite.
The total mass of ***pentlantite = 93,14 g***

Summary of anode

To fabricate the required material, which the anode contains, the following minerals and chemicals are used:

Substance output	Mass [g]
NiO	0,65
ZrO2	0,4485
Y2O3	0,0715
SO2	0,372
Tailing and slag	204,1706

Substance input	Mass [g]
Pentlandite	93,14
Baddeleyite	112,13
Xenotime	0,1176
O2	0,325

Cathode

For the fuel cell, fabricated in Jülich, 0,55 g La(0,84)Sr(0,16)MnO3, was needed.
For the fuel cell fabricated in Risoe, the material required has the following composition:

1,3 g La(0,85)Sr(0,15)Mn(1,1)O3 and,
0,13 g YSZ

Starting with the YSZ, leads to the requirement of the mineral ***Xenotime = 0,0294 g***
Baddeleyite = 28,03 g

To produce 0,55 g La(0,84)Sr(0,16)MnO3 in Jülich, the mass quantities of the required nitrates, can be seen below:

Substance	Mass [g]
La(NO3)3*6H2O	0,85
Sr(NO3)2	0,08
Mn(NO3)2*4H2O	0,59

Since the cathode material is very similar for Risoe and Jülich, an assumption is made, that the mass quantities for the nitrates are directly proprtional to the mass quantity of the cathode material. The following mass quntities for the nitrates required in Risoe are shown below:

Substance	Mass [g]	Nbr
La(NO3)3*6H2O	2,010	1
Sr(NO3)2	0,189	2
Mn(NO3)2*4H2O	1,395	3

Nbr 1: Lanthanum
When producing the lanthanum-nitrate, the raw material is La2O3, and the following reaction takes place:

La2O3 + 6HNO3 + 9H2O --> 2La(NO3)3*6H2O

Substance	Molar mass [g/mol]	Input [g]	Output [g]
La2O3	325,809	0,756	
HNO3	63,013	0,878	
H2O	18,015	0,376	
La(NO3)3*6H2O	433,011		2,010

Since the mineral Bastnasite contains 33,2 mass % of La2O3, the required mass to produce the cathode is ***2,28 g Bastnasite***.

Nbr 2: Strontium

At the production of Sr(NO3)3 a couple of reactions must take place:

$$SrCO3 + 2HNO3 \rightarrow Sr(NO3)2 + H2CO3$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
SrCO3	147,629	0,132	
HNO3	63,013	0,112	
Sr(NO3)2	211,630		0,189
H2CO3	62,025		0,055

Since SrCO3 is not to be found in any mineral :

$$SrS + CO2 + H2O \rightarrow SrCO3 + H2S$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
SrS	119,680	0,107	
CO2	44,010	0,039	
H2O	18,015	0,016	
SrCO3	147,629		0,132
H2S	34,076		0,030

Before the reaction above, the strontium sulphide has been extracted from 75 to 100% => m(SrS) = 0,143 g

Yet another reaction must take place:

$$SrSO4 + 2C \rightarrow SrS + 2CO2$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
SrSO4	183,678	0,219	
C	12,011	0,029	
SrS	119,68		0,143
CO2	44,010		0,105

The mineral Celestite contains 95 mass % of SrSO4, so the need of ***Celestite is 0,23 g***

Nbr 3: Manganese

To fabricate Mn(NO3)2*4H2O, four reactions must take place:

$$MnO + 2HNO3 + 3H2O \rightarrow Mn(NO3)2*4H2O$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
MnO	70,937	0,395	
HNO3	63,013	0,700	
H2O	18,015	0,300	
Mn(NO3)2*4H2O	251,008		1,395

To fabricate MnO the following reaction is required:

$$Mn3O4 + H2 \rightarrow 3MnO + H2O$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
Mn3O4	228,812	0,424	
H2	2,016	0,004	
MnO	70,937		0,395
H2O	18,015		0,033

The next reaktion that is needed is:

$$3Mn2O3 + H2 \rightarrow 2Mn3O4 + H2O$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
Mn2O3	157,874	0,439	
H2	2,016	0,002	
Mn3O4	228,812		0,424
H2O	18,015		0,017

Since the mineral Pyrolusite contains the substance MnO2 the reaction required is:

$$2MnO2 + H2 \rightarrow Mn2O3 + H2O$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
MnO2	86,937	0,483	
H2	2,016	0,006	
Mn2O3	157,974		0,439
H2O	18,015		0,050

In Pyrolusite, manganese oxide stands for approximately 70 mass %. This means that to produce 1,395 g Mn(NO3)2*4H2O, there is a need of ***0,69 g Pyrolusite.***

Summary of Cathode:
To fabricate the three required nitrates, which the cathode contains, the following minerals and chemicals are used:

Substance output	Mass [g]
La(NO3)2*6H2O	2,010
Sr(NO3)2	0,189
Mn(NO3)2*4H2O	1,395
H2CO3	0,055
CO2	0,105
H2O	0,100
H2S	0,030
Tailing and slag	29,821

Substance input	massa [g]
Baddeleyite	28,030
Bastnasite	2,280
Celestite	0,230
Pyrolusite	0,690
Xenotime	0,0294
HNO3	1,690
H2O	0,676
CO2	0,039
C	0,029
H2	0,012

Interconnect
The same calculatons done for the cathode are here done for the interconnect. Another assumption done for the interconnect is that the vanadine in the Risoe interconnect is negligent in the mass quantity calculations. The total mass of the interconnect from Risoe = 123,75g.

Substance	Mass, Risoe [g]	Mass, Jülich [g]	Nbr
La(NO3)2*6H2O	187,481	37,875	1
Sr(NO3)2	22,894	4,625	2
Cr(NO3)3*9H2O	216,563	43,750	3

Nr 1: Lanthanum
The same procedure is done for the interconnect as the cathode.

$$La2O3 + 6HNO3 + 9H2O \rightarrow 2La(NO3)3*6H2O$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
La2O3	325,809	70,532	
HNO3	63,013	81,849	
H2O	18,015	35,1	
La(NO3)3*6H2O	433,01		187,481

Since the mineral Bastnasite contains 33,2 mass % of La2O3, the required mass to fabricate the cathode is ***212,45 g Bastnasite.***

Nr 2: Strontium

Here also the calculations is the same as for the cathode, exept that the mass for the Sr(NO3)2 is 22,894 g.

$$SrCO3 + 2HNO3 \rightarrow Sr(NO3)2 + H2CO3$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
SrCO3	147,629	15,97	
HNO3	63,013	13,632	
Sr(NO3)2	211,63		22,894
H2CO3	62,025		6,71

$$SrS + CO2 + H2O \rightarrow SrCO3 + H2S$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
SrS	119,68	12,947	
CO2	44,01	4,76	
H2O	18,015	1,949	
SrCO3	147,629		15,97
H2S	34,076		3,686

Extraction of SrS => m(SrS) = 17,263 g

$$SrSO4 + 2C \rightarrow SrS + 2CO2$$

Substance	Molmassa [g/mol]	Input [g]	Output [g]
SrSO4	183,678	26,494	
C	12,011	3,465	
SrS	119,68		17,263
CO2	44,01		12,696

Since Celestite contains 95 mass % of SrSO4 => mass qantity of ***celestite = 27,89 g***

Nbr 3: Chromium

Chromium-nitrate is fabricated in four reactions:

$$Cr2O3 + 6HNO3 + 15H2O \rightarrow 2Cr(NO3)3*9H2O$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
Cr2O3	151,99	41,129	
HNO3	63,013	102,31	
H2O	18,015	73,124	
Cr(NO3)3*9H2O	400,146		216,563

To create Cr2O3 the following reaction is used:

$$Na2Cr2O7 + S \rightarrow Cr2O3 + Na2SO4$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
Na2Cr2O7	261,967	70,889	
S	32,06	8,676	
Cr2O3	151,99		41,129
Na2SO4	142,037		38,436

The next reaction required:

$$2Na2CrO4 + H2SO4 \rightarrow Na2Cr2O7 + Na2SO4 + H2O$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
Na2CrO4	161,973	87,661	
H2SO4	98,073	26,539	
Na2Cr2O7	261,967		70,889
Na2SO4	142,037		38,436
H2O	18,015		4,875

The mineral chromite, FeCr2O4 can through a reaction create Na2CrO4:

$$4FeOCr2O3 + 7O2 + 8Na2CO3 \rightarrow 8Na2CrO4 + 8CO2 + 2Fe2O3$$

Substance	Molar mass [g/mol]	Input [g]	Output [g]
FeCr2O4	223,837	60,571	
O2	31,999	15,154	
Na2CO3	105,989	57,362	
Na2CrO4	161,973		87,661
CO2	44,01		23,819
Fe2O3	159,692		21,607

From the mineral chromite, 80% of the Na2CrO4, can be extracted. This means that the mass quantity of ***chromite*** = ***75,71 g*** .

Summary of interconnect

To fabricate the three required nitrates, which the interconnect contains, the following minerals and chemicals are used:

Substance output	Mass [g]
La(NO3)2*6H2O	187,481
Sr(NO3)2	22,894
Cr(NO3)3*9H2O	216,563
H2CO3	6,71
H2S	3,686
CO2	36,515
Na2SO4	76,872
H2O	4,875
Fe2O3	21,607
Tailing and slag	162,767

Substance input	Mass [g]
Chromite	75,71
Celestite	27,89
Bastnasite	212,45
O2	15,154
Na2CO3	57,362
H2SO4	26,539
S	8,676
HNO3	197,791
H2O	110,173
C	3,465
CO2	4,76

Appendix B - Calculations of energy, required during the production of the SOFC raw materials

In the Ph.D. thesis by Zapp P. [9], the energy required to convert the minerals to raw materials is shown in table B-1.

Table B-1. Required energy to convert minerals to raw materials [9].

Mineral	Raw material	Energy [MJ/kg]
Bastnasite	La2O3	75
Celestite	SrCO3	?
Pyrolusite	MnO	11,62
Baddeleyite	ZrO2	?
Xenotime	Y2O3	75
Pentlandite	NiO	232,23
Chromite	Cr2O3	110,379

Since no values of energy for the two minerals celestite and baddeleyite have been found, two different values will be used during the calculations. One minimum value = 11,62 MJ/kg, and a maximum value = 232,23 MJ/kg. The calculations will be carried out for each SOFC component.

The distribution of energy carrier, when mining and processing the raw materials, is assumed to be 50 % diesel and 50 % UCPTE* electricity. This due to the need of devices when mining that are considered to use diesel as fuel. The electricity is required when refining the raw material to nitrates.

* UCPTE electricity = mix of average electricities in certain countries, see appendix D.

Electrolyte

As mentioned in appendix A the two minerals, that is required to produce yttria-srabilized zirconia (YSZ), is baddeleyite and xenotime.

Baddeleyite

For each unit cell (uc), 3,45 kg of baddeleyite is required to produce 13,18 g ZrO2. As shown in table B-1, no energy value has been found for baddeleyite. Therefore two different values will be used to decide the contribution to the environmental impact.

E(min) = 11,62 MJ/kg

E(min,diesel) = 0,5*11,62 = 5,81 MJ/kg
E(min,electricity) = 0,5*11,62 = 5,81 MJ/kg

Total: E(min,diesel,E) = 5,81*0,0138 = 0,080 MJ/uc
E(min,electricity,E) = 5,81*0,0138 = 0,080 MJ/uc

E(max) = 232,23 MJ/kg E(max,diesel) = E(max,electricity) = 116,12 MJ/kg

Total: E(av,diesel,E) = E(av,electricity,E) = 1,602 MJ/uc

Xenotime

Each unit cell requires 3,62 g xenotime for the production of 2,20 g Y2O3. The energy required to process the material is 75 MJ/kg Y2O3 according to table B-1.

E(diesel) = E(electricity) = 37,5 MJ/kg

Total: E(diesel) = E(electricity) = 37,5*0,0022 = 0,0825MJ/uc

Summary - electrolyte

The total energy required when mining and processing the raw materials, required for the electrolyte is summarized in table B-2:

Table B-2, Total energy need to process materials, required for the electrolyte

	Minimum Energy MJ/uc	Maximum energy MJ/uc
Diesel (e)	0,163	1,685
UCPTE Electricity	0,163	1,685

Anode

In the anode three minerals are required, pentlandite, baddeleyite and xenotime. The same calculations will be done for the anode as for the electrolyte.

Pentlandite

Every unit cell requires 93,14 g pentlandite as raw material to fabricate 0,65 g NiO. According to table B-1, the used energy is 232,23 MJ/kg NiO.

E(diesel) = E(electricity) = 116,12 MJ/kg

Total: E(diesel) = E(electricity) = 37,5*0,00065 = 0,024MJ/uc

Baddeleyite

To fabricate 0,4485 g ZrO2, there is a requirement of 112,13 g baddeleyite. The energy value to do this is unknown, so the minimum and the maximum values are used.

E(min) = 11,62 MJ/kg E(min,diesel) = E(min,electricity) = 5,81 MJ/kg

Total: E(min,diesel,E) = 5,81*0,0004485 = 0,003 MJ/uc

E(max) = 232,23 MJ/kg E(max,diesel) = E(max,electricity) = 116,12 MJ/kg

Total: E(av,diesel,E) = E(av,electricity,E) = 0,052 MJ/uc

Xenotime
Each unit cell requires 0,1176 g xenotime for the production of 0,0715 g Y2O3. The energy needed to process the material is 75 MJ/kg Y2O3 according to table B-1.

E(diesel) = E(electricity) = 37,5 MJ/kg
Total: E(diesel) = E(electricity) = 37,5*0,0000715 = 0,003 MJ/uc

Summary - anode
The total energy need when mining and processing the raw materials, required for the anode is summarized in table B-3:

Table B-3, Total energy need to process materials, required for the anode.

	Minimum Energy MJ/uc	**Maximum energy MJ/uc**
Diesel (e)	0,03	0,079
UCPTE Electricity	0,03	0,079

Cathode

The minerals, required to form the cathode is baddeleyite, xenotime, bastnasite, celestite and pyrolusite.

Baddeleyite
To produce 0,112 g ZrO2, there is a requirement of 28,03 g baddeleyite. The energy value to do this is unknown.

E(min) = 11,62 MJ/kg
E(min,diesel) = E(min,electricity) = 5,81 MJ/kg
Total: E(min,diesel,E) = 5,81*0,000112 = 0,0007 MJ/uc

E(max) = 232,23 MJ/kg
E(max,diesel) = E(max,electricity) = 116,12 MJ/kg
Total: E(av,diesel,E) = E(av,electricity,E) = 0,0130 MJ/uc

Xenotime
Each unit cell requires 0,0294 g xenotime for the production of 0,0179 g Y2O3. The energy need to process the material is 75 MJ/kg Y2O3 according to table B-1.

E(diesel) = E(electricity) = 37,5 MJ/kg
Total: E(diesel) = E(electricity) = 37,5*0,0000179 = 0,0007 MJ/uc

Bastnasite
In the cathode 2,28 g batsnasite is needed to fabricate 0,756 g La2O3. This process requires 75 MJ/kg La2O3.

E(diesel) = E(electricity) = 37,5 MJ/kg

Total: E(diesel) = E(electricity) = 37,5*0,000756 = 0,0284 MJ/uc

Celestite

The raw material enriched out of 0,23 g celestite, is 0,132 g SrCO3. The energy used to accomplish this process is unknown.

E(min) = 11,62 MJ/kg

E(min,diesel) = E(min,electricity) = 5,81 MJ/kg

Total: E(min,diesel,E) = 5,81*0,000132 = 0,0008 MJ/uc

E(max) = 232,23 MJ/kg

E(max,diesel) = E(max,electricity) = 116,12 MJ/kg

Total: E(av,diesel,E) = E(av,electricity,E) = 0,0153 MJ/uc

Pyrolusite

In the cathode, there is a need of 0,69 g pyrolusite to fabricate 0,395 g MnO. Energy required for this is 11,62 MJ/kg MnO.

E(diesel) = E(electricity) = 5,81 MJ/kg

Total: E(diesel) = E(electricity) = 5,81*0,000395 = 0,0023 MJ/uc

Summary - cathode

The total energy need when mining and processing the raw materials, required for the cathode is summarized in table B-4:

Table B-4, Total energy need to process materials, required for the cathode.

	Minimum Energy MJ/uc	**Maximum energy MJ/uc**
Diesel (e)	0,033	0,06
UCPTE Electricity	0,033	0,06

Interconnect

The interconnect, who is the most material demanding component, uses the minerals bastnasite, celestite and chromite.

Bastnasite

The interconnect needs 70,532 g La2O3, which is processed out of 212,45 g bastnasite. This process requires 75 MJ/kg La2O3.

E(diesel) = E(electricity) = 37,5 MJ/kg

Total: E(diesel) = E(electricity) = 37,5*0,0705 = 2,644 MJ/uc

Celestite

The raw material enriched out of 27,89 g celestite, is 15,97 g SrCO3. The energy used to accomplish this process is unknown.

Appendix C - Presentation of energy carriers and transport data

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Energy carriers

UCPTE[1] electricity

Table D-1. Emissions and utilisation of resources at the production of 1 MJ UCPTE electricity [23].

Substance	Amount	Unit	Impactmedia	Category
Aldehydes	0,0007	g	Air	Emission
BOD	0	g	Water	Emission
CO	0,097	g	Air	Emission
CO2	112,7	g	Air	Emission
COD	0,0001	g	Water	Emission
Dissolved solids	0,0879	g	Water	Emission
F- (aq)	0,0004	g	Water	Emission
Fe2+ (aq)	0	g	Water	Emission
Flourides	0	g	Water	Emission
HC	0,587	g	Air	Emission
N2O	0,0196	g	Air	Emission
Na+ (aq)	0,0001	g	Water	Emission
NH3	0,0001	g	Air	Emission
NH3 (aq)	0,0002	g	Water	Emission
Nitrates	0,0001	g	Water	Emission
NOx	0,343	g	Air	Emission
Oil (aq)	0,0013	g	Water	Emission
Radioactive waste h [cm3]	0,0009	ml	Ground	Emission
Radioactive waste l [cm3]	0,0101	ml	Ground	Emission
Radioactive waste m [cm3]	0,0101	ml	Ground	Emission
SO2	0,695	g	Air	Emission
SO42- (aq)	0,0001	g	Water	Emission
Dust	0,0546	g	Air	Emission
Suspended solids	0	g	Water	Emission
Coal (r)	28,71	g	Resource	Resource
Natural gas (r)	5,84	g	Resource	Resource
Natural uranium (r)	0,0021	g	Resource	Resource
Crude oil (r)	7,4	g	Resource	Resource
Water power [MJel] (r)	0,202	MJ	Resource	Resource

UCPTE mix of electricity (per MJ electricity)[2]:

- Water power: 0,202 MJ
- Nuclear power: 0,369 MJ
- Coal[3]: 0,256 MJ
- Oil[3]: 0,093 MJ
- Natural gas: 0,080 MJ

[1] Electricity mix from the countries; Austria, Belgium, France, Germany, Greece, Holland, Italy, Luxembourg, Portugal, Spain, Switzerland and the former Yugoslavia.
[2] Ecobalance of packaging materials state of 1990, Swiss Federal Office of Environment Forests and Landscape (FOEFL), Berne (1991).
[3] Efficiency: 0,33.

Danish average electricity

Table D-2. Emissions and utilisation of resources at the production of 1 MJ Danish average electricity [23].

Substance	Amount	Unit	Impactmedia	Category
CH4	0,595	g	Air	Emission
CO	0,0489	g	Air	Emission
CO2	219	g	Air	Emission
Phenol	0,000000643	g	Air	Emission
HC	0,0218	g	Air	Emission
N2O	0,0064	g	Air	Emission
NH3	0,00000404	g	Air	Emission
NOx	0,131	g	Air	Emission
Rn-222 [Bq]	0,89	Bq	Air	Emission
SO2	0,216	g	Air	Emission
Dust	0,106	g	Air	Emission
Ashes	0,0261	g	Ground	Waste
COD	0,000143	g	Air	Emission
Dissolved solids	0,00196	g	Air	Emission
NH3 [aq]	0,00000384	g	Air	Emission
Oil (aq)	0,0000775	g	Air	Emission
Tot-N (aq)	0,0000215	g	Air	Emission
Biomass (r)	1,17	g	Resource	Resource
Coal (r)	74,6	g	Resource	Resource
Natural gas (r)	1,9	g	Resource	Resource
Crude oil (r)	3,24	g	Resource	Resource
Water power [MJel] (r)	0,00055	MJ	Resource	Resource

Danish mix of electricity (per MJ electricity)[4,5]

- Water power: 0,0006 MJ
- Coal: 0,9100 MJ
- Oil: 0,0370 MJ
- Natural gas: 0,0240 MJ
- Other solid fuels: 0,0070 MJ
- Sun and wind: 0,0212 MJ

[4] Energifakta, Svensk Energiförsörjning AB, Stockholm, Sverige, 1991.
[5] Efficiency: 33%

Swedish average electricity

Table D-3. Emissions and utilisation of resources at the production of 1 MJ Swedish average[6] electricity [23].

Substance	Amount	Unit	Impactmedia	Category
Ashes	0,07	g	Air	Emission
CH4	0,0001	g	Air	Emission
CO	0,0021	g	Air	Emission
CO2	3,8	g	Air	Emission
HC	0,0011	g	Air	Emission
N2O	0,0004	g	Air	Emission
NOx	0,0092	g	Air	Emission
Radioactive waste, high activity	0,0014	ml	Ground	Emission
Radioactive waste, low activity	0,0153	ml	Ground	Emission
Radioactive waste, average act.	0,0154	ml	Ground	Emission
SO2	0,0067	g	Air	Emission
Dust	0,0001	g	Air	Emission
Biomass (r)	2,35	g	Resource	Resource
Coal (r)	1,48	g	Resource	Resource
Natural gas (r)	0,297	g	Resource	Resource
Natural uranium (r)	0,0032	g	Resource	Resource
Crude oil (r)	1,66	g	Resource	Resource
Peat (r)	0,21	g	Resource	Resource
Water power [MJel] (r)	0,47	MJ	Resource	Resource

Swedish mix of electricity (per MJ electricity)[6]

- Water power: 0,47 MJ
- Nuclear power: 0,56 MJ
- Coal[7]: 0,013 MJ
- Oil[7]: 0,021 MJ
- Natural gas[7]: 0,0031 MJ
- Peat[7]: 0,0015 MJ
- Renewable fuel[7]: 0,015 MJ

[6] Lundgren L., "Hur påverkas miljön av en kilowattimme använd el i Sverige?", Vattenfall Research (1992)
[7] Efficiency 0,33.

Australian average electricity

Table D-4. Emissions and utilisation of resources at the production of 1 MJ Australian average electricity [23].

Substance	Amount	Unit	Impactmedia	Category
Ashes	0,0343	g	Water	Emission
CH4	0,549	g	Air	Emission
CO	0,0348	g	Air	Emission
CO2	200	g	Air	Emission
COD	0,0001	g	Water	Emission
Dissolved solids	0,0017	g	Water	Emission
Phenol (aq,)	0	g	Water	Emission
HC	0,0183	g	Air	Emission
N2O	0,0056	g	Air	Emission
NH3	0	g	Air	Emission
NH3 (aq,)	0	g	Water	Emission
NOx	0,141	g	Air	Emission
Oil (aq,)	0,0001	g	Water	Emission
Rn-222 [Bq]	0,768	Bq	Air	Emission
SO2	0,182	g	Air	Emission
Dust	0,0932	g	Air	Emission
Tot-N (aq,)	0	g	Water	Emission
Biomass (r)	0,568	g	Resource	Resource
Coal (r)	64,4	g	Resource	Resource
Natural gas (r)	5,8	g	Resource	Resource
Crude oil (r)	2,5	g	Resource	Resource
Water power [MJel] (r)	0,1	MJ	Resource	Resource

Australian mix of electricity (per MJ electricity):

- Water power: 0,100 MJ
- Coal: 0,786 MJ
- Oil[8]: 0,028 MJ
- Natural gas[8]: 0,082 MJ

[8] Efficiency: 0.33.

Diesel, working machines

The emissions and utilisation of resources for the working machines, used during the life cycle, have been approximated with data for an excavator[9]. Table D-5 presents the emission data for the use of 1 MJ diesel at the operation of the excavator.

Table D-5. Emissions at the use of 1 MJ diesel at the operation of an excavator [23].

Substance	Amount	Unit	Impactmedia	Category
CO	0.077	g	Air	Emission
CO2	71.93	g	Air	Emission
COD	0.0011	g	Water	Emission
Phenol (aq)	0.0005	g	Water	Emission
HC	0.047	g	Air	Emission
N2O	0.0015	g	Air	Emission
Nox	0.78	g	Air	Emission
Oil (aq)	0.0004	g	Water	Emission
SO2	0.035	g	Air	Emission
Dust	0.026	g	Air	Emission
Tot-N	0.0002	g	Water	Emission
Oil Eo1 (e)	1	MJ	Diesel U	Energy

Diesel

Production

Table D-6. Emissions and utilisation of resources at the production of 1 MJ diesel [23].

Substance	Amount	Unit	Impactmedia	Category
CH4	0,028	g	Air	Emission
CO	0,008	g	Air	Emission
CO2	10,4	g	Air	Emission
COD	0,0012	g	Water	Emission
Phenol	0	g	Air	Emission
HC	0,014	g	Air	Emission
N2O	0,0045	g	Air	Emission
NOx	0,044	g	Air	Emission
Oil (aq)	0,0004	g	Water	Emission
SO2	0,021	g	Air	Emission
Dust	0,0005	g	Air	Emission
Tot-N	0,0002	g	Water	Emission
Natural gas (r)	0,58	g	Resource	Resource
Crude oil (r)	25,1	g	Resource	Resource

Energy value (diesel) = 42,5 MJ/kg
Density (diesel) = 840 kg/m^3

[9] Stripple, H., "Livscykelanalys av väg, En modellstudie för inventering", Göteborg (1995)

Use of diesel for boat

Table D-7. Emissions at the use of 1 MJ diesel for boat [23].

Substance	Amount	Unit	Impactmedia	Category
CO	0,14	g	Air	Emission
CO2	74,6	g	Air	Emission
HC	0,0996	g	Air	Emission
NOx	0,74	g	Air	Emission
SO2	1,28	g	Air	Emission
Dust	0,11	g	Air	Emission

Coal

Production

Table D-8. *Emissions and utilisation of resources at the production of 1 MJ coal [23].*

Substance	Amount	Unit	Impactmedia	Category
CH4	0,286	g	Air	Emission
CO	0,0022	g	Air	Emission
CO2	1,66	g	Air	Emission
COD	0	g	Water	Emission
Dissolved solids	0,001	g	Water	Emission
HC	0,0072	g	Air	Emission
N2O	0,0002	g	Air	Emission
NH3	0	g	Air	Emission
NH3 (aq,)	0	g	Water	Emission
NOx	0,0087	g	Air	Emission
Oil (aq,)	0	g	Water	Emission
Rn-222 [Bq]	0,44	Bq	Ground	Emission
SO2	0,0082	g	Air	Emission
Dust	0,0101	g	Air	Emission
Coal (r)	37	g	Resource	Resource
Natural gas (r)	0,054	g	Resource	Resource
Crude oil (r)	0,165	g	Resource	Resource

Use

Table D-9. Emissions at the use of 1 MJ coal [23].

Substance	Amount	Unit	Impactmedia	Category
Ashes	0.3	g	Air	Emission
CO	0.017	g	Air	Emission
CO2	91.4	g	Air	Emission
HC	0.01	g	Air	Emission
NOx	0.15	g	Air	Emission
SO2	0.38	g	Air	Emission
Dust	0.013	g	Air	Emission

Energy value (Coal) = 34 MJ/kg

Gasoline

Production

Table D-10. Emissions and utilisation of resources at the production of 1 MJ gasoline [23].

Substance	Amount	Unit	Impactmedia	Category
CH4	0,026	g	Air	Emission
CO	0,005	g	Air	Emission
CO2	15,4	g	Air	Emission
HC	0,05	g	Air	Emission
N2O	0,002	g	Air	Emission
NOx	0,048	g	Air	Emission
SO2	0,027	g	Air	Emission
Natural gas (r)	3,4	g	Resource	Resource
Crude oil (r)	24	g	Resource	Resource

Use

Table D-11. Emissions at the use of 1 MJ gasoline [23].

Substance	Amount	Unit	Impactmedia	Category
CH4	0,03	g	Air	Emission
CO	1,76	g	Air	Emission
CO2	75	g	Air	Emission
HC	0,25	g	Air	Emission
N2O	0,02	g	Air	Emission
Nox	0,12	g	Air	Emission
SO2	0,005	g	Air	Emission
Gasoline	1	MJ		Raw material

Energy value (gasoline 95-oktane, free of lead) = 43,1-43,9 MJ/kg
Density (gasoline 95-oktane, free of lead) = 750 kg/m^3

LP-gas (propane)

Production

Table D-10. Emissions and utilisation of resources at the production of 1 MJ LP-gas [23].

Substance	Amount	Unit	Impactmedia	Category
CH4	0,051	g	Air	Emission
CO	0,01	g	Air	Emission
CO2	13,9	g	Air	Emission
HC	0,017	g	Air	Emission
N2O	0,0003	g	Air	Emission
NOx	0,096	g	Air	Emission
SO2	0,025	g	Air	Emission
LP-gas (r)	21,7	g	Resource	Resource
Natural gas (r)	0,58	g	Resource	Resource
Crude oil (r)	3,1	g	Resource	Resource

Use

Table D-11. Emissions at the use of 1 MJ LP-gas [23].

Substance	Amount	Unit	Impactmedia	Category
CO	55,2	g	Air	Emission
CO2	0,15	g	Air	Emission
LP-gas (propame) (e)	1	MJ		Resource

Natural gas

Production [10]

Table D-12. Emissions and utilisation of resources at the production of 1 MJ natural gas [23].

Substance	Amount	Unit	Impactmedia	Category
CH4	0,188	g	Air	Emission
CO	0,0053	g	Air	Emission
CO2	8	g	Air	Emission
HC	0,0028	g	Air	Emission
N2O	0,0005	g	Air	Emission
NOx	0,016	g	Air	Emission
Natural gas (r)	22,4	g	Resource	Resource

Use [11]

Table D-13. Emissions at the use of 1 MJ natural gas [23].

Substance	Amount	Unit	Impactmedia	Category
CO	0,001	g	Air	Emission
CO2	55,2	g	Air	Emission
HC	0	g	Air	Emission
NOx	0,15	g	Air	Emission
SO2	0,002	g	Air	Emission
Dust	0,013	g	Air	Emission
Natural gas (e)	1	MJ		Raw material

Energy value (Danish natural gas) = 39,6 GJ/1000 m^3
Density (Danish natural gas) = 1 kg/m^3

[10] Life of fuels, Ecotraffic AB, Stockholm, Sverige, 1992.
[11] Tillman et al, Packaging and environment, CIT, Göteborg, Sverige, 1992.

Oil Eo1

Production [12]

Table D-14. Emissions and utilisation of resources at the production of 1 MJ oil (Eo1) [23].

Substance	Amount	Unit	Impactmedia	Category
CO	0,0001	g	Air	Emission
CO2	3,99	g	Air	Emission
COD	0,0012	g	Water	Emission
Phenol	0,0001	g	Air	Emission
HC	0,0084	g	Air	Emission
NOx	0,0041	g	Air	Emission
Oil (aq)	0,0004	g	Water	Emission
SO2	0,014	g	Air	Emission
Dust	0,0005	g	Air	Emission
Tot-N	0,0002	g	Water	Emission
Natural gas (r)	1,35	g	Resource	Resource
Crude oil (r)	25,8	g	Resource	Resource

Use [13]

Table D-15. Emissions at the use of 1 MJ oil (Eo1) [23].

Substance	Amount	Unit	Impactmedia	Category
CO	0,01	g	Air	Emission
CO2	81	g	Air	Emission
HC	0,01	g	Air	Emission
NOx	0,15	g	Air	Emission
SO2	0,2	g	Air	Emission
Dust	0,02	g	Air	Emission
Oil Eo1 (e)	1	MJ		Raw material

Energy value (Eo1) = 42,3 MJ/kg
Density (Eo1) = 845 kg/m^3

[12] Tillman et al. Packaging and environment, CIT, Göteborg, Sverige, 1992.
[13] Swan, Göran. STORA Teknik Säffle, Sverige, 1993.

Oil Eo5

Production [14]

Table D-16. Emissions and utilisation of resources at the production of 1 MJ oil (Eo5) [23].

Substance	Amount	Unit	Impactmedia	Category
CO	0,0001	g	Air	Emission
CO2	3,99	g	Air	Emission
COD	0,0012	g	Water	Emission
Phenol	0	g	Air	Emission
HC	0,0084	g	Air	Emission
NOx	0,0041	g	Air	Emission
Oil (aq)	0,0004	g	Water	Emission
SO2	0,014	g	Air	Emission
Dust	0,0005	g	Air	Emission
Tot-N	0,0002	g	Water	Emission
Natural gas (r)	1,35	g	Resource	Resource
Crude oil (r)	25,8	g	Resource	Resource

Use [15]

Table D-17. Emissions at the use of 1 MJ oil (Eo5) [23].

Substance	Amount	Unit	Impactmedia	Category
Ashes	0,007	g	Air	Emission
CO	0,013	g	Air	Emission
CO2	81	g	Air	Emission
HC	0,01	g	Air	Emission
NOx	0,15	g	Air	Emission
SO2	0,38	g	Air	Emission
Dust	0,03	g	Air	Emission
Oil Eo1 (e)	1	MJ		Raw material

Energy value (Eo5) = 40,6 MJ/kg
Density (Eo5) = 940 kg/m^3

[14] Tillman et al. Packaging and the environment, CIT, Göteborg, Sverige, 1992.
[15] Swan, Göran, STORA Teknik Säffle, Sverige, 1993.

Transport data

For all transports, the reference is (Tillman et al, CIT Ekologik, Göteborg, Sverige) except for the data for the road transports.

Boat (ocean going)

Table D-18. Emissions and utilisation of resources per ton and km for ocean going transport by boat [Detaljrapport Sydkraft].

Substance	Amount	Unit	Impactmedia	Category
CO	0,0658	g	Air	Emission
CO2	35,062	g	Air	Emission
HC	0,0468	g	Air	Emission
NOx	0,3478	g	Air	Emission
SO2	0,6016	g	Air	Emission
Dust	0,0517	g	Air	Emission
Diesel (e)	0,47	MJ	Energy	Resource

Boat (coast going)

Table D-19. Emissions and utilisation of resources per ton and km for coast going transport by boat [Detaljrapport Sydkraft].

Substance	Amount	Unit	Impactmedia	Category
CO	0,028	g	Air	Emission
CO2	14,92	g	Air	Emission
HC	0,0199	g	Air	Emission
NOx	0,148	g	Air	Emission
SO2	0,256	g	Air	Emission
Dust	0,022	g	Air	Emission
Diesel (e)	0,2	MJ	Energy	Resource

Train (UCPTE electricity)

Table D-20. Utilisation of resources per ton and km at European average transport by train [Detaljrapport Sydkraft].

Substance	Amount	Unit	Impactmedia	Category
Electricity UCPTE (e)	0.3	MJ	Energy	Resource

Road, short distance <100 km (0 % load) [16]

Table D-21. Emissions and utilisation of resources per ton and km for road, short distance transport [Detaljrapport Sydkraft].

Substance	Amount	Unit	Impactmedia	Category
CO	0,1851	g	Air	Emission
CO2	46,7	g	Air	Emission
HC	0,1328	g	Air	Emission
NOx	0,6442	g	Air	Emission
SO2	0,0083	g	Air	Emission
Dust	0,0374	g	Air	Emission
Diesel (e)	0,6426	MJ	Energy	Resource

Road, short distance <100 km (100 % load) [13]

Table D-22. Emissions and utilisation of resources per ton and km for road, short distance transport [Detaljrapport Sydkraft].

Substance	Amount	Unit	Impactmedia	Category
CO	0,3321	g	Air	Emission
CO2	56,04	g	Air	Emission
HC	0,1004	g	Air	Emission
NOx	0,6442	g	Air	Emission
SO2	0,01	g	Air	Emission
Dust	0,0374	g	Air	Emission
Diesel (e)	0,7711	MJ	Energy	Resource

Road, long distance (100 % load) [13]

Table D-23. Emissions and utilisation of resources per ton and km for road, long distance transport [Detaljrapport Sydkraft].

Substance	Amount	Unit	Impactmedia	Category
CO	0.1998	g	Air	Emission
CO2	51.2267	g	Air	Emission
HC	0.0563	g	Air	Emission
NOx	0.7811	g	Air	Emission
SO2	0.0091	g	Air	Emission
Dust	0.0168	g	Air	Emission
Diesel (e)	0.7007	MJ	Energy	Resource

[16] Eriksson, Elin et al, "Transporters miljöpåverkan i ett livscykelperspektiv", REFORSK FoU 126, 1995.

Appendix D - Presentation of inventory results

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Production of the SOFC-module (200 kW)

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
Aldehydes	Emission	Air	g	0.00005223
CH4	Emission	Air	g	0.76746656
CO	Emission	Air	g	0.09667077
CO2	Emission	Air	g	298.04327321
Phenol	Emission	Air	g	0.00000136
Fluorides	Emission	Air	g	0.00000020
HC	Emission	Air	g	0.08721886
N2O	Emission	Air	g	0.01004367
NH3	Emission	Air	g	0.00092269
Nitrates	Emission	Air	g	0.00000638
NOx	Emission	Air	g	0.30844865
Rn-222[Bq]	Emission	Air	Bq	1.14061180
SO2	Emission	Air	g	0.37631767
Dust	Emission	Air	g	0.14953028
VOC	Emission	Air	g	0.00212016
Ashes	Emission	Ground	g	0.03344941
Radioactive waste h [cm3]	Emission	Ground	ml	0.00006458
Radioactive waste l [cm3]	Emission	Ground	ml	0.00073375
Radioactive waste m [cm3]	Emission	Ground	ml	0.00073375
(SO4)2-	Emission	Water	g	0.00000569
BOD	Emission	Water	g	0.00000299
COD	Emission	Water	g	0.00029949
Dissolved solids	Emission	Water	g	0.00889771
F-	Emission	Water	g	0.00002695
Fe2+	Emission	Water	g	0.00000006
Na+	Emission	Water	g	0.00000442
NH3	Emission	Water	g	0.00001742
N-tot	Emission	Water	g	0.00004506
Oil waste	Emission	Water	g	0.00023187
Suspended solids	Emission	Water	g	0.00000299
De-ionized water	Raw Material		g	0.79816117
C	Raw Material		g	0.02515469
CO2	Raw Material		g	0.03454991
H2SO4	Raw Material		g	0.19106480
Na2CO3	Raw Material		g	0.41297182
O2	Raw Material		g	0.10909967
Sulphur	Raw Material		g	0.06246197
Oxide gas	Raw Material		g	0.00233980
Hydrogen gas	Raw Material		g	0.00008639
Diesel (e)	Resource	Energy	MJ	0.06187162
Diesel, boat (e)	Resource	Energy	MJ	0.02954300
El, Denmark av. (e)	Resource	Energy	MJ	1.28158631
El, UCPTE (e)	Resource	Energy	MJ	0.07264846
Natural gas (e)	Resource	Energy	MJ	0.00082045
Oil EO1 (e)	Resource	Energy	MJ	0.00021785
Oil EO5 (e)	Resource	Energy	MJ	0.00000051
Baddeleyite (r)	Resource	Resource	g	25.84698815
Bastnasite (r)	Resource	Resource	g	1.54592656
Celestite (r)	Resource	Resource	g	0.20244705

Chromite (r)	Resource	Resource	g	0.54506636
Pentlandite (r)	Resource	Resource	g	0.67055186
Pyrolusite (r)	Resource	Resource	g	0.00496758
Xenotime (r)	Resource	Resource	g	0.02712013
Biomass (r)	Resource	Resource	g	1.49945593
Coal (r)	Resource	Resource	g	97.69207378
Natural gas (r)	Resource	Resource	g	2.93097424
Natural gas, feedstock (r)	Resource	Resource	g	0.26144502
Natural uranium (r)	Resource	Resource	g	0.00015184
Crude oil (r)	Resource	Resource	g	6.99007895
Water powert [MJe] (r)	Resource	Resource	MJ	0.01537986
De-ionized water	Waste		g	0.03581700
CO2	Waste		g	0.26364193
Fe2O3	Waste		g	0.15555738
H2CO3	Waste		g	0.04870392
H2S	Waste		g	0.02675296
LaSrCrVO	Waste		g	0.35097062
LSM	Waste		g	0.00237580
Na2SO4	Waste		g	0.55343208
NiO-YSZ	Waste		g	0.00194384
NO3Sub	Waste		g	2.19928620
SO2	Waste		g	0.00267818
TailingA	Waste		g	0.12645598
TailingBadd	Waste		g	23.26228934
TailingBast	Waste		g	1.03267892
TailingCel	Waste		g	0.01012235
TailingChro	Waste		g	0.10901328
TailingCLa	Waste		g	0.00000691
TailingCSr	Waste		g	0.00025558
TailingCYSZ	Waste		g	0.01945565
TailingE	Waste		g	2.39466349
TailingILa	Waste		g	0.00046364
TailingISr	Waste		g	0.03106900
TailingPent	Waste		g	0.61690769
TailingPyro	Waste		g	0.00149028
YSZ	Waste		g	0.07199397

Production of raw material for the electrolyte, yttria-stabilized zirconia (YSZ)

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
Aldehydes	Emission	Air	g	0.00001299
CH4	Emission	Air	g	0.00075610
CO	Emission	Air	g	0.00775323
CO2	Emission	Air	g	4.51279792
Phenol	Emission	Air	g	0.00000015
Fluorides	Emission	Air	g	0.00000005
HC	Emission	Air	g	0.01493301
N2O	Emission	Air	g	0.00047572
NH3	Emission	Air	g	0.00000246
Nitrates	Emission	Air	g	0.00000159
NOx	Emission	Air	g	0.03438252
SO2	Emission	Air	g	0.03341482
Dust	Emission	Air	g	0.00384538
Radioactive waste h [cm3]	Emission	Ground	ml	0.00001607
Radioactive waste l [cm3]	Emission	Ground	ml	0.00018252
Radioactive waste m [cm3]	Emission	Ground	ml	0.00018252
(SO4)2-	Emission	Water	g	0.00000142
BOD	Emission	Water	g	0.00000074
COD	Emission	Water	g	0.00003384
Dissolved solids	Emission	Water	g	0.00158850
F-	Emission	Water	g	0.00000670
Fe2+	Emission	Water	g	0.00000002
Na+	Emission	Water	g	0.00000110
NH3	Emission	Water	g	0.00000311
N-tot	Emission	Water	g	0.00000516
Oil wastage	Emission	Water	g	0.00003466
Suspended solids	Emission	Water	g	0.00000074
Diesel (e)	Resource	Energy	MJ	0.01252341
Diesel, boat (e)	Resource	Energy	MJ	0.01448022
Electricity, UCPTE (e)	Resource	Energy	MJ	0.01807163
Baddeleyite (r)	Resource	Resource	g	25.64140815
Xenotime (r)	Resource	Resource	g	0.02690490
Coal (r)	Resource	Resource	g	0.51883657
Natural gas (r)	Resource	Resource	g	0.12120044
Natural uranium (r)	Resource	Resource	g	0.00003777
Crude oil (r)	Resource	Resource	g	0.81152116
Water power [MJe] (r)	Resource	Resource	MJ	0.00365047
TailingBadd	Waste		g	23.07726733
TailingE	Waste		g	2.47212900

Production of raw material for the anode, yttria-stabilized zirconia with nickel-oxide (NiO-YSZ)

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
Aldehydes	Emission	Air	g	0.00000058
CH4	Emission	Air	g	0.00002906
CO	Emission	Air	g	0.00032339
CO2	Emission	Air	g	0.18786915
Phenol	Emission	Air	g	0.00000001
Fluorides	Emission	Air	g	0.00000000
HC	Emission	Air	g	0.00065167
N2O	Emission	Air	g	0.00002059
NH3	Emission	Air	g	0.00000011
Nitrates	Emission	Air	g	0.00000007
NOx	Emission	Air	g	0.00141082
SO2	Emission	Air	g	0.00126642
Stoft	Emission	Air	g	0.00015335
Radioactive waste h [cm3]	Emission	Ground	ml	0.00000072
Radioactive waste l [cm3]	Emission	Ground	ml	0.00000820
Radioactive waste m [cm3]	Emission	Ground	ml	0.00000820
(SO4)2-	Emission	Water	g	0.00000006
BOD	Emission	Water	g	0.00000003
COD	Emission	Water	g	0.00000132
Dissolved solids	Emission	Water	g	0.00007138
F-	Emission	Water	g	0.00000030
Fe2+	Emission	Water	g	0.00000000
Na+	Emission	Water	g	0.00000005
NH3	Emission	Water	g	0.00000014
N-tot	Emission	Water	g	0.00000020
Oil waste	Emission	Water	g	0.00000149
Suspended solids	Emission	Water	g	0.00000003
Oxygen gas	Raw Material		g	0.00233980
Diesel (e)	Resource	Energy	MJ	0.00056875
Diesel, boat (e)	Resource	Energy	MJ	0.00046926
El, UCPTE (e)	Resource	Energy	MJ	0.00081201
Baddeleyite (r)	Resource	Resource	g	0.80726706
Pentlandite (r)	Resource	Resource	g	0.67055074
Xenotime (r)	Resource	Resource	g	0.00084665
Coal (r)	Resource	Resource	g	0.02331279
Natural gas (r)	Resource	Resource	g	0.00534418
Natural uranium (r)	Resource	Resource	g	0.00000170
Crude oil (r)	Resource	Resource	g	0.03206298
Water power [MJe] (r)	Resource	Resource	MJ	0.00016403
SO2	Waste		g	0.00267817
TailingA	Waste		g	0.12645577
TailingBadd	Waste		g	0.72654037
TailingPent	Waste		g	0.61690667

Production of raw material for the cathode, doped lanthanum (LSM)

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
Aldehydes	Emission	Air	g	0.00000037
CH4	Emission	Air	g	0.00048016
CO	Emission	Air	g	0.00028569
CO2	Emission	Air	g	0.28951108
Phenol	Emission	Air	g	0.00000000
Fluorides	Emission	Air	g	0.00000000
HC	Emission	Air	g	0.00044209
N2O	Emission	Air	g	0.00001804
NH3	Emission	Air	g	0.00000778
Nitrates	Emission	Air	g	0.00000005
NOx	Emission	Air	g	0.00112731
Rn-222[Bq]	Emission	Air	Bq	0.00065997
SO2	Emission	Air	g	0.00091302
Dust	Emission	Air	g	0.00017924
VOC	Emission	Air	g	0.00001794
Ashes	Emission	Ground	g	0.00001935
Radioactive waste h [cm3]	Emission	Ground	ml	0.00000046
Radioactive waste l [cm3]	Emission	Ground	ml	0.00000526
Radioactive waste m [cm3]	Emission	Ground	ml	0.00000526
(SO4)2-	Emission	Water	g	0.00000004
BOD	Emission	Water	g	0.00000002
COD	Emission	Water	g	0.00000096
Dissolved solids	Emission	Water	g	0.00004720
F-	Emission	Water	g	0.00000019
Fe2+	Emission	Water	g	0.00000000
Na+	Emission	Water	g	0.00000003
NH3	Emission	Water	g	0.00000009
N-tot	Emission	Water	g	0.00000015
Oil wastage	Emission	Water	g	0.00000102
Suspended solids	Emission	Water	g	0.00000002
De-ionized water	Raw Material		g	0.00498197
C	Raw Material		g	0.00020878
CO2	Raw Material		g	0.00028078
Hydrogen gas	Raw Material		g	0.00008639
Diesel (e)	Resource	Energy	MJ	0.00043196
Diesel, boat (e)	Resource	Energy	MJ	0.00024319
El, Denmark av. (e)	Resource	Energy	MJ	0.00074154
El, UCPTE (e)	Resource	Energy	MJ	0.00052049
Natural gas (e)	Resource	Energy	MJ	0.00000692
Oil EO1 (e)	Resource	Energy	MJ	0.00000187
Oil EO5 (e)	Resource	Energy	MJ	0.00000000
Baddeleyite (r)	Resource	Resource	g	0.20179878
Bastnasite (r)	Resource	Resource	g	0.01641460
Celestite (r)	Resource	Resource	g	0.00165586
Pyrolusite (r)	Resource	Resource	g	0.00496758
Xenotime (r)	Resource	Resource	g	0.00021166
Biomass (r)	Resource	Resource	g	0.00086760

Coal (r)	Resource	Resource	g	0.07026199
Natural gas (r)	Resource	Resource	g	0.00499763
Natural gas, feedstock (r)	Resource	Resource	g	0.00221631
Natural uranium (r)	Resource	Resource	g	0.00000109
Crude oil (r)	Resource	Resource	g	0.02324893
Water power [MJe] (r)	Resource	Resource	MJ	0.00010555
De-ionized water	Waste		g	0.00071994
CO2	Waste		g	0.00075594
H2CO3	Waste		g	0.00039597
H2S	Waste		g	0.00021598
NO3Sub	Waste		g	0.01651539
TailingBadd	Waste		g	0.18161889
TailingBast	Waste		g	0.01096495
TailingCel	Waste		g	0.00008279
TailingCLa	Waste		g	0.00000691
TailingCSr	Waste		g	0.00025558
TailingCYSZ	Waste		g	0.01945562
TailingPyro	Waste		g	0.00149027

Production of raw material for the interconnect, LSCV*

**LSCV = La(0,8)Sr(0,2)Cr(0,97)V(0,03)O3*

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
Aldehydes	Emission	Air	g	0.00003869
CH4	Emission	Air	g	0.04228942
CO	Emission	Air	g	0.02905482
CO2	Emission	Air	g	26.73756773
Phenol	Emission	Air	g	0.00000042
Fluorides	Emission	Air	g	0.00000015
HC	Emission	Air	g	0.04513592
N2O	Emission	Air	g	0.00175736
NH3	Emission	Air	g	0.00090750
Nitrates	Emission	Air	g	0.00000472
NOx	Emission	Air	g	0.11321774
Rn-222[Bq]	Emission	Air	Bq	0.05709040
SO2	Emission	Air	g	0.07896354
Dust	Emission	Air	g	0.01650279
VOC	Emission	Air	g	0.00210222
Ashes	Emission	Ground	g	0.00167423
Radioactive waste h [cm3]	Emission	Ground	ml	0.00004784
Radioactive waste l [cm3]	Emission	Ground	ml	0.00054349
Radioactive waste m [cm3]	Emission	Ground	ml	0.00054349
(SO4)2-	Emission	Water	g	0.00000421
BOD	Emission	Water	g	0.00000221
COD	Emission	Water	g	0.00009045
Dissolved solids	Emission	Water	g	0.00485567
F-	Emission	Water	g	0.00001996
Fe2+	Emission	Water	g	0.00000004
Na+	Emission	Water	g	0.00000327
NH3	Emission	Water	g	0.00000950
N-tot	Emission	Water	g	0.00001356
Oil waste	Emission	Water	g	0.00010151
Suspended solids	Emission	Water	g	0.00000221
De-ionized water	Raw Material		g	0.79317787
C	Raw Material		g	0.02494587
CO2	Raw Material		g	0.03426908
H2SO4	Raw Material		g	0.19106448
Na2CO3	Raw Material		g	0.41297114
O2	Raw Material		g	0.10909949
Sulphur	Raw Material		g	0.06246186
Diesel (e)	Resource	Energy	MJ	0.04873984
Diesel, boat (e)	Resource	Energy	MJ	0.01480405
El, Denmark av. (e)	Resource	Energy	MJ	0.06414652
El, UCPTE (e)	Resource	Energy	MJ	0.05381051
Natural gas (e)	Resource	Energy	MJ	0.00081353
Oil EO1 (e)	Resource	Energy	MJ	0.00021598
Oil EO5 (e)	Resource	Energy	MJ	0.00000050
Bastnasite (r)	Resource	Resource	g	1.52950939
Celestite (r)	Resource	Resource	g	0.20079085
Chromite (r)	Resource	Resource	g	0.54506546
Biomass (r)	Resource	Resource	g	0.07505143

Coal (r)	Resource	Resource	g	6.33023025
Natural gas (r)	Resource	Resource	g	0.49150259
Natural gas, feedstock (r)	Resource	Resource	g	0.25922827
Natural uranium (r)	Resource	Resource	g	0.00011246
Crude oil (r)	Resource	Resource	g	2.20656958
Water power [MJe] (r)	Resource	Resource	MJ	0.01090500
De-ionized water	Waste		g	0.03509700
CO2	Waste		g	0.26288556
Fe2O3	Waste		g	0.15555712
H2CO3	Waste		g	0.04830788
H2S	Waste		g	0.02653694
Na2SO4	Waste		g	0.55343116
NO3Sub	Waste		g	2.18276715
TailingBast	Waste		g	1.02171225
TailingCel	Waste		g	0.01003954
TailingChro	Waste		g	0.10901309
TailingILa	Waste		g	0.00046364
TailingISr	Waste		g	0.03106895

Production of gas tubine

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
Aldehydes	Emission	Air	g	0.00000005
As	Emission	Air	g	0.00000001
Waste, scrap	Emission	Air	g	0.00008039
Ca	Emission	Air	g	0.00000001
Cd	Emission	Air	g	0.00000000
CF4 + C2F6	Emission	Air	g	0.00000001
CH4	Emission	Air	g	0.00005716
CO	Emission	Air	g	0.00001789
CO2	Emission	Air	g	0.02178090
Cr	Emission	Air	g	0.00000000
Cu	Emission	Air	g	0.00000012
Dioxin	Emission	Air	g	0.00000000
F (g)	Emission	Air	g	0.00000001
F (part)	Emission	Air	g	0.00001340
Fe	Emission	Air	g	0.00000270
Phenol	Emission	Air	g	0.00000000
Fluorides	Emission	Air	g	0.00000000
HC	Emission	Air	g	0.00004662
HCl	Emission	Air	g	0.00000045
HF	Emission	Air	g	0.00000060
Hg	Emission	Air	g	0.00000000
Mn	Emission	Air	g	0.00000000
N2O	Emission	Air	g	0.00000156
NH3	Emission	Air	g	0.00000031
NH4NO3	Emission	Air	g	0.00000016
Ni	Emission	Air	g	0.00000000
Nitrates	Emission	Air	g	0.00000001
NOx	Emission	Air	g	0.00006855
PAH	Emission	Air	g	0.00000000
Pb	Emission	Air	g	0.00000021
Rn-222[Bq]	Emission	Air	Bq	0.00008558
SO2	Emission	Air	g	0.00012050
SOx	Emission	Air	g	0.00000033
Dust	Emission	Air	g	0.00002544
V	Emission	Air	g	0.00000001
VOC	Emission	Air	g	0.00000050
Zn	Emission	Air	g	0.00000022
Ashes	Emission	waste	g	0.00000024
Waste, rock	Emission	waste	g	0.34091506
Waste, dust	Emission	waste	g	0.00015606
Waste, hazardous waste	Emission	waste	g	0.00000003
Waste, granite	Emission	waste	g	0.01196997
Waste, non-magnetc	Emission	waste	g	0.00030334
Waste, coal	Emission	waste	g	0.00000022
Waste, copper-ore	Emission	waste	g	0.05221249
Waste, LD- dust	Emission	waste	g	0.00000901
Waste, LD-cinder	Emission	waste	g	0.00051568
Waste, LD-sludge	Emission	waste	g	0.00040800
Waste, oxide	Emission	waste	g	0.00004728

Waste, sand	Emission	waste	g	0.44684572
Waste, scrap	Emission	waste	g	0.00004728
Waste, cinder	Emission	waste	g	0.00046057
Waste, sludge	Emission	waste	g	0.00012630
Waste, brick	Emission	waste	g	0.00000693
Bauxite-remains	Emission	waste	g	0.00003069
Dross fines	Emission	waste	g	0.00000006
Solid waste	Emission	waste	g	0.00002988
Inert remains	Emission	waste	g	0.00000319
NaOH	Emission	waste	g	0.00022716
Radioactive waste h [cm3]	Emission	waste	ml	0.00000000
Radioactive waste l [cm3]	Emission	waste	ml	0.00000005
Radioactive waste m [cm3]	Emission	waste	ml	0.00000005
Skimmings/dross	Emission	waste	g	0.00000035
Scrap (steel)	Emission	waste	g	0.00000018
Coal	Emission	nftg*	g	0.00000048
Ashes	Emission	Ground	g	0.00000150
Radioactive waste h [cm3]	Emission	Ground	ml	0.00000009
Radioactive waste l [cm3]	Emission	Ground	ml	0.00000102
Radioactive waste m [cm3]	Emission	Ground	ml	0.00000103
(SO4)2-	Emission	Water	g	0.00000001
As	Emission	Water	g	0.00000001
BOD	Emission	Water	g	0.00000000
Cd	Emission	Water	g	0.00000000
Cl-	Emission	Water	g	0.00000001
CN-tot	Emission	Water	g	0.00000000
Co	Emission	Water	g	0.00000000
COD	Emission	Water	g	0.00000034
Cr	Emission	Water	g	0.00000000
Cu	Emission	Water	g	0.00000001
Cu	Emission	Water	mg	0.00000001
Dissolved solids	Emission	Water	g	0.00000624
F-	Emission	Water	g	0.00000003
Fe2+	Emission	Water	g	0.00000000
Phenol	Emission	Water	g	0.00000000
Fluorider	Emission	Water	g	0.00000000
F-tot	Emission	Water	g	0.00000000
H2SO4	Emission	Water	g	0.00000002
Hg	Emission	Water	g	0.00000000
HNO3	Emission	Water	g	0.00000001
Chlorides	Emission	Water	g	0.00000008
Mn	Emission	Water	g	0.00000003
Na+	Emission	Water	g	0.00000000
NaCl	Emission	Water	g	0.00000003
NH3	Emission	Water	g	0.00000002
NH4+	Emission	Water	g	0.00000008
NH4-N	Emission	Water	g	0.00000038
NH4NO3	Emission	Water	g	0.00000056
Ni	Emission	Water	g	0.00000000
Nitrates	Emission	Water	g	0.00000000
NO2-	Emission	Water	g	0.00000001
NO3-	Emission	Water	g	0.00000004
NO3-N	Emission	Water	g	0.00000042
N-tot	Emission	Water	g	0.00000046
Oil wastage	Emission	Water	g	0.00000012
Organic coal	Emission	Water	g	0.00000000

PAH	Emission	Water	g	0.00000000
Pb	Emission	Water	g	0.00000001
P-tot	Emission	Water	g	0.00000000
Suspended solids	Emission	Water	g	0.00000012
Zn	Emission	Water	g	0.00000004
Zn	Emission	Water	mg	0.00000011
Gasoline (e)	Resource	Energy	MJ	0.00000003
Diesel (e)	Resource	Energy	MJ	0.00002104
Diesel, boat (e)	Resource	Energy	MJ	0.00001240
El, Australia av. (e)	Resource	Energy	MJ	0.00000012
El, Sweden av. (e)	Resource	Energy	MJ	0.00002462
El, UCPTE (e)	Resource	Energy	MJ	0.00006890
Coal (e)	Resource	Energy	MJ	0.00000001
LP-gas (propane) (e)	Resource	Energy	MJ	0.00000042
Natural gas (e)	Resource	Energy	MJ	0.00000035
Oil EO1 (e)	Resource	Energy	MJ	0.00001060
Oil EO5 (e)	Resource	Energy	MJ	0.00000062
Bauxite (r)	Resource	Resource	g	0.00010713
Bentonite (r)	Resource	Resource	g	0.00007813
Biomass (r)	Resource	Resource	g	0.00005793
Pyrolusite (r)	Resource	Resource	g	0.00036452
Caliche (r)	Resource	Resource	g	0.00002033
Caliche (r)	Resource	Resource	g	0.00056486
Dolomite (r)	Resource	Resource	g	0.00000002
Dolomite (r)	Resource	Resource	g	0.00010903
Felspar (r)	Resource	Resource	g	0.00000002
Iron-ore (r)	Resource	Resource	g	0.02578142
Limestone (r)	Resource	Resource	g	0.00183659
Coal (r)	Resource	Resource	g	0.00202242
Coal, feedstock (r)	Resource	Resource	g	0.00719145
Copper-ore[0,35%Cu](r)	Resource	Resource	g	0.45088502
LP-gas (r)	Resource	Resource	g	0.00000674
Na2CO3 (r)	Resource	Resource	g	0.00000003
Natural gas (r)	Resource	Resource	g	0.00045651
Natural gas, feedstock (r)	Resource	Resource	g	0.00006197
Natural uranium (r)	Resource	Resource	g	0.00000022
Olivine (r)	Resource	Resource	g	0.00056103
Crude oil (r)	Resource	Resource	g	0.00162106
Crude oil, feedstock (r)	Resource	Resource	g	0.00002894
Salt	Resource	Resource	g	0.00001474
Sand (r)	Resource	Resource	g	0.00000020
Seawater (r)	Resource	Resource	g	0.00003443
Peat (r)	Resource	Resource	g	0.00000517
Water (r)	Resource	Resource	g	0.01140605
Water power [MJe] (r)	Resource	Resource	MJ	0.00002550
Aluminum hydroxide	Resource	Resurs nttc**	g	0.00000034
Aluminum powder	Resource	Resurs nttc**	g	0.00000970
Waste, refractory	Resource	Resurs nttc**	g	0.00000033
Emulsifier	Resource	Resurs nttc**	g	0.00000013
Emulsifier	Resource	Resurs nttc**	g	0.00000310
Calcium fluride	Resource	Resurs nttc**	g	0.00000074
Coal	Resource	Resurs nttc**	g	0.00000089
Micro ballons	Resource	Resurs nttc**	g	0.00000734
Na2SO4	Resource	Resurs nttc**	g	0.00000000
Refractory mat.	Resource	Resurs nttc**	g	0.00000025
Scrap (steel)	Resource	Resurs nttc**	g	0.00451199

Solvey soda	Resource	Resurs nttc**	g	0.00000003
Steel	Resource	Resurs nttc**	g	0.00000018
Sulphuric acid	Resource	Resurs nttc**	g	0.00000086

*nftg = not followed to the grave

**nttc = not traced to the cradle

Construction of the SOFC/GT process

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
CH4	Emission	Air	g	0.00011425
CO	Emission	Air	g	0.00037743
CO2	Emission	Air	g	0.33562404
Phenol	Emission	Air	g	0.00000002
HC	Emission	Air	g	0.00025481
N2O	Emission	Air	g	0.00002405
NOx	Emission	Air	g	0.00340745
SO2	Emission	Air	g	0.00009636
SOx	Emission	Air	g	0.00013695
Dust	Emission	Air	g	0.00010845
COD	Emission	Water	g	0.00000905
Phenol	Emission	Water	g	0.00000203
N-tot	Emission	Water	g	0.00000146
Oil wastage	Emission	Water	g	0.00000306
Asphalt	Raw Material		g	1.58377004
Concrete	Raw Material		g	6.42819977
Copper, virgin	Raw Material		g	0.00438457
SOFC-module	Raw Material		g	0.56247251
Steel	Raw Material		g	0.03069290
Steel, demolition	Raw Material		g	0.16768999
Diesel (e)	Resource	Energy	MJ	0.00406727
Diesel, boat (e)	Resource	Energy	MJ	0.00000687
Natural gas (r)	Resource	Resource	g	0.00236301
Crude oil (r)	Resource	Resource	g	0.10226110

Demolition of the SOFC/GT process

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
CH4	Emission	Air	g	0.00011710
CO	Emission	Air	g	0.00036087
CO2	Emission	Air	g	0.34377080
Phenol	Emission	Air	g	0.00000002
HC	Emission	Air	g	0.00025651
N2O	Emission	Air	g	0.00002486
NOx	Emission	Air	g	0.00344212
SO2	Emission	Air	g	0.00008792
SOx	Emission	Air	g	0.00014550
Dust	Emission	Air	g	0.00011110
SOFC/GT, landfill	Emission	Ground	g	0.63481498
COD	Emission	Water	g	0.00000943
Phenol	Emission	Water	g	0.00000216
N-tot	Emission	Water	g	0.00000152
Oil wastege	Emission	Water	g	0.00000319
Diesel (e)	Resource	Energy	MJ	0.00417536
Natural gas (r)	Resource	Resource	g	0.00242171
Crude oil (r)	Resource	Resource	g	0.10480147

Operation of SOFC/GT process

The results are calculated per produced kWh electricity

Substance	Category	Impactmedia	Unit	Per produced kWh
Waste, dumping & comb.	Emission		g	0.00115161
Waste, hazardous waste	Emission		g	0.00248996
Waste, recycling	Emission		g	0.00037816
Aldehydes	Emission	Air	g	0.00000000
As	Emission	Air	g	0.00000000
Waste, scrap	Emission	Air	g	0.00011444
B(a)P	Emission	Air	g	0.00000000
Benzene	Emission	Air	g	0.00000000
Lead	Emission	Air	g	0.00000000
Ca	Emission	Air	g	0.00000001
Cd	Emission	Air	g	0.00000000
CF4 + C2F6	Emission	Air	g	0.00000002
CH4	Emission	Air	g	0.08268878
CO	Emission	Air	g	0.02426769
CO2	Emission	Air	g	297.30978031
Cr	Emission	Air	g	0.00000001
Cu	Emission	Air	g	0.00000000
F (g)	Emission	Air	g	0.00000002
F (part)	Emission	Air	g	0.00001907
Fe	Emission	Air	g	0.00000379
Phenol	Emission	Air	g	0.00000016
Fluorides	Emission	Air	g	0.00000000
H2O	Emission	Air	g	320.65853402
HC	Emission	Air	g	0.00128093
HCl	Emission	Air	g	0.00000064
HF	Emission	Air	g	0.00000086
Hg	Emission	Air	g	0.00000000
Mn	Emission	Air	g	0.00000000
N2O	Emission	Air	g	0.00060321
NH3	Emission	Air	g	0.00000007
NH4NO3	Emission	Air	g	0.00000004
Ni	Emission	Air	g	0.00000000
Nitrates	Emission	Air	g	0.00000000
NMVOC	Emission	Air	g	0.01125363
NOx	Emission	Air	g	0.10674104
PAH	Emission	Air	g	0.00000000
Pb	Emission	Air	g	0.00000002
Rn-222[Bq]	Emission	Air	Bq	0.00047760
SO2	Emission	Air	g	0.00257524
SOx	Emission	Air	g	0.00000023
Dust	Emission	Air	g	0.00009825
TCCD	Emission	Air	g	0.00000000
Tl	Emission	Air	g	0.00000000
V	Emission	Air	g	0.00000001
VOC	Emission	Air	g	0.00000012
Zn	Emission	Air	g	0.00000015
Ashes	Emission	Waste	g	0.00000035
Waste, rock	Emission	Waste	g	0.00000129
Waste, dust	Emission	Waste	g	0.00005118

Waste, hazardous waste	Emission	Waste	g	0.00000483
Waste, granite	Emission	Waste	g	0.01701742
Waste, non-magnetic	Emission	Waste	g	0.00043184
Waste, coal	Emission	Waste	g	0.00000032
Waste, quartzite	Emission	Waste	g	0.00003084
Waste, LD- dust	Emission	Waste	g	0.00001283
Waste, LD-cinder	Emission	Waste	g	0.00073413
Waste, LD-sludge	Emission	Waste	g	0.00058083
Waste, oil	Emission	Waste	g	0.00006593
Waste, oxide	Emission	Waste	g	0.00004039
Waste, scrap	Emission	Waste	g	0.00004039
Waste, cinder	Emission	Waste	g	0.00039341
Waste, sludge	Emission	Waste	g	0.01250242
Waste, brick	Emission	Waste	g	0.00000592
Bauxite-remains	Emission	Waste	g	0.00004370
Dross fines	Emission	Waste	g	0.00000009
Solid Waste	Emission	Waste	g	0.00004254
Inert remains	Emission	Waste	g	0.00000454
NaOH	Emission	Waste	g	0.00032339
Noise	Emission	Waste	g	0.00000000
Radioactive Waste h [cm3]	Emission	Waste	ml	0.00000001
Radioactive Waste l [cm3]	Emission	Waste	ml	0.00000007
Radioactive Waste m [cm3]	Emission	Waste	ml	0.00000007
Skimmings/dross	Emission	Waste	g	0.00000050
Scrap (steel)	Emission	Waste	g	0.00000025
Benzene	Emission	nftg*	g	0.00000390
Coke-gas	Emission	nftg*	g	0.00000003
Coke dust	Emission	nftg*	g	0.00003735
Coal	Emission	nftg*	g	0.00000069
Sulphur	Emission	nftg*	g	0.00000074
Tar	Emission	nftg*	g	0.00001348
Ashes	Emission	Ground	g	0.00001240
Waste, hazardous Waste	Emission	Ground	g	0.04824300
Waste, domestic	Emission	Ground	g	0.05446790
Waste, industrial	Emission	Ground	g	0.01089358
Sewage	Emission	Ground	m3	0.00000140
Radioactive Waste h [cm3]	Emission	Ground	ml	0.00000004
Radioactive Waste l [cm3]	Emission	Ground	ml	0.00000049
Radioactive Waste m [cm3]	Emission	Ground	ml	0.00000050
(SO4)2-	Emission	Water	g	0.00000000
As	Emission	Water	g	0.00000000
Barite, bentonite	Emission	Water	g	0.14006031
BOD	Emission	Water	g	0.00000000
Cd	Emission	Water	g	0.00000000
Cl-	Emission	Water	g	0.00000000
CN-tot	Emission	Water	g	0.00000000
Co	Emission	Water	g	0.00000000
COD	Emission	Water	g	0.00003343
Cr	Emission	Water	g	0.00000000
Cu	Emission	Water	g	0.00000000
Cu	Emission	Water	mg	0.00000000
Dissolved solids	Emission	Water	g	0.00000123
F-	Emission	Water	g	0.00000000
Fe2+	Emission	Water	g	0.00000000
Phenol	Emission	Water	g	0.00000000
F-tot	Emission	Water	g	0.00000000

H2SO4	Emission	Water	g	0.00000003
HNO3	Emission	Water	g	0.00000000
Chlorides	Emission	Water	g	0.00000011
Mn	Emission	Water	g	0.00000004
Na+	Emission	Water	g	0.00000000
NaCl	Emission	Water	g	0.00000001
NH3	Emission	Water	g	0.00000000
NH4+	Emission	Water	g	0.00000012
NH4-N	Emission	Water	g	0.00000008
NH4NO3	Emission	Water	g	0.00000011
Ni	Emission	Water	g	0.00000000
Nitrates	Emission	Water	g	0.00000000
NO2-	Emission	Water	g	0.00000001
NO3-	Emission	Water	g	0.00000007
NO3-N	Emission	Water	g	0.00000008
N-tot	Emission	Water	g	0.00000587
Oil wastage	Emission	Water	g	0.00001131
Organic coal	Emission	Water	g	0.00000000
PAH	Emission	Water	g	0.00000000
Pb	Emission	Water	g	0.00000001
Prod. chemicals	Emission	Water	g	0.00155623
P-tot	Emission	Water	g	0.00000001
Radioactivity, L6	Emission	Water	g	0.00000000
Suspended solids	Emission	Water	g	0.00000018
Zn	Emission	Water	g	0.00000000
Zn	Emission	Water	mg	0.00000002
Other drill-chemicals	Emission	Water	g	0.04201810
Gasoline (e)	Resource	Energy	MJ	0.00000004
Diesel (e)	Resource	Energy	MJ	0.02813710
Diesel, boat (e)	Resource	Energy	MJ	0.00001952
El, Australia av. (e)	Resource	Energy	MJ	0.00000018
El, Denmark av. (e)	Resource	Energy	MJ	0.00039217
El, Sweden av. (e)	Resource	Energy	MJ	0.00003915
El, UCPTE (e)	Resource	Energy	MJ	0.00000204
Coal (e)	Resource	Energy	MJ	0.00000001
LP-gas (propane) (e)	Resource	Energy	MJ	0.00000059
Natural gas (e)	Resource	Energy	MJ	0.31121429
Oil EO1 (e)	Resource	Energy	MJ	0.00001057
Oil EO5 (e)	Resource	Energy	MJ	0.00000133
Use of acreage	Resource	Ground	m2	0.00000001
Isolation zone	Resource	Ground	m2	0.00000063
THT (C4H8S)	Resource	R ej v	g	0.00233434
Bauxite (r)	Resource	Resource	g	0.00015251
Bentonite (r)	Resource	Resource	g	0.00011122
Biomass (r)	Resource	Resource	g	0.00054215
Pyrolusite (r)	Resource	Resource	g	0.00045931
Caliche (r)	Resource	Resource	g	0.00003270
Caliche (r)	Resource	Resource	g	0.00011255
CaSO4 x 2 H20 (r)	Resource	Resource	g	0.00025565
Dolomite (r)	Resource	Resource	g	0.00000004
Dolomite (r)	Resource	Resource	g	0.00009313
Felspar (r)	Resource	Resource	g	0.00000003
Iron-ore (r)	Resource	Resource	g	0.03670266
Limestone (r)	Resource	Resource	g	0.30982716
Coal (r)	Resource	Resource	g	0.02938126
Coal, feedstock (r)	Resource	Resource	g	0.01080033

Quartzite (r)	Resource	Resource	g	0.00021439
LP-gas (r)	Resource	Resource	g	0.00000959
N2	Resource	Resource	g	0.00264558
Na2CO3 (r)	Resource	Resource	g	0.00000004
Natural gas (r)	Resource	Resource	g	6.98832069
Natural gas (r)	Resource	Resource	g	130.72295525
Natural gas, feedstock (r)	Resource	Resource	g	0.00001521
Natural uranium (r)	Resource	Resource	g	0.00000012
Olivine (r)	Resource	Resource	g	0.00079869
Crude oil (r)	Resource	Resource	g	0.70830030
Crude oil, feedstock (r)	Resource	Resource	g	0.00002435
Salt	Resource	Resource	g	0.00002099
Sand (r)	Resource	Resource	g	0.00000031
Seavatten (r)	Resource	Resource	g	0.00565018
Glass sand and such-like	Resource	Resource	g	0.00155623
Rock (r)	Resource	Resource	g	0.03360492
Peat (r)	Resource	Resource	g	0.00000743
Water (r)	Resource	Resource	g	0.00372994
Water power [MJel] (r)	Resource	Resource	MJ	0.00001729
Additives	Resource	Resource, nttc**	g	0.00007657
Aluminum hydroxide	Resource	Resource, nttc**	g	0.00000049
Aluminum powder	Resource	Resource, nttc**	g	0.00000248
Waste, refractory	Resource	Resource, nttc**	g	0.00000047
Emulsifier	Resource	Resource, nttc**	g	0.00000021
Emulsifier	Resource	Resource, nttc**	g	0.00000062
Grind.aid	Resource	Resource, nttc**	g	0.00000005
Grind.media 1	Resource	Resource, nttc**	g	0.00000022
Grind.media 2	Resource	Resource, nttc**	g	0.00000021
Gravel	Resource	Resource, nttc**	g	0.12449806
Gravel	Resource	Resource, nttc**	g	0.16600260
Iron sulphate	Resource	Resource, nttc**	g	0.00004892
Calcium fluoride	Resource	Resource, nttc**	g	0.00000105
Coal	Resource	Resource, nttc**	g	0.00000127
Acreage	Resource	Resource, nttc**	m2	0.00000104
Micro ballons	Resource	Resource, nttc**	g	0.00000146
Na2SO4	Resource	Resource, nttc**	g	0.00000000
Refractory mat.	Resource	Resource, nttc**	g	0.00000036
Silica fume	Resource	Resource, nttc**	g	0.00017866
Scrap (steel)	Resource	Resource, nttc**	g	0.00385399
Solvey soda	Resource	Resource, nttc**	g	0.00000004
Steel	Resource	Resource, nttc**	g	0.00000025
Sulphuric acid	Resource	Resource, nttc**	g	0.00000122

*nftg = not followed to the grave

**nttc = not traced to the cradle

Production of Natural gas

The results are calculated per produced m3 natural gas

Substance	Category	Impactmedia	Unit	Per m3
Waste, dumping &comb.	Emission		g	0.00740000
Waste, hazardous waste	Emission		g	0.01600000
Waste, recycling	Emission		g	0.00243000
Aldehydes	Emission	Air	g	0.00000001
As	Emission	Air	g	0.00000002
Waste, scrap	Emission	Air	g	0.00073536
B(a)P	Emission	Air	g	0.00000000
Benzene	Emission	Air	g	0.00000001
Lead	Emission	Air	g	0.00000000
Ca	Emission	Air	g	0.00000004
Cd	Emission	Air	g	0.00000000
CF4 + C2F6	Emission	Air	g	0.00000011
CH4	Emission	Air	g	0.53134184
CO	Emission	Air	g	0.15593942
CO2	Emission	Air	g	161.29568100
Cr	Emission	Air	g	0.00000004
Cu	Emission	Air	g	0.00000002
F (g)	Emission	Air	g	0.00000013
F (part)	Emission	Air	g	0.00012254
Fe	Emission	Air	g	0.00002438
Phenol	Emission	Air	g	0.00000103
Fluorides	Emission	Air	g	0.00000000
HC	Emission	Air	g	0.00823100
HCl	Emission	Air	g	0.00000410
HF	Emission	Air	g	0.00000553
Hg	Emission	Air	g	0.00000000
Mn	Emission	Air	g	0.00000000
N2O	Emission	Air	g	0.00387614
NH3	Emission	Air	g	0.00000044
NH4NO3	Emission	Air	g	0.00000026
Ni	Emission	Air	g	0.00000003
Nitrates	Emission	Air	g	0.00000000
NMVOC	Emission	Air	g	0.07231363
NOx	Emission	Air	g	0.68589693
PAH	Emission	Air	g	0.00000003
Pb	Emission	Air	g	0.00000013
Rn-222[Bq]	Emission	Air	Bq	0.00306898
SO2	Emission	Air	g	0.01654796
SOx	Emission	Air	g	0.00000148
Dust	Emission	Air	g	0.00063134
TCCD	Emission	Air	g	0.00000000
Tl	Emission	Air	g	0.00000000
V	Emission	Air	g	0.00000006
VOC	Emission	Air	g	0.00000079
Zn	Emission	Air	g	0.00000096
Ashes	Emission	Waste	g	0.00000225
Waste, rocks	Emission	Waste	g	0.00000830
Waste, dust	Emission	Waste	g	0.00032886
Waste, hazardous waste	Emission	Waste	g	0.00003104

Waste, granite	Emission	Waste	g	0.10935060
Waste, non-magnetic	Emission	Waste	g	0.00277493
Waste, coal	Emission	Waste	g	0.00000202
Waste, quartzite	Emission	Waste	g	0.00019817
Waste, LD- dust	Emission	Waste	g	0.00008242
Waste, LD-cinder	Emission	Waste	g	0.00471738
Waste, LD-sludge	Emission	Waste	g	0.00373228
Waste, oil	Emission	Waste	g	0.00042368
Waste, oxide	Emission	Waste	g	0.00025951
Waste, scrap	Emission	Waste	g	0.00025951
Waste, cinder	Emission	Waste	g	0.00252795
Waste, sludge	Emission	Waste	g	0.08033811
Waste, brick	Emission	Waste	g	0.00003803
Bauxite-remains	Emission	Waste	g	0.00028078
Dross fines	Emission	Waste	g	0.00000056
Solid waste	Emission	Waste	g	0.00027333
Inert remains	Emission	Waste	g	0.00002915
NaOH	Emission	Waste	g	0.00207805
Noise	Emission	Waste	g	0.00000000
Radioactive waste h [cm3]	Emission	Waste	ml	0.00000004
Radioactive waste l [cm3]	Emission	Waste	ml	0.00000044
Radioactive waste m [cm3]	Emission	Waste	ml	0.00000044
Skimmings/dross	Emission	Waste	g	0.00000320
Scrap (steel)	Emission	Waste	g	0.00000162
Benzene	Emission	nftg*	g	0.00002504
Coke-gas	Emission	nftg*	g	0.00000017
Coke dust	Emission	nftg*	g	0.00023999
Coal	Emission	nftg*	g	0.00000442
Sulphur	Emission	nftg*	g	0.00000473
Tar	Emission	nftg*	g	0.00008665
Ashes	Emission	Ground	g	0.00007969
Waste, hazardous waste	Emission	Ground	g	0.31000000
Waste, domestic	Emission	Ground	g	0.34999999
Waste, industrial	Emission	Ground	g	0.07000000
Sewage	Emission	Ground	m3	0.00000900
Radioactive waste h [cm3]	Emission	Ground	ml	0.00000028
Radioactive waste l [cm3]	Emission	Ground	ml	0.00000317
Radioactive waste m [cm3]	Emission	Ground	ml	0.00000319
(SO4)2-	Emission	Water	g	0.00000000
As	Emission	Water	g	0.00000000
Barite, bentonite	Emission	Water	g	0.89999998
BOD	Emission	Water	g	0.00000000
Cd	Emission	Water	g	0.00000000
Cl-	Emission	Water	g	0.00000001
CN-tot	Emission	Water	g	0.00000003
Co	Emission	Water	g	0.00000000
COD	Emission	Water	g	0.00021481
Cr	Emission	Water	g	0.00000000
Cu	Emission	Water	g	0.00000001
Cu	Emission	Water	g	0.00000000
Dissolved solids	Emission	Water	g	0.00000791
F-	Emission	Water	g	0.00000000
Fe2+	Emission	Water	g	0.00000000
Phenol	Emission	Water	g	0.00000001
F-tot	Emission	Water	g	0.00000000
H2SO4	Emission	Water	g	0.00000021

HNO3	Emission	Water	g	0.00000001
Chlorides	Emission	Water	g	0.00000071
Mn	Emission	Water	g	0.00000026
Na+	Emission	Water	g	0.00000000
NaCl	Emission	Water	g	0.00000003
NH3	Emission	Water	g	0.00000002
NH4+	Emission	Water	g	0.00000077
NH4-N	Emission	Water	g	0.00000049
NH4NO3	Emission	Water	g	0.00000072
Ni	Emission	Water	g	0.00000000
Nitrates	Emission	Water	g	0.00000000
NO2-	Emission	Water	g	0.00000005
NO3-	Emission	Water	g	0.00000044
NO3-N	Emission	Water	g	0.00000054
N-tot	Emission	Water	g	0.00003775
Oil wastage	Emission	Water	g	0.00007268
Organic coal	Emission	Water	g	0.00000000
PAH	Emission	Water	g	0.00000001
Pb	Emission	Water	g	0.00000004
Prod. chemicals	Emission	Water	g	0.01000000
P-tot	Emission	Water	g	0.00000004
Radioactivity, L6	Emission	Water	g	0.00000001
Suspended solids	Emission	Water	g	0.00000113
Zn	Emission	Water	g	0.00000000
Zn	Emission	Water	g	0.00000000
Other drill-chemicals	Emission	Water	g	0.27000001
Gasoline (e)	Resource	Energy	MJ	0.00000025
Diesel (e)	Resource	Energy	MJ	0.18080345
Diesel, boat (e)	Resource	Energy	MJ	0.00012544
El, Australia av. (e)	Resource	Energy	MJ	0.00000114
El, Denmark av. (e)	Resource	Energy	MJ	0.00252000
El, Sweden av. (e)	Resource	Energy	MJ	0.00025156
El, UCPTE (e)	Resource	Energy	MJ	0.00001309
Coal (e)	Resource	Energy	MJ	0.00000007
LP-gas (propane) (e)	Resource	Energy	MJ	0.00000380
Natural gas (e)	Resource	Energy	MJ	1.99980174
Oil EO1 (e)	Resource	Energy	MJ	0.00006793
Oil EO5 (e)	Resource	Energy	MJ	0.00000857
Use of acreage	Resource	Ground	m2	0.00000006
Isolation zone	Resource	Ground	m2	0.00000406
THT (C4H8S)	Resource	R ej v	g	0.01500000
Bauxite (r)	Resource	Resource	g	0.00097997
Bentonite (r)	Resource	Resource	g	0.00071468
Biomass (r)	Resource	Resource	g	0.00348378
Pyrolusite (r)	Resource	Resource	g	0.00295146
Caliche (r)	Resource	Resource	g	0.00021012
Caliche (r)	Resource	Resource	g	0.00072325
CaSO4 x 2 H20 (r)	Resource	Resource	g	0.00164277
Dolomite (r)	Resource	Resource	g	0.00000025
Dolomite (r)	Resource	Resource	g	0.00059843
Felspar (r)	Resource	Resource	g	0.00000018
Iron-ore (r)	Resource	Resource	g	0.23584406
Limestone (r)	Resource	Resource	g	1.99088834
Coal (r)	Resource	Resource	g	0.18879818
Coal, feedstock (r)	Resource	Resource	g	0.06940077
Quartzite (r)	Resource	Resource	g	0.00137763

LP-gas (r)	Resource	Resource	g	0.00006162
N2	Resource	Resource	g	0.01700000
Na2CO3 (r)	Resource	Resource	g	0.00000029
Natural gas (r)	Resource	Resource	g	44.90557290
Natural gas (r)	Resource	Resource	g	839.99997377
Natural gas, feedstock (r)	Resource	Resource	g	0.00009774
Natural uranium (r)	Resource	Resource	g	0.00000076
Olivine (r)	Resource	Resource	g	0.00513225
Crude oil (r)	Resource	Resource	g	4.55139827
Crude oil, feedstock (r)	Resource	Resource	g	0.00015645
Salt	Resource	Resource	g	0.00013487
Sand (r)	Resource	Resource	g	0.00000202
Seawater (r)	Resource	Resource	g	0.03630694
Glass and such-like	Resource	Resource	g	0.01000000
Rock (r)	Resource	Resource	g	0.21593859
Peat (r)	Resource	Resource	g	0.00004777
Water (r)	Resource	Resource	g	0.02396785
Water power [MJe] (r)	Resource	Resource	MJ	0.00011108
Additives	Resource	Resurs nttc**	g	0.00049201
Aluminum hydroxide	Resource	Resurs nttc**	g	0.00000312
Aluminum powder	Resource	Resurs nttc**	g	0.00001591
Waste, refractory	Resource	Resurs nttc**	g	0.00000304
Emulsifier	Resource	Resurs nttc**	g	0.00000137
Emulsifier	Resource	Resurs nttc**	g	0.00000397
Grind.aid	Resource	Resurs nttc**	g	0.00000034
Grind.media 1	Resource	Resurs nttc**	g	0.00000139
Grind.media 2	Resource	Resurs nttc**	g	0.00000137
Gravel	Resource	Resurs nttc**	g	0.80000001
Gravel	Resource	Resurs nttc**	g	1.06670003
Iron sulphate	Resource	Resurs nttc**	g	0.00031434
Calcium fluoride	Resource	Resurs nttc**	g	0.00000677
Coal	Resource	Resurs nttc**	g	0.00000818
Acreage	Resource	Resurs nttc**	m2	0.00000667
Micro ballons	Resource	Resurs nttc**	g	0.00000939
Na2SO4	Resource	Resurs nttc**	g	0.00000001
Refractory mat.	Resource	Resurs nttc**	g	0.00000229
Silica fume	Resource	Resurs nttc**	g	0.00114803
Scrap (steel)	Resource	Resurs nttc**	kg	0.02476497
Solvey soda	Resource	Resurs nttc**	g	0.00000029
Steel	Resource	Resurs nttc**	g	0.00000162
Sulphuric acid	Resource	Resurs nttc**	g	0.00000783

*nftg = not followed to the grave

**nttc = not traced to the cradle

Production of 1 kg sulphuric acid (HNO3)

Substance	Category	Impactmedia	Unit	Per produced kg HNO3
Aldehydes	Emission	Air	g	0.00014500
CH4	Emission	Air	g	1.64686729
CO	Emission	Air	g	2.47520990
CO2	Emission	Air	g	474.60884610
Phenol	Emission	Air	g	0.00000875
Fluorides	Emission	Air	g	0.00000056
HC	Emission	Air	g	0.14556602
N2O	Emission	Air	g	0.00830068
NH3	Emission	Air	g	0.63212740
Nitrates	Emission	Air	g	0.00001771
NOx	Emission	Air	g	6.47742342
SO2	Emission	Air	g	0.56445086
Dust	Emission	Air	g	0.16466890
VOC	Emission	Air	g	1.47490003
Ashes	Emission	Ground	g	0.00000247
Radioactive waste h [cm3]	Emission	Ground	ml	0.00017928
Radioactive waste l [cm3]	Emission	Ground	ml	0.00203687
Radioactive waste m [cm3]	Emission	Ground	ml	0.00203687
(SO4)2-	Emission	Water	g	0.00001579
BOD	Emission	Water	g	0.00000829
COD	Emission	Water	g	0.00020503
Dissolved solids	Emission	Water	g	0.01772679
F-	Emission	Water	g	0.00007482
Fe2+	Emission	Water	g	0.00000017
Na+	Emission	Water	g	0.00001226
NH3	Emission	Water	g	0.00003469
N-tot	Emission	Water	g	0.00002939
Oil wastage	Emission	Water	g	0.00032775
Suspended solids	Emission	Water	g	0.00000829
Electricity, UCPTE (e)	Resource	Energy	MJ	0.20167001
Natural gas (e)	Resource	Energy	MJ	0.56889000
Oil EO1 (e)	Resource	Energy	MJ	0.15351000
Oil EO5 (e)	Resource	Energy	MJ	0.00035300
Coal (r)	Resource	Resource	g	5.78994566
Natural gas (r)	Resource	Resource	g	14.12860360
Natural gas, feedstock (r)	Resource	Resource	g	182.04479972
Natural uranium (r)	Resource	Resource	g	0.00042149
Crude oil (r)	Resource	Resource	g	5.46202327
Water power [MJe] (r)	Resource	Resource	MJ	0.04073734

Production of 1 kg copper (50/80)

Here 50 % of the input material into the process is recycled copper

Substance	Category	Impactmedia	Unit	Per produced kg copper
Aldehydes	Emission	Air	g	0.01041480
As	Emission	Air	g	0.00246681
Cd	Emission	Air	g	0.00083480
CH4	Emission	Air	g	0.17511205
CO	Emission	Air	g	2.21919228
CO2	Emission	Air	g	2052.08501840
Cr	Emission	Air	g	0.00000296
Cu	Emission	Air	g	0.02570884
Dioxine	Emission	Air	g	0.00000000
Phenol	Emission	Air	g	0.00005625
Fluorides	Emission	Air	g	0.00004027
HC	Emission	Air	g	8.94417441
Hg	Emission	Air	g	0.00017124
N2O	Emission	Air	g	0.29370024
NH3	Emission	Air	g	0.06625471
NH4NO3	Emission	Air	g	0.03312125
Nitrates	Emission	Air	g	0.00127179
NOx	Emission	Air	g	8.26557419
Pb	Emission	Air	g	0.04121883
SO2	Emission	Air	g	19.66416648
SOx	Emission	Air	g	0.06960946
Dust	Emission	Air	g	1.29511330
VOC	Emission	Air	g	0.10412799
Zn	Emission	Air	g	0.01893530
Waste, rock	Emission	Waste	g	73188.95374283
Waste, dust	Emission	Waste	g	20.64042844
Waste. copper-ore	Emission	Waste	g	11209.20690443
Waste, sand	Emission	Waste	g	95930.81006148
Ashes	Emission	Ground	g	0.00495663
Radioactive waste h [cm3]	Emission	Ground	ml	0.01297285
Radioactive waste l [cm3]	Emission	Ground	ml	0.14738298
Radioactive waste m [cm3]	Emission	Ground	ml	0.14739006
(SO4)2-	Emission	Water	g	0.00113418
As	Emission	Water	g	0.00304590
BOD	Emission	Water	g	0.00059534
Cd	Emission	Water	g	0.00013710
Cl-	Emission	Water	g	0.00171671
COD	Emission	Water	g	0.00528456
Cu	Emission	Water	g	0.00267992
Cu	Emission	Water	g	0.00000321
Dissolved solids	Emission	Water	g	1.27324127
F-	Emission	Water	g	0.00537398
Fe2+	Emission	Water	g	0.00001207
Fluorides	Emission	Water	g	0.00044737
Hg	Emission	Water	g	0.00007632
HNO3	Emission	Water	g	0.00166305
Na+	Emission	Water	g	0.00088069
NaCl	Emission	Water	g	0.00568325
NH3	Emission	Water	g	0.00335790

NH4-N	Emission	Water	g	0.08137081
NH4NO3	Emission	Water	g	0.12111847
Ni	Emission	Water	g	0.00038023
NO3-N	Emission	Water	g	0.09105060
N-tot	Emission	Water	g	0.02658461
Oil wastage	Emission	Water	g	0.02031297
Pb	Emission	Water	g	0.00020182
Suspended solids	Emission	Water	g	0.00059534
Zn	Emission	Water	g	0.00783426
Zn	Emission	Water	g	0.00002299
Diesel (e)	Resource	Energy	MJ	2.06810906
Diesel, boat (e)	Resource	Energy	MJ	0.03307851
Electricity, Sweden av. (e)	Resource	Energy	MJ	0.07080772
Electricity, UCPTE (e)	Resource	Energy	MJ	14.48511154
Natural gas (e)	Resource	Energy	MJ	0.04016365
Oil EO1 (e)	Resource	Energy	MJ	0.76518733
Oil EO5 (e)	Resource	Energy	MJ	0.00001279
Biomass (r)	Resource	Resource	g	0.16639814
Caliche (r)	Resource	Resource	g	121.11847706
Coal (r)	Resource	Resource	g	415.97233462
Copper-ore [0,35%Cu](r)	Resource	Resource	g	96797.98453387
Natural gas (r)	Resource	Resource	g	87.92092337
Natural gas, feedstock (r)	Resource	Resource	g	12.85236842
Natural uranium (r)	Resource	Resource	g	0.03050047
Crude oil (r)	Resource	Resource	g	179.78933898
Crude oil, feedstock (r)	Resource	Resource	g	2.97111131
Peat (r)	Resource	Resource	g	0.01486962
Water (r)	Resource	Resource	g	2394.32185178
Water power [MJe] (r)	Resource	Resource	MJ	2.95927226
Aluminum powder	Resource	Resurs nttc*	g	1.99645839
Emulsifier	Resource	Resurs nttc*	g	0.66548613
Micro ballons	Resource	Resurs nttc*	g	1.57296721

*nttc = not trced to the cradle

Recycling process of 1 kg copper (50/80)

Here 80 % of the copper out of the process is recycled

Substance	Category	Impactmedia	Unit	Per recycled kg copper
As	Emission	Air	g	0.00151056
Bly	Emission	Air	g	0.02984000
Cd	Emission	Air	g	0.00084803
CH4	Emission	Air	g	0.00330589
CO	Emission	Air	g	0.03553955
CO2	Emission	Air	g	73.51246728
Cu	Emission	Air	g	0.02167859
Dioxine	Emission	Air	g	0.00000000
Phenol	Emission	Air	g	0.00004388
HC	Emission	Air	g	0.01930437
Hg	Emission	Air	g	0.00023964
N2O	Emission	Air	g	0.00090169
NOx	Emission	Air	g	0.38446819
Pb	Emission	Air	g	0.01244156
SO2	Emission	Air	g	5.80486560
Dust	Emission	Air	g	0.27247917
Zn	Emission	Air	g	0.02576085
Waste, dust	Emission	Waste	g	12.28399982
Copper, landfill	Emission	Waste	g	349.99999404
Ashes	Emission	Ground	g	0.06522447
Radioactive waste h [cm3]	Emission	Ground	ml	0.00125790
Radioactive waste l [cm3]	Emission	Ground	ml	0.01425621
Radioactive waste m [cm3]	Emission	Ground	ml	0.01434938
Cl-	Emission	Water	g	0.00254800
COD	Emission	Water	g	0.00102002
Fluorides	Emission	Water	g	0.00066400
N-tot	Emission	Water	g	0.00016652
Oil wastage	Emission	Water	g	0.00034872
Diesel (e)	Resource	Energy	MJ	0.11340849
Electricity, Sweden av. (e)	Resource	Energy	MJ	0.93177818
Oil EO1 (e)	Resource	Energy	MJ	0.75839999
Biomass (r)	Resource	Resource	g	2.18967862
Coal (r)	Resource	Resource	g	1.37903172
Natural gas (r)	Resource	Resource	g	1.36635504
Natural uranium (r)	Resource	Resource	g	0.00298169
Crude oil (r)	Resource	Resource	g	23.96002418
Peat (r)	Resource	Resource	g	0.19567341
Water power [MJe] (r)	Resource	Resource	MJ	0.43793574

Production of 1 kg steel (25/70)

Here 25 % of the input material into the process is recycled steel

Substance	Category	Impactmedia	Unit	Per produced kg steel
Aldehydes	Emission	Air	g	0.00003129
As	Emission	Air	g	0.00006586
Waste, scrap	Emission	Air	g	2.46529491
Ca	Emission	Air	g	0.00025000
Cd	Emission	Air	g	0.00000562
CF4 + C2F6	Emission	Air	g	0.00035724
CH4	Emission	Air	g	1.71964430
CO	Emission	Air	g	0.14775724
CO2	Emission	Air	g	350.19384109
Cr	Emission	Air	g	0.00014338
Cu	Emission	Air	g	0.00007161
F (g)	Emission	Air	g	0.00044654
F (part)	Emission	Air	g	0.41082045
Fe	Emission	Air	g	0.08286625
Phenol	Emission	Air	g	0.00000249
Fluorides	Emission	Air	g	0.00000012
HC	Emission	Air	g	0.12273447
HCl	Emission	Air	g	0.01375289
HF	Emission	Air	g	0.01854483
Hg	Emission	Air	g	0.00000000
Mn	Emission	Air	g	0.00000824
N2O	Emission	Air	g	0.00451840
NH3	Emission	Air	g	0.00014693
NH4NO3	Emission	Air	g	0.00020415
Ni	Emission	Air	g	0.00008985
Nitrates	Emission	Air	g	0.00000382
NOx	Emission	Air	g	0.58150158
PAH	Emission	Air	g	0.00009382
Pb	Emission	Air	g	0.00051828
Rn-222[Bq]	Emission	Air	Bq	2.62468646
SO2	Emission	Air	g	0.87641523
SOx	Emission	Air	g	0.00027263
Dust	Emission	Air	g	0.58743421
V	Emission	Air	g	0.00019996
VOC	Emission	Air	g	0.00057190
Zn	Emission	Air	g	0.00410861
Ashes	Emission	Waste	g	0.00747544
Waste, rock	Emission	Waste	g	0.02784121
Waste, dust	Emission	Waste	g	1.83749998
Waste, hazardous waste	Emission	Waste	g	0.00090269
Waste, granite	Emission	Waste	g	367.09830681
Waste, non-magnetic	Emission	Waste	g	9.30299934
Waste, coal	Emission	Waste	g	0.00678747
Waste, LD- dust	Emission	Waste	g	0.27629907
Waste, LD-cinder	Emission	Waste	g	15.81509888
Waste, LD-sludge	Emission	Waste	g	12.51253394
Waste, oxide	Emission	Waste	g	1.45000005
Waste, scrap	Emission	Waste	g	1.45000005
Waste, cinder	Emission	Waste	g	14.12500000

Waste, sludge	Emission	Waste	g	3.87337586
Waste, brick	Emission	Waste	g	0.21250001
Bauxite-remains	Emission	Waste	g	0.94131520
Dross fines	Emission	Waste	g	0.00187548
Solid waste	Emission	Waste	g	0.91634545
Inert remains	Emission	Waste	g	0.09773010
NaOH	Emission	Waste	g	6.96671235
Radioactive waste h [cm3]	Emission	Waste	ml	0.00013034
Radioactive waste l [cm3]	Emission	Waste	ml	0.00147592
Radioactive waste m [cm3]	Emission	Waste	ml	0.00148550
Skimmings/dross	Emission	Waste	g	0.01071705
Scrap (steel)	Emission	Waste	g	0.00544784
Coal	Emission	nftg*	g	0.01482526
Ashes	Emission	Ground	g	0.04540168
Radioactive waste h [cm3]	Emission	Ground	ml	0.00091427
Radioactive waste l [cm3]	Emission	Ground	ml	0.01036284
Radioactive waste m [cm3]	Emission	Ground	ml	0.01042770
(SO4)2-	Emission	Water	g	0.00000341
As	Emission	Water	g	0.00000069
BOD	Emission	Water	g	0.00000179
Cd	Emission	Water	g	0.00000000
Cl-	Emission	Water	g	0.00001926
CN-tot	Emission	Water	g	0.00008107
Co	Emission	Water	g	0.00000107
COD	Emission	Water	g	0.00923434
Cr	Emission	Water	g	0.00001189
Cu	Emission	Water	g	0.00003376
Cu	Emission	Water	g	0.000000001
Dissolved solids	Emission	Water	g	0.00962330
F-	Emission	Water	g	0.00001615
Fe2+	Emission	Water	g	0.00000004
Phenol	Emission	Water	g	0.00004261
F-tot	Emission	Water	g	0.00000089
H2SO4	Emission	Water	g	0.00071447
HNO3	Emission	Water	g	0.00001026
Chlorides	Emission	Water	g	0.00238504
Mn	Emission	Water	g	0.00086081
Na+	Emission	Water	g	0.00001491
NaCl	Emission	Water	g	0.00000099
NH3	Emission	Water	g	0.00002419
NH4+	Emission	Water	g	0.00242794
NH4-N	Emission	Water	g	0.00001417
NH4NO3	Emission	Water	g	0.00002109
Ni	Emission	Water	g	0.00000993
Nitrates	Emission	Water	g	0.00000000
NO2-	Emission	Water	g	0.00015508
NO3-	Emission	Water	g	0.00135133
NO3-N	Emission	Water	g	0.00001586
N-tot	Emission	Water	g	0.01013491
Oil wastage	Emission	Water	g	0.00056669
Organic coal	Emission	Water	g	0.00000179
PAH	Emission	Water	g	0.00001786
Pb	Emission	Water	g	0.00013360
P-tot	Emission	Water	g	0.00012414
Suspended solids	Emission	Water	g	0.00366446
Zn	Emission	Water	g	0.00001457

Zn	Emission	Water	g	0.000000004
Gasoline (e)	Resource	Energy	MJ	0.00083641
Diesel (e)	Resource	Energy	MJ	0.05473044
Diesel, boat (e)	Resource	Energy	MJ	0.37563046
Electricity, Australia av. (e)	Resource	Energy	MJ	0.00381445
Electricity, Sweden av. (e)	Resource	Energy	MJ	0.74489811
Electricity, UCPTE (e)	Resource	Energy	MJ	0.04365590
Coal (e)	Resource	Energy	MJ	0.00023171
LP-gas (propane) (e)	Resource	Energy	MJ	0.01275189
Natural gas (e)	Resource	Energy	MJ	0.00495182
Oil EO1 (e)	Resource	Energy	MJ	0.21581064
Oil EO5 (e)	Resource	Energy	MJ	0.01912194
Bauxite (r)	Resource	Resource	g	3.28538021
Bentonite (r)	Resource	Resource	g	2.39597346
Biomass (r)	Resource	Resource	g	1.75298857
Pyrolusite (r)	Resource	Resource	g	11.17920462
Caliche (r)	Resource	Resource	g	0.62343229
Caliche (r)	Resource	Resource	g	0.02109183
Dolomite (r)	Resource	Resource	g	0.00073317
Dolomite (r)	Resource	Resource	g	3.34375002
Felspar (r)	Resource	Resource	g	0.00054149
Iron-ore (r)	Resource	Resource	g	790.67124259
Limestone (r)	Resource	Resource	g	56.32492013
Coal (r)	Resource	Resource	g	2.60119859
Coal, feedstock (r)	Resource	Resource	g	220.54940642
LP-gas (r)	Resource	Resource	g	0.20657156
Na2CO3 (r)	Resource	Resource	g	0.00085776
Natural gas (r)	Resource	Resource	g	1.26935095
Natural gas, feedstock (r)	Resource	Resource	g	0.06453345
Natural uranium (r)	Resource	Resource	g	0.00247502
Olivine (r)	Resource	Resource	g	17.20596378
Crude oil (r)	Resource	Resource	g	16.62439055
Crude oil, feedstock (r)	Resource	Resource	g	0.46319140
Salt	Resource	Resource	g	0.45215708
Sand (r)	Resource	Resource	g	0.00598993
Seawater (r)	Resource	Resource	g	1.05577728
Peat (r)	Resource	Resource	g	0.15642381
Water (r)	Resource	Resource	g	7.76831020
Water power [MJel] (r)	Resource	Resource	MJ	0.35933714
Aluminum hydroxide	Resource	Resource nttc**	g	0.01044913
Aluminum powder	Resource	Resource nttc**	g	0.01218378
Waste, refractory	Resource	Resource nttc**	g	0.01018120
Emulsifier	Resource	Resource nttc**	g	0.00407316
Emulsifier	Resource	Resource nttc**	g	0.00011589
Calcium fluoride	Resource	Resource nttc**	g	0.02268443
Coal	Resource	Resource nttc**	g	0.02741780
Micro ballons	Resource	Resource nttc**	g	0.00027392
Na2SO4	Resource	Resource nttc**	g	0.00004423
Refractory mat.	Resource	Resource nttc**	g	0.00768056
Scrap (steel)	Resource	Resource nttc**	g	138.37499917
Solvey soda	Resource	Resource nttc**	g	0.00085776
Steel	Resource	Resource nttc**	g	0.00544784
Sulphuric acid	Resource	Resource nttc**	g	0.02625678

*nftg = not followed to the grave
**nttc = not traced to the cradle

Recycling process of 1 kg steel (25/70)

Here 70 % of the steel out of the process is recycled

Substance	Category	Impactmedia	Unit	Per recycled kg steel
Aldehydes	Emission	Air	g	0.00000021
As	Emission	Air	g	0.00000008
Ca	Emission	Air	g	0.00063234
CH4	Emission	Air	g	0.02659314
CO	Emission	Air	g	0.05612992
CO2	Emission	Air	g	17.70727335
Cr	Emission	Air	g	0.00004437
Cu	Emission	Air	g	0.00004451
Fe	Emission	Air	g	0.00734026
Phenol	Emission	Air	g	0.00000106
Fluorides	Emission	Air	g	0.00000000
HC	Emission	Air	g	0.02044672
N2O	Emission	Air	g	0.00106820
NH3	Emission	Air	g	0.00000198
NH4NO3	Emission	Air	g	0.00000092
Ni	Emission	Air	g	0.00000664
Nitrates	Emission	Air	g	0.00000003
NOx	Emission	Air	g	0.22649857
PAH	Emission	Air	g	0.00000037
Pb	Emission	Air	g	0.00064815
Rn-222[Bq]	Emission	Air	Bq	0.03282903
SO2	Emission	Air	g	0.02505057
SOx	Emission	Air	g	0.00000128
Dust	Emission	Air	g	0.01475756
V	Emission	Air	g	0.00000002
VOC	Emission	Air	g	0.00000290
Zn	Emission	Air	g	0.00556507
Waste, dust	Emission	Waste	g	4.64769637
Waste, hazardous waste	Emission	Waste	g	0.00014462
Waste, granite	Emission	Waste	g	3.15252918
Waste, oxide	Emission	Waste	g	3.66757009
Waste, scrap	Emission	Waste	g	3.66757009
Waste, cinder	Emission	Waste	g	35.72719021
Waste, sludge	Emission	Waste	g	0.37226244
Waste, brick	Emission	Waste	g	0.53748872
Steel, landfill	Emission	Waste	g	524.99997616
Ashes	Emission	Ground	g	0.03547247
Radioactive waste h [cm3]	Emission	Ground	ml	0.00068435
Radioactive waste l [cm3]	Emission	Ground	ml	0.00775603
Radioactive waste m [cm3]	Emission	Ground	ml	0.00780670
(SO4)2-	Emission	Water	g	0.00000002
BOD	Emission	Water	g	0.00000001
CN-tot	Emission	Water	g	0.00000056
COD	Emission	Water	g	0.00028166
Cr	Emission	Water	g	0.00000003
Cu	Emission	Water	g	0.00000004
Cu	Emission	Water	g	0.00000000009
Dissolved solids	Emission	Water	g	0.00009789
F-	Emission	Water	g	0.00000011

Fe2+	Emission	Water	g	0.00000000
Phenol	Emission	Water	g	0.00000006
HNO3	Emission	Water	g	0.00000005
Na+	Emission	Water	g	0.00000002
NaCl	Emission	Water	g	0.00000016
NH3	Emission	Water	g	0.00000022
NH4+	Emission	Water	g	0.00000917
NH4-N	Emission	Water	g	0.00000226
NH4NO3	Emission	Water	g	0.00000337
Ni	Emission	Water	g	0.00000002
NO3-N	Emission	Water	g	0.00000253
N-tot	Emission	Water	g	0.00009623
Oil wastage	Emission	Water	g	0.00019394
Pb	Emission	Water	g	0.00000001
P-tot	Emission	Water	g	0.00000068
Suspended solids	Emission	Water	g	0.00029944
Zn	Emission	Water	g	0.00001296
Zn	Emission	Water	g	0.0000000006
Diesel (e)	Resource	Energy	MJ	0.18267330
Diesel, boat (e)	Resource	Energy	MJ	0.00232989
Electricity, Sweden av. (e)	Resource	Energy	MJ	0.50673837
Electricity, UCPTE (e)	Resource	Energy	MJ	0.00029034
Natural gas (e)	Resource	Energy	MJ	0.00000112
Oil EO1 (e)	Resource	Energy	MJ	0.00003607
Oil EO5 (e)	Resource	Energy	MJ	0.00011221
Biomass (r)	Resource	Resource	g	1.19083513
Pyrolusite (r)	Resource	Resource	g	8.12177009
Caliche (r)	Resource	Resource	g	0.00336947
Dolomite (r)	Resource	Resource	g	8.45754288
Limestone (r)	Resource	Resource	g	8.79584459
Coal (r)	Resource	Resource	g	0.75830854
Coal, feedstock (r)	Resource	Resource	g	2.76062289
Natural gas (r)	Resource	Resource	g	0.26375729
Natural gas, feedstock (r)	Resource	Resource	g	0.00035755
Natural uranium (r)	Resource	Resource	g	0.00162217
Crude oil (r)	Resource	Resource	g	5.50305076
Crude oil, feedstock (r)	Resource	Resource	g	0.00008266
Seawater (r)	Resource	Resource	g	0.16915086
Peat (r)	Resource	Resource	g	0.10641506
Water (r)	Resource	Resource	g	0.00026710
Water power [MJe] (r)	Resource	Resource	MJ	0.23822568
Aluminum powder	Resource	Resource nttc*	g	0.00005554
Emulsifier	Resource	Resource nttc*	g	0.00001851
Micro ballons	Resource	Resource nttc*	g	0.00004376

* nttc = not traced to cradle

Appendix E - Presentation of impact assessment results

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Impact Assessment of the total production of the SOFC-module

Here the energy used during the production of the SOFC raw materials is maximum (see appendix B)

Substance	Category	Impactmedia	AOD	AP	GWP 100	POCP
Aldehydes	Emission	Air				0.00002314
CH4	Emission	Air			16.11679770	0.00537227
CO	Emission	Air				0.00309346
CO2	Emission	Air			298.04327321	
Phenol	Emission	Air				0.00000056
HC	Emission	Air				0.03628304
N2O	Emission	Air			3.11353886	
NH3	Emission	Air		0.00005444		
NOx	Emission	Air		0.00678587		
SO2	Emission	Air		0.01166585		
VOC	Emission	Air				0.00079930
COD	Emission	Water	0.00029949			
N-tot	Emission	Water	0.00090112			
		Total:	**0.00120**	**0.01851**	**317.27361**	**0.04557**

Impact Assessment of the total production of the gas turbine

Substance	Category	Impactmedia	AOD	AP	GWP 100	POCP
Aldehyder	Emission	Air				0.00000002
CF4 + C2F6	Emission	Air			0.00005242	
CH4	Emission	Air			0.00120031	0.00000040
CO	Emission	Air				0.00000057
CO2	Emission	Air			0.02178090	
Fenol	Emission	Air				0.00000000
HC	Emission	Air				0.00001939
HCl	Emission	Air		0.00000001		
N2O	Emission	Air			0.00048319	
NH3	Emission	Air		0.00000002		
NOx	Emission	Air		0.00000151		
SO2	Emission	Air		0.00000374		
VOC	Emission	Air				0.00000019
COD	Emission	Water	0.00000034			
N-tot	Emission	Water	0.00000912			
		Total:	**0.00001**	**0.00001**	**0.02352**	**0.00002**

Impact Assessment of the total construction phase

Substance	Category	Impactmedia	AOD	AP	GWP 100	POCP
CH4	Emission	Air			0.00239935	0.00000080
CO	Emission	Air				0.00001208
CO2	Emission	Air			0.33562404	
Fenol	Emission	Air				0.00000001
HC	Emission	Air				0.00010600
N2O	Emission	Air			0.00745445	
NOx	Emission	Air		0.00007496		
SO2	Emission	Air		0.00000299		
COD	Emission	Water	0.00000905			
N-tot	Emission	Water	0.00002910			
		Total:	**0.00004**	**0.00008**	**0.34548**	**0.00012**

Impact Assessment of the total demolition phase

Substance	Category	Impactmedia	AOD	AP	GWP 100	POCP
CH4	Emission	Air			0.00245909	0.00000082
CO	Emission	Air				0.00001155
CO2	Emission	Air			0.34377080	
Fenol	Emission	Air				0.00000001
HC	Emission	Air				0.00010671
N2O	Emission	Air			0.00770608	
NOx	Emission	Air		0.00007573		
SO2	Emission	Air		0.00000273		
COD	Emission	Water	0.00000943			
N-tot	Emission	Water	0.00007357			
		Total:	**0.00008**	**0.00008**	**0.35394**	**0.00012**

Impact Assessment of the total operation phase

Substance	Category	Impactmedia	AOD	AP	GWP 100	POCP
Aldehyder	Emission	Air				0.00000000
Bensen	Emission	Air				0.00000000
CF4 + C2F6	Emission	Air			0.00007462	
CH4	Emission	Air			1.73646446	0.00057882
CO	Emission	Air				0.00077657
CO2	Emission	Air			297.30978031	
Fenol	Emission	Air				0.00000007
HC	Emission	Air				0.00053287
HCl	Emission	Air		0.00000002		
N2O	Emission	Air			0.18699651	
NH3	Emission	Air		0.00000000		
NOx	Emission	Air		0.00234830		
SO2	Emission	Air		0.00007983		
VOC	Emission	Air				0.00000005
COD	Emission	Water	0.00003343			
N-tot	Emission	Water	0.00011749			
		Total:	**0.00015**	**0.00243**	**299.2333**	**0.00189**